



MSA University
Faculty of Engineering

ECE552 Mobile Communication Systems

Chapter 1 Principles of Wireless Communication Systems

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1.1 Wireless Communication Systems

Depending on the **communication channel**, the communication system is categorized as follows:

1. Wired (Line communication)

Parallel wire communication

Twisted wire communication

Coaxial cable communication

Optical fiber communication

2. Wireless (Space communication)

Ground wave communication

Skywave communication

Space wave communication

Satellite communication

WIRED VS. WIRELESS

1. Wired System

Advantages

- Security: Security issues prior to that of wireless networks.
- Speed: It is a bit faster than wireless networks, as they are less likely to be affected by walls, other devices, and other forms of interference.
- Little interference and signal attenuation.
- Higher data rates.

Disadvantages

- Maintenance: To repair issues, you must first determine which wires, or devices are causing the issues before they can begin working.
- Cable Clutter: Without proper cable management, you could find yourself in an office cluttered with cables.
- Mobility: No mobility exists.

2. Wireless System

Advantages

- BYOD (Bring Your Own Device) policies in the workplace are becoming more and more popular.
- Mobility: It exists for the devices.
- Organization: Plain and simple, there are fewer wires
- Maintenance is easier since most of it is software-based.

Disadvantages

- Security: It is easier for unauthorized individuals to hack your signal.
- Speed: Wireless networks are more affected by outside interference, including walls, other devices, etc. As a result, wireless connectivity speeds tend to be slower than wired network speeds.
- High noise, interference, and signal attenuation.

1.1 Wireless Communication Systems

1.1.1 Types of Wireless Communication Systems

- **Wireless communications** is a type of data communication that is performed and delivered wirelessly. This is a broad term that incorporates all procedures and forms of connecting and communicating between two or more devices using a wireless signal through wireless communication technologies and devices.

- Wireless Communication System is any electrical communication system that uses a naturally occurring communication channel, such as air or water.
- There are different wireless communication systems, which include:
  - **TV and Radio** broadcasting.
  - **Cordless** phone.
  - **Satellite** Communications.
  - **Radar**.
  - **Mobile** Telephone System (Cellular Communication)
  - Global Positioning System **(GPS).**
  - Wireless Personal Area Network **(WPAN): Infrared** and **Bluetooth** Comm.
  - Wireless Local Area Network **(WLAN)** such as Wireless Fidelity **(WiFi).**
  - Wireless Wide Area Network **(WWAN)** such as Worldwide Interoperability for Microwave Access **(WIMAX).**

- **Mobile** is wireless communication systems focus to communicate over the air via radio-waves and support some forms of user mobility.

1.1 Wireless Communication Systems

1.1.2 Block Diagram of Typical Wireless digital communication system

- The general block diagram of a digital communication system is shown in Fig. 1.1. It consists of the following elements:

1. Transmitter (Tx) includes an input sensor to convert messages into electric signal (baseband) then the high frequency part converts baseband signal into form appropriate for channel transmission (bandpass signal).

So, the transmitters format, source encode, encrypt, channel encode, multiplex, pulse modulate, bandpass modulate, frequency spread, multiple access and finally transmit the signal over the specified high frequency carrier.

2. Channel is the medium between a transmitter and a receiver in which the bandpass signal is going to pass through. Several factors affecting the channel capacity such as: attenuation, noise, interference, fading, .. Etc

3. Receiver (Rx) processes the received signal to estimate the baseband signal and then an output sensor that convert baseband signal to a message similar to the transmitted one.

- The ***modulate*** and ***demodulate/detect*** blocks together are called a ***modem***. The term "modem" often encompasses several of the signal processing steps shown in Figs. 1.1 and 1.2; when this is the case, the modem can be thought of as the "brain" of the system.

1.1 Wireless Communication Systems

1.1.2 Block Diagram of Typical Wireless digital communication system

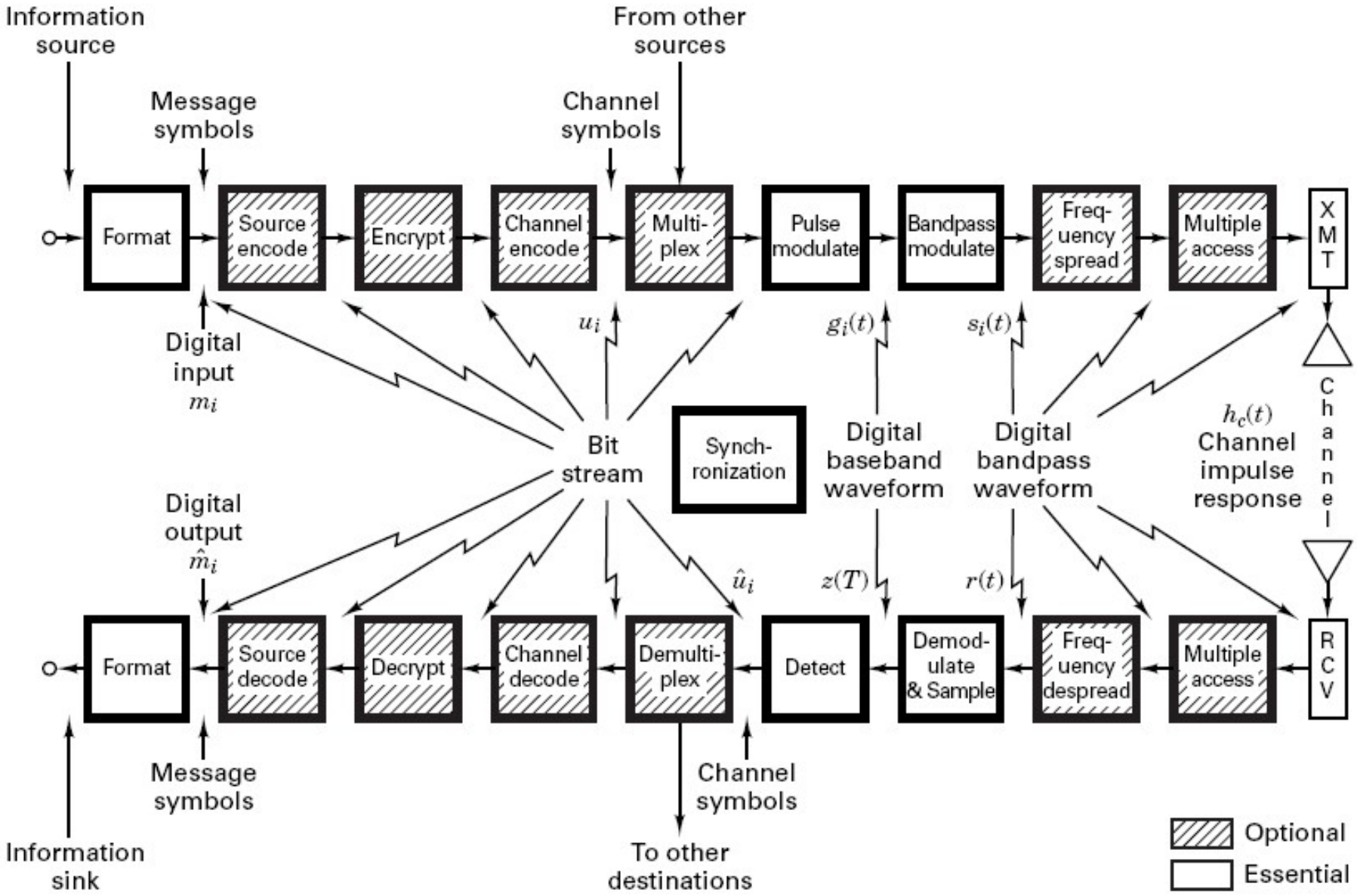


Fig. 1.1 Block diagram of a typical digital communication system.

Elements of Communication Systems

The definitions of the terms used in the communication system are discussed below.

- **Information**

Message or information is the entity that is to be transmitted. It can be in the form of audio, video, temperature, picture, pressure, etc.

- **Signal**

The single-valued function of time carries the information. The information is converted into an electrical form for transmission.

- **Transducer**

It is a device or an arrangement that **converts one form of energy to the other.** An electrical transducer converts physical variables such as pressure, force, and temperature into corresponding electrical signal variations. For example, **a microphone converts audio signals into electrical signals**. The **photodetector converts light signals into electrical signals**.

- **Amplifier**

The electronic circuit or device that increases the amplitude or the strength of the transmitted signal is called an amplifier. When the signal strength becomes less than the required value, amplification can be done anywhere between the transmitter and receiver. A DC power source will be provided for the amplification.

- **Source coding and channel coding**

Source coding compresses data to maximize efficiency by removing redundancy, while channel coding adds controlled redundancy to safeguard data integrity against transmission errors over noisy channels

- **Modulator**

As the original message signal cannot be transmitted over a large distance because of their low frequency and amplitude, **they are superimposed with high frequency and amplitude waves called carrier** waves. This phenomenon of superimposing message signals with a carrier wave is called **modulation**, and the resultant wave is a modulated wave that is to be transmitted.

There are different types of modulation, and they are explained below:

1.2 General Wireless Communication Background

Digital Bandpass Modulation Techniques

- Digital modulation is the process by which digital symbols are transformed into waveforms that are compatible with the characteristics of the channel. In general, any signal, s(t) can be described as:

$$s(t) = A(t)\cos(\omega(t)t + \phi(t))$$

Where A(t) is the signal amplitude, ω(t) and ϕ(t) are its frequency and phase respectively.

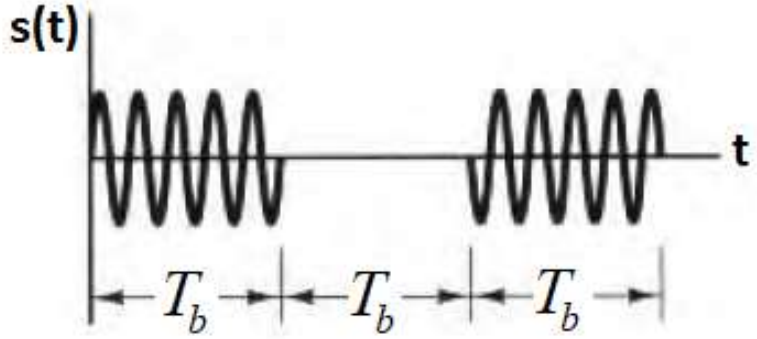
1. Amplitude Shift Keying (ASK) [Fig. 1.11]

- The amplitude term, A(t) has M discrete values while the frequency term, ω_0 and phase term, ϕ are arbitrary constants.

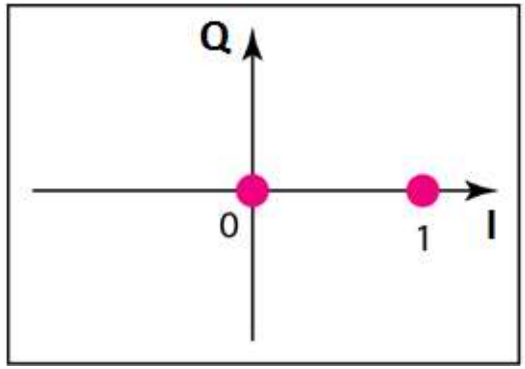
$$s_i(t) = \sqrt{2E_{bi}(t)/T_b}\cos(\omega_0 t + \phi$$

$$\begin{cases} 0 \le t \le T_b \\ i = 1, \ldots., M \end{cases}$$

(a)



(b)



(c) M = 2

Fig. 1.2 ASK (a) The analytic expression; (b) Time waveform; (c) Constellation Diagram

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1.2 General Wireless Communication Background

Digital Bandpass Modulation Techniques

2. Frequency Shift Keying [Fig. 1.12]

- The frequency term ω_i has M discrete values while the amplitude term, A and phase term, ϕ are arbitrary constants.

$$s_i(t) = \sqrt{\frac{2E_b}{T_b}}\cos(\omega_i t + \phi) \quad \begin{cases} 0 \le t \le T_b \\ i = 1, \ ...., \ M \end{cases}$$

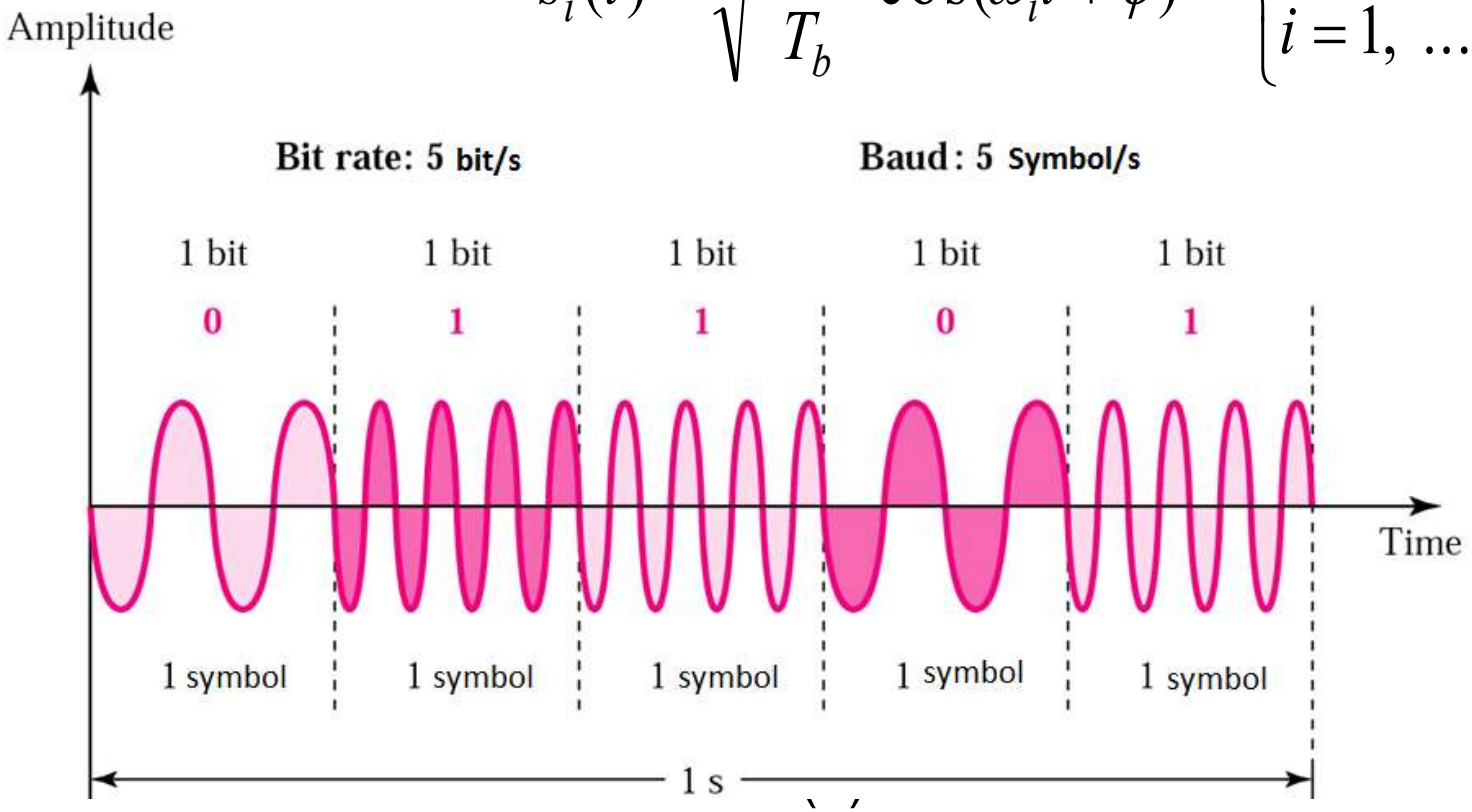


Fig. 1.3 FSK time waveform

1.2 General Wireless Communication Background

Digital Bandpass Modulation Techniques

3. Phase Shift Keying (PSK) [Fig. 1.13]

- The phase term $\phi(t)$, will have M discrete values while the amplitude term, A and frequency term, ω_0 are arbitrary constants.

$$s_i(t) = \sqrt{2E_b / T_b}\cos(\omega_o t + \pi(i-1))$$

$$\begin{cases} 0 \le t \le T_b \\ i = 1,\ 2 \end{cases} \qquad \text{(a)}$$

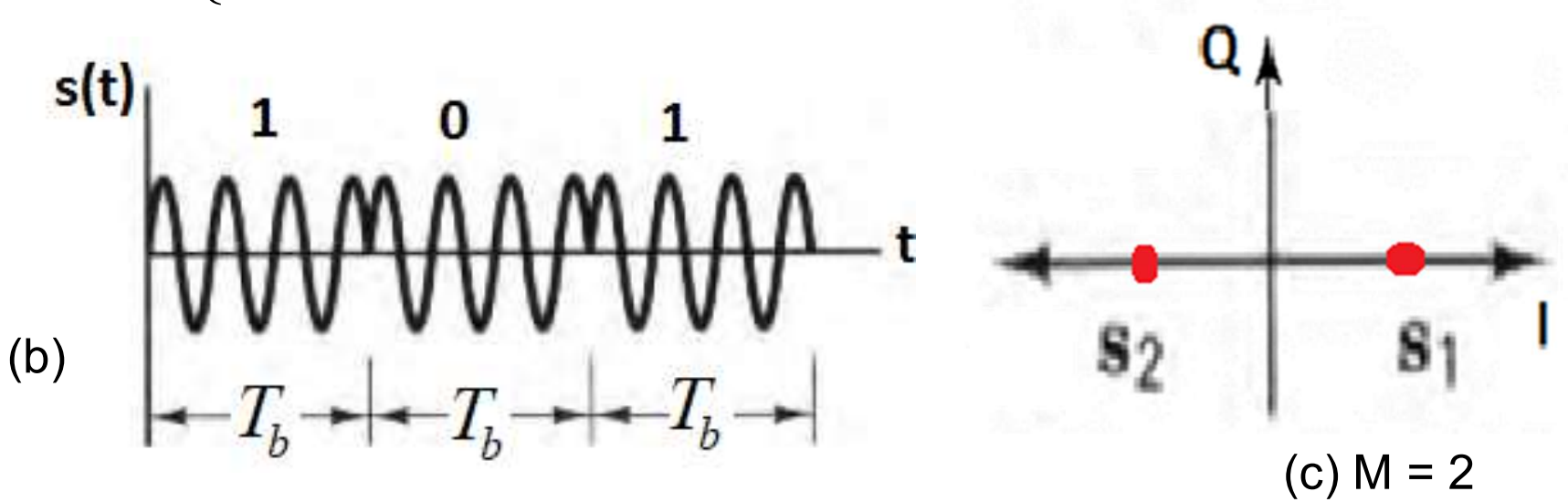


Fig. 1.4 PSK (a) The analytic expression; (b) Time waveform; (c) Constellation Diagram

1.2 General Wireless Communication Background

Digital Bandpass Modulation Techniques

4. Quadrature Phase Shift Keying (QPSK) [Fig. 1.14]

QPSK is a multilevel modulation technique characterized by:

1. Two bits per symbol.
2. More spectrally efficient, two times more bandwidth efficient than BPSK.
3. More complex receiver.

$$s(t) = \begin{cases} A\cos\left(2\pi f_c t + \frac{\pi}{4}\right) \\ A\cos\left(2\pi f_c t + \frac{3\pi}{4}\right) \\ A\cos\left(2\pi f_c t - \frac{3\pi}{4}\right) \\ A\cos\left(2\pi f_c t - \frac{\pi}{4}\right) \end{cases}$$

(a)

Input	I	Q
11	0.707	0.707
01	- 0.707	0.707
00	- 0.707	- 0.707
10	0.707	- 0.707

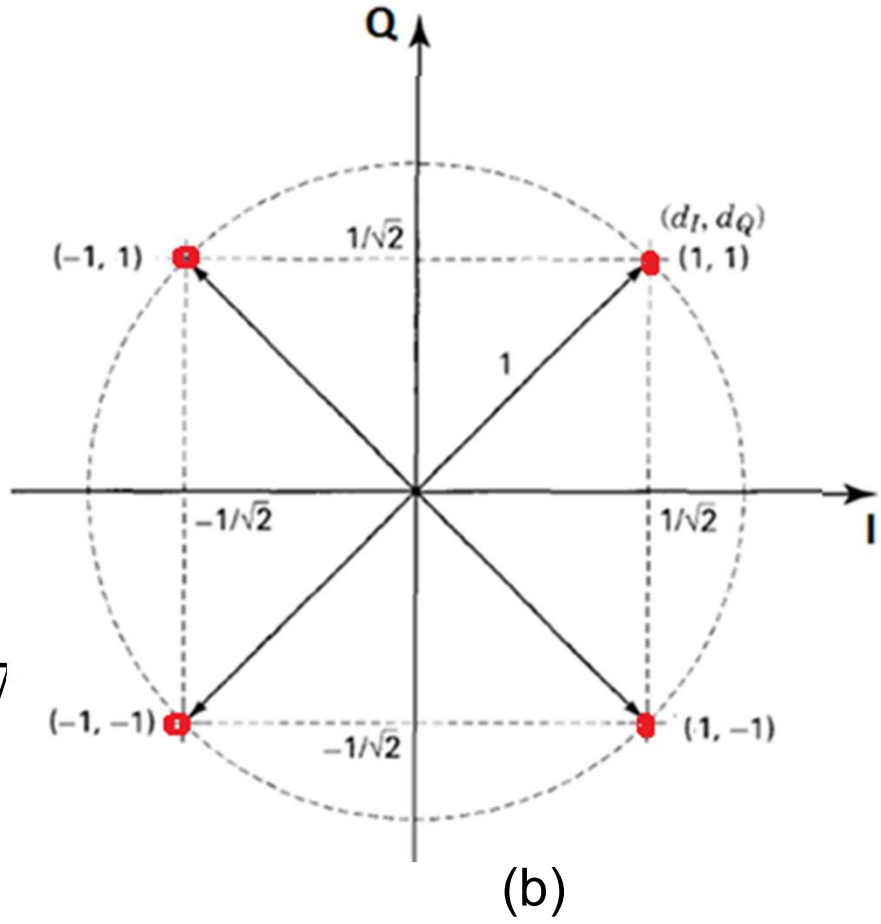


Fig. 1.5 QPSK (a) The analytic expression and (b) Constellation Diagram

1.2 General Wireless Communication Background

Digital Bandpass Modulation Techniques

5. Quadrature Amplitude Modulation (QAM) [Fig. 1.15]

- Quadrature amplitude modulation is a combination of ASK and PSK so that a maximum contrast between each signal unit (bit, 2 bit, 3 bit, and so on) is achieved. The 4-QAM and 8-QAM constellation diagrams are shown in Fig. 1.15.

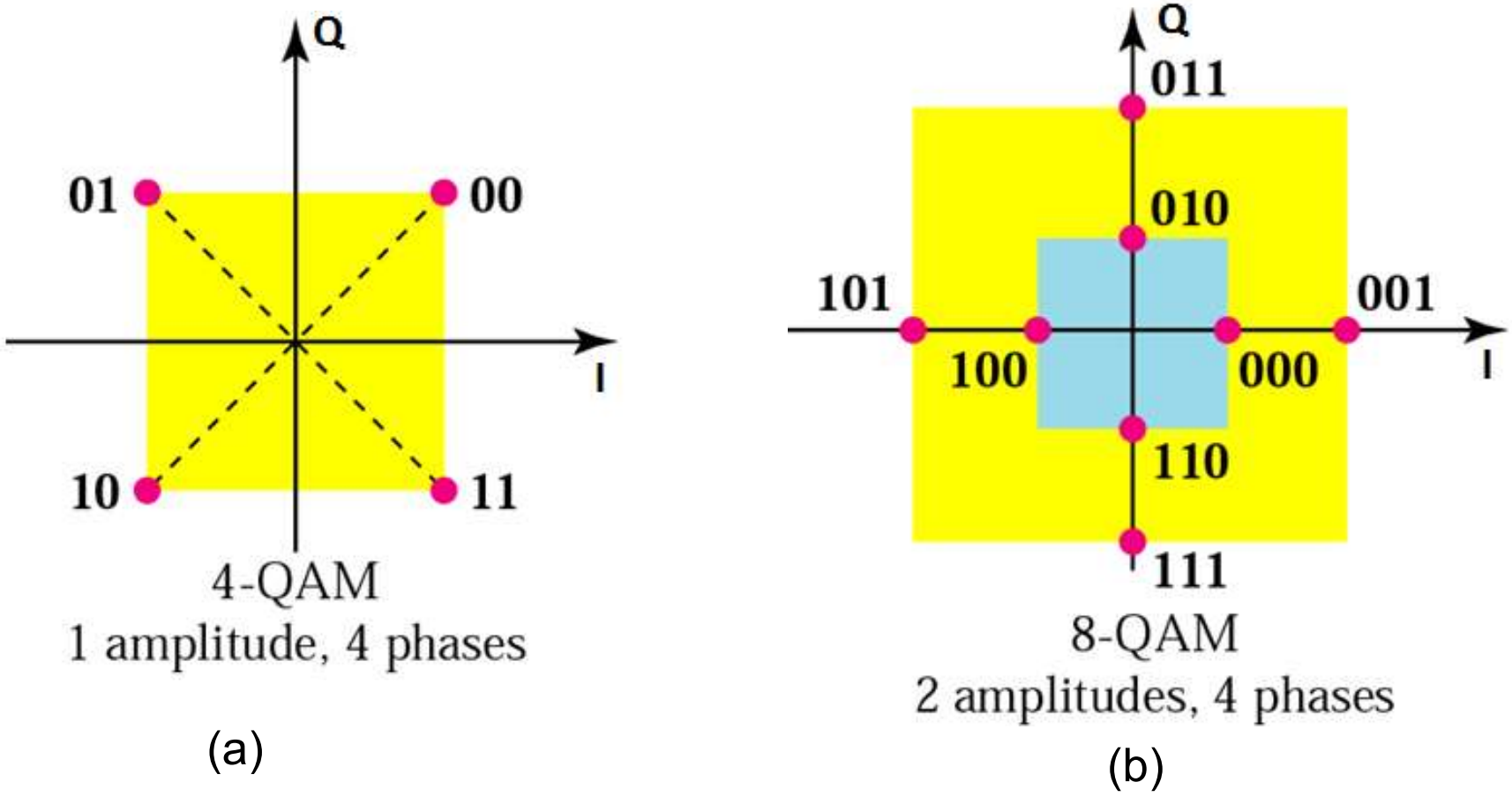


Fig. 1.6 Constellation diagrams of: (a) 4-QAM; (b) 8-QAM.

- **Transmitter**

It is the arrangement that processes the message signal into a suitable form for transmission and, subsequently, reception.

- **Antenna**

An antenna is a structure or a device that will **radiate and receive electromagnetic waves**. So, they are used in both transmitters and receivers. An antenna is a metallic object, often a collection of wires. The electromagnetic waves are polarized according to the position of the antenna.

- **Channel**

A channel refers to a physical medium such as wire, cables, or space through which the signal is passed from the transmitter to the receiver. Many channel impairments affect channel performance to a pronounced level. Noise, attenuation, and distortion, to mention the major impairments.

- **Attenuation**

Attenuation is a problem caused by the medium. When the signal is propagating for a longer distance through a medium, depending on the length of the medium, the initial power decreases. The loss in initial power is directly proportional to the length of the medium. Using amplifiers, the signal power is strengthened or amplified to reduce attenuation. Also, digital signals are comparatively less prone to attenuation than analog signals.

- **Encoding**

The process of communication is initiated the moment the sender gets some thought in his brain. To share his ideas with others, the thoughts must be converted into meaningful content by careful selection of words.

- **Distortion**

It is also another type of channel problem. When the signal is distorted, the distorted signal may have a frequency and bandwidth different from the transmitted signal. The variation in the signal frequency can be linear or non-linear.

- **Receiver**

An arrangement that extracts the message or information from the transmitted signal at the output end of the channel and reproduces it in a suitable form as the original message signal is a receiver.

- **Demodulator**

It is the inverse phenomenon of modulation, i.e., the process of separation of the message signal from the carrier wave takes place in the demodulator. The information is retrieved from the modulated wave.

- **Repeaters**

Repeaters are **placed at different locations between the transmitter and receiver**. A repeater receives the transmitted signal, amplifies it, and sends it to the next repeater without distorting the original signal.

1.1 Wireless Communication Systems

1.1.2 Block Diagram of Typical Wireless digital communication system

- The high frequency components or the transceiver has a Transmit side (Tx) and a Receive side (Rx), which are connected to the antenna through a duplexer that can be realized as a switch or a filter, as shown in Fig. 1.2.
- So, the transmitter consists of a frequency up-conversion stage to a Radio Frequency (RF), a high-power amplifier, and an antenna. While the receiver consists of an antenna, a low-noise amplifier (LNA) and frequency down-conversion.

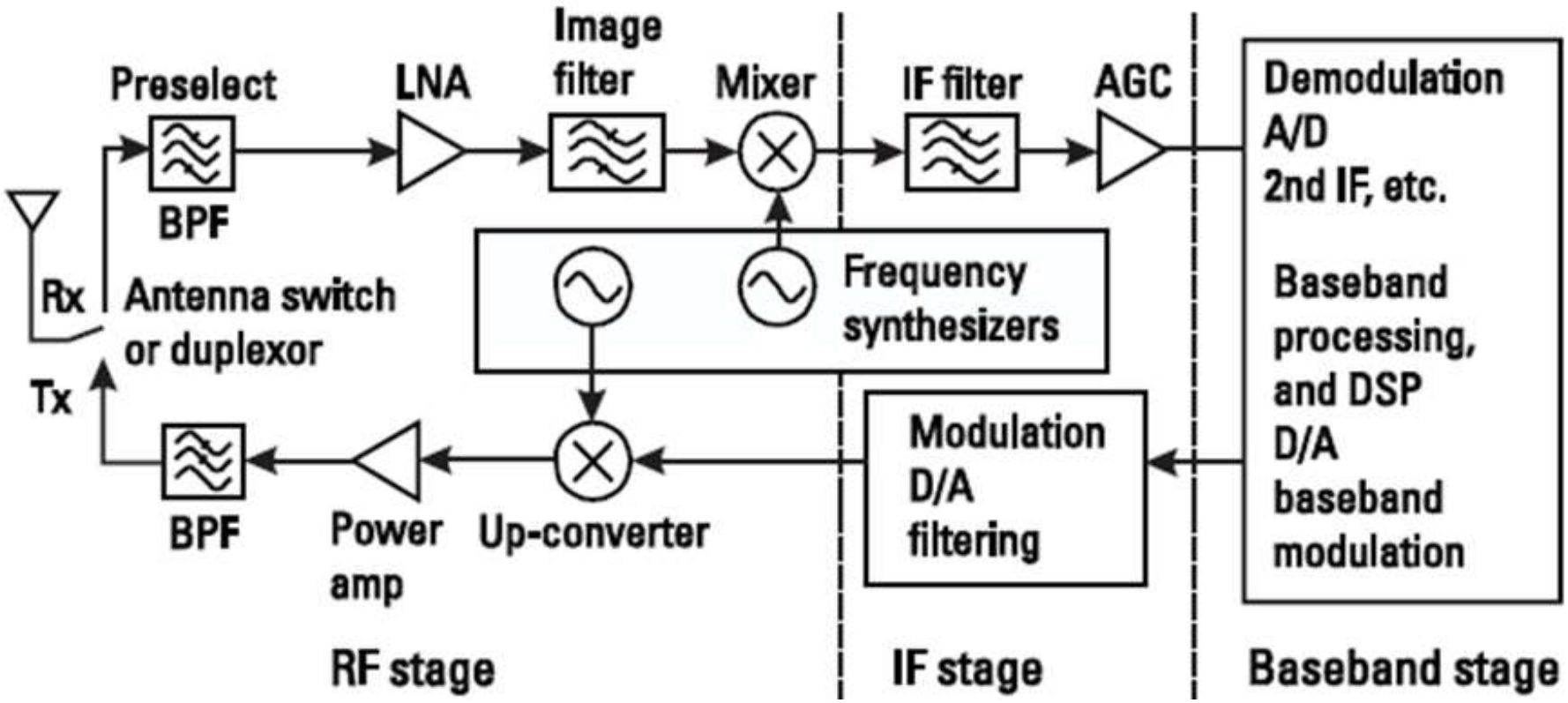
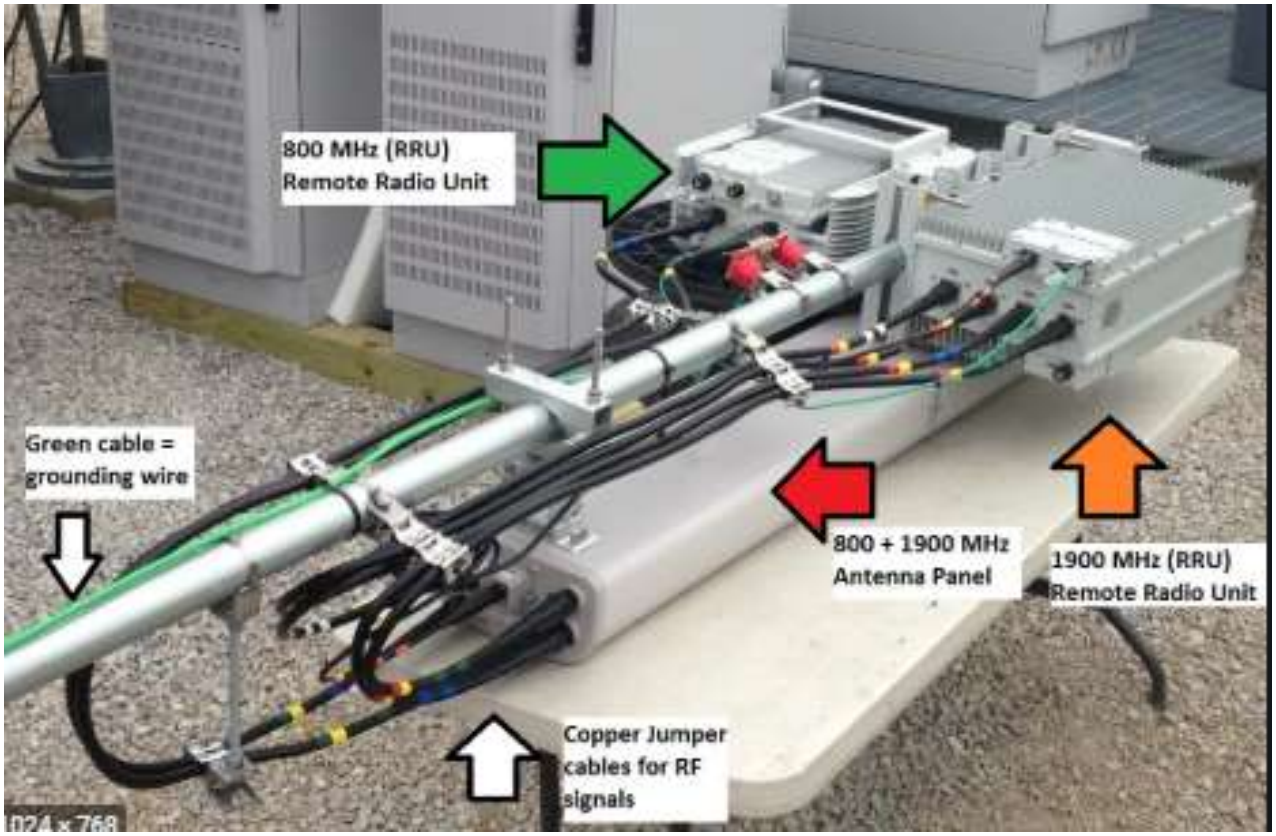


Fig. 1.7(a) Typical transceiver block diagram.

1.1 Wireless Communication Systems

1.1.2 Block Diagram of Typical Wireless digital communication system

Mobile Hardware Subsystem



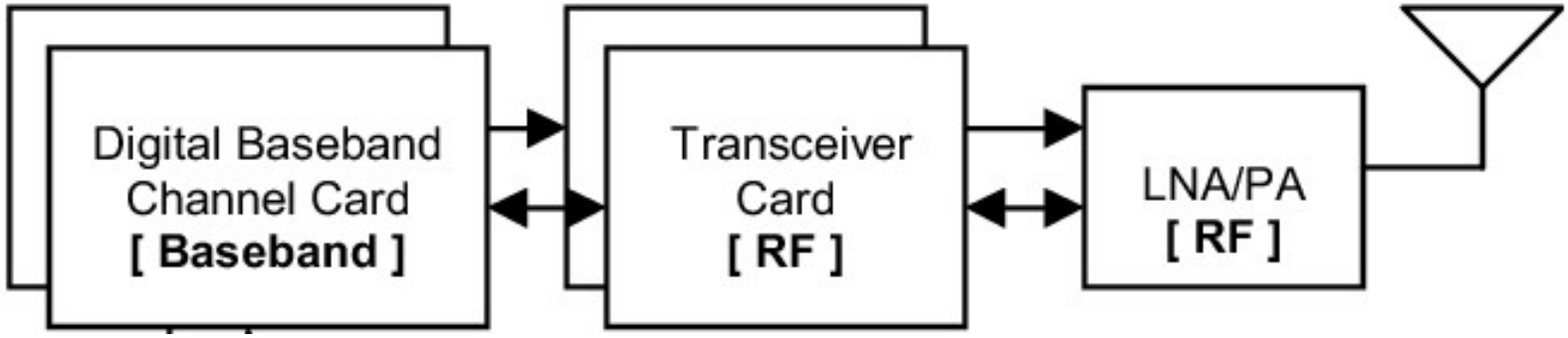


Fig. 1.7(b) Typical transceiver block diagram.

1.1 Wireless Communication Systems

1.1.3 Transmission modes in wireless Systems

1. Simplex RF Communication Systems [Refer to Fig. 1.4(a)]

A radio technology that allows only one-way communication from a transmitter to a receiver.

- Examples: FM radio, Pagers, TV, Adaptive Multi-Rate (AMR) systems.

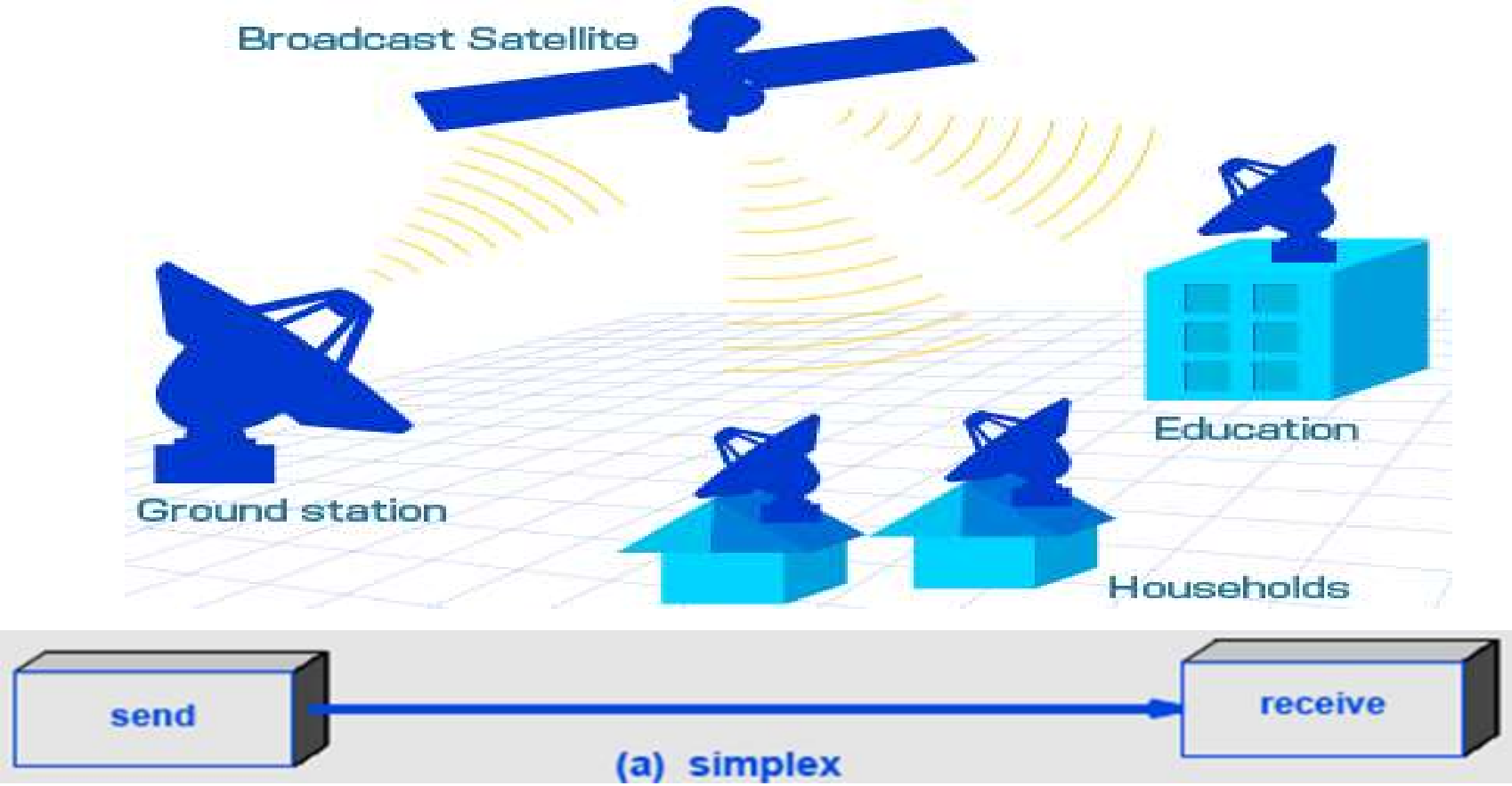


Fig. 1.8(a) Simplex RF Communication Systems.

1.1 Wireless Communication Systems

1.1.3 Transmission modes in wireless Systems

2. Half-duplex RF Communication Systems [Refer to Fig. 1.4(b)]

- Two-way communication but not simultaneous (Push-to-talk) Time Division Duplex (TDD). Use any desired frequency channel for transmitter and receiver. Send transmit and receive signals at different times.
  Advantage: switch has low insertion loss relative to duplexer.
  Disadvantage: receiver more sensitive to transmitted signals from other user.
  Examples: Walkie-talkie, wireless keyboard mouse.

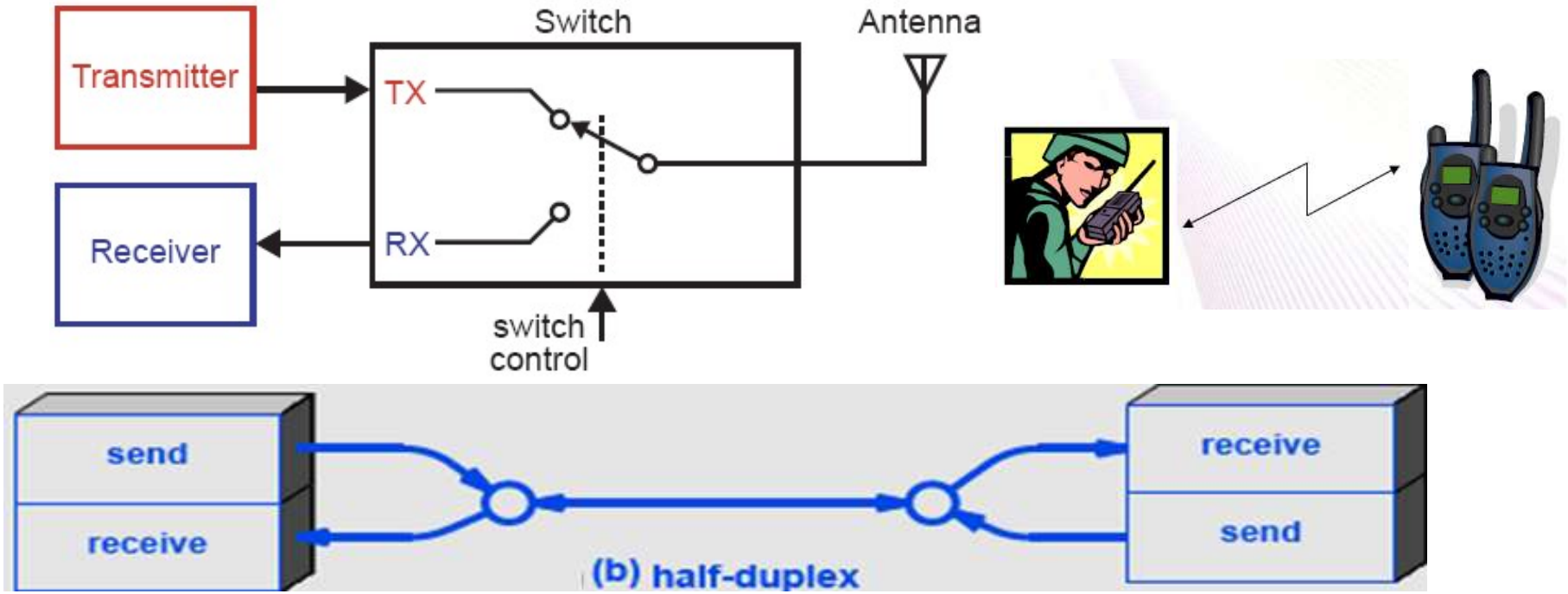


Fig. 1.8(b) Half-duplex Communication.

1.1 Wireless Communication Systems

1.1.3 Transmission modes in wireless Systems

3. Full-duplex Communication Systems [Refer to Fig. 1.4(c)]

- Two-way communication Simultaneous Frequency Division Duplex (FDD) and Time Division Duplex (TDD). Separate frequency channels into transmit and receive bands allows simultaneous transmission and reception.
- Isolation of receiver from transmitter achieved with duplexer. Cannot communicate directly between users, only between handsets and base station.

Advantage: isolates users.

Disadvantages: - takes up twice the bandwidth.

Examples: Cellular phones, satellite communication.

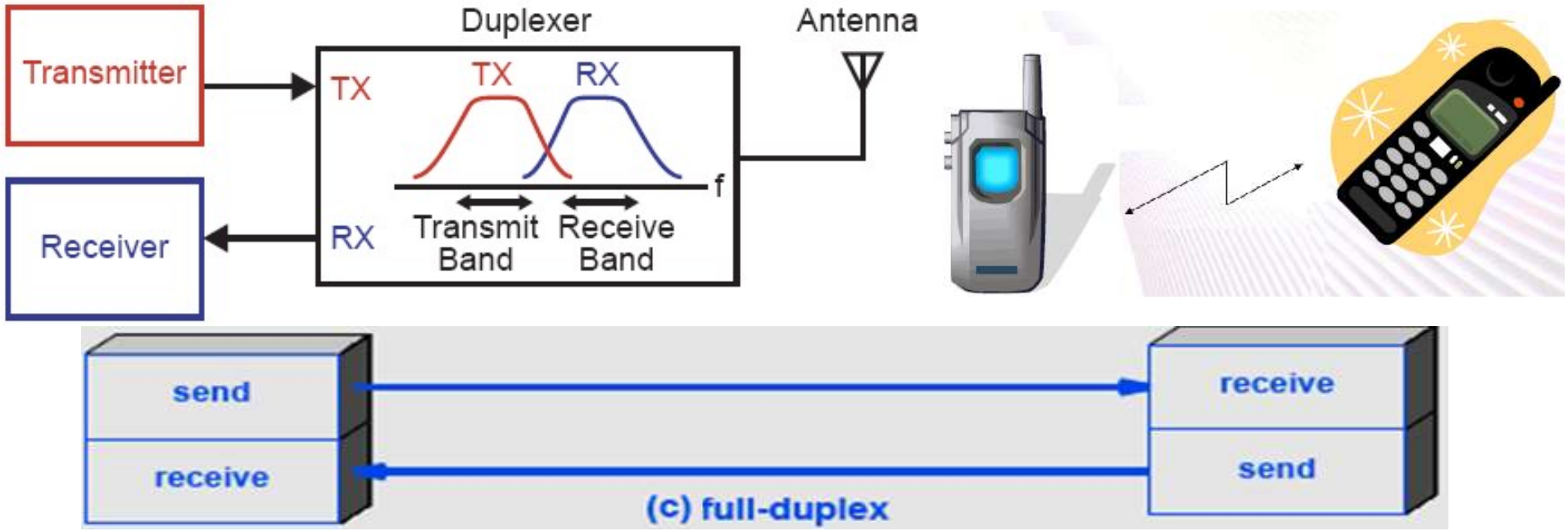


Fig. 1.8(c) Full-duplex Communication.

1.1 Wireless Communication Systems (Continued)

1.1.4 Gain and Power Units (Definitions)

dB: (Decibel) = $10\log_{10}(P_{out}/P_{in})$

- Log-ratio of two signal levels. Named after Alexander Graham Bell. For example, a cable has 6 dB losses or an amplifier has 15 dB of gain. System gains and losses can be added/subtracted, especially when changes are in several orders of magnitude.

dBm (dBmilliWatt):

- Relative to 1 mW, i.e. 0 dBm is 1 mW (milliWatt). It is commonly used for amplifiers. There's also **dBW** (Referenced to 1 WATT), but you generally only use those when dealing with "Big Stuff", as 30 dBW is 1 kW.

$$P_{dBm} = 10\log_{10}\left(\frac{P_{watt}}{1\ mW}\right)$$

P_{watt}	P_{dBm}
2.5 nW	–56
250 nW	–36
25 μW	–16
1 mW	0
2.5 mW	+4
250 mW	+24

dBc – relative to carrier [Refer to Fig. 1.9(a)]

- The signal bandwidth can be defined in terms of the relative amount of power that appears in an adjacent channel. This signal exhibits an Adjacent Channel Interference Ratio (ACIR). In general, if powers of the sideband and the carrier are given as P_2 and P_C, respectively, the ACPR in dBc is found as:

$$ACIR = 10\log_{10}\left(P_2 / P_C\right)\ dBc$$

1.1 Wireless Communication Systems (Continued)

1.1.4 Gain and Power Units (Definitions)

dBi (dB isotropic) for EIRP (Effective Isotropic Radiated Power):

- The gain of an antenna as referenced to an isotropic source. An isotropic source is the perfect omnidirectional radiator, a true "Point Source", and does not exist in nature as shown in Fig. 1.9(b).

dBd (dBdipole):

- The gain of an antenna has over a dipole antenna at the same frequency. A dipole antenna is the smallest and least gain practical antenna that can be made. A dipole antenna has 2.14 dBi gain over a 0dBi isotropic antenna. Thus, a simple dipole antenna has a gain of 2.14 dBi or 0 dBd and is used as a standard for calibration.

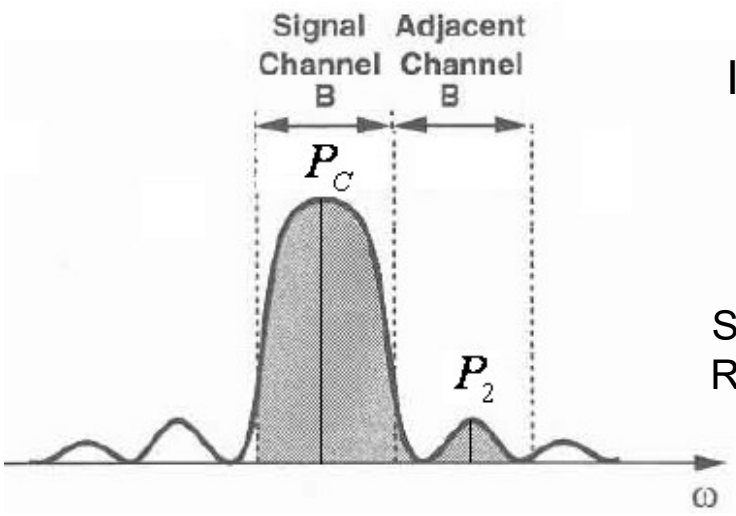
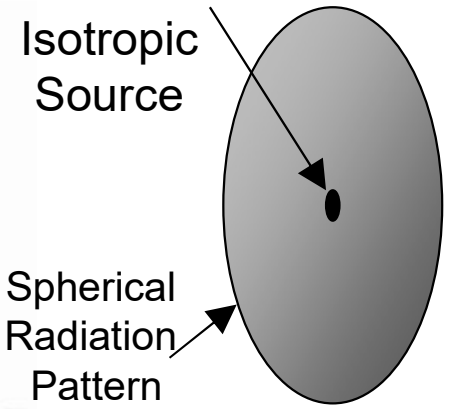


Fig. 1.9(a) Adjacent channel power



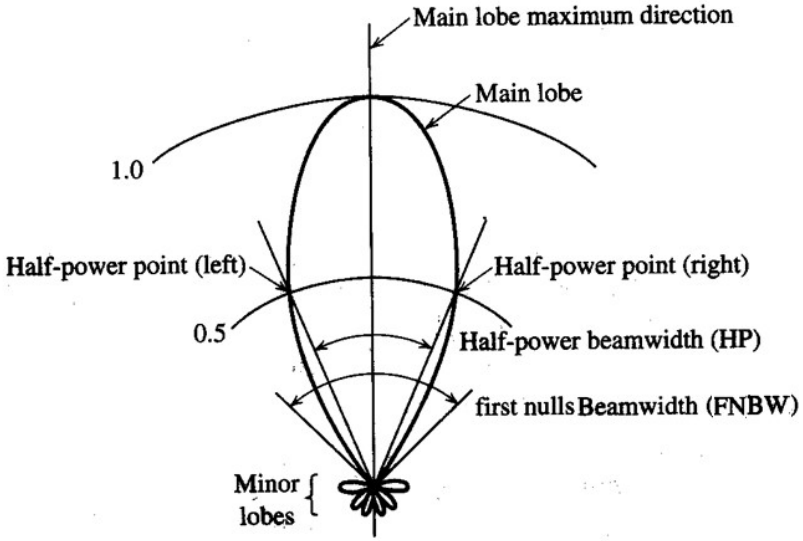


Fig. 1.9(b) Antenna radiation pattern

1.2 Cellular System Concepts

- Radio Communication applied to mobile telephony has the following limitations:
  - Powerful transmitters and huge equipments are required.
  - Number of users is limited to the frequency band allocated.
- This limits the possible number of concurrent users due to interference. So replacing the powerful transmitters with smaller low power ones, each covering a small area called a “cell” solved the issue. This solution results a cellular systems as shown in Fig. 1.16.
- As the demand increases, the number of Base Stations (BSs) or Base Transmitter Stations (BTSs) may be increased with decreasing transmission power thus providing additional users without increasing radio BW.

- During planning of the network (cell planning), the set of frequencies allocated to a network operator have to be distributed over a very large number of cells.
- The frequencies are divided among a small grouping of cells (cluster) and this grouping can be repeated throughout the network to provide necessary coverage.
- Each cell has its own BTS using low power transmitters allow the efficient reuse of the available frequencies with a small geographical area.
- In this way, coverage of the entire network is provided without co-channel interference.

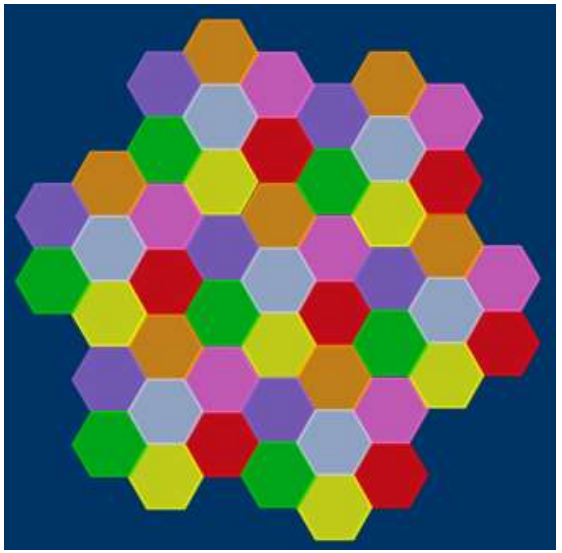
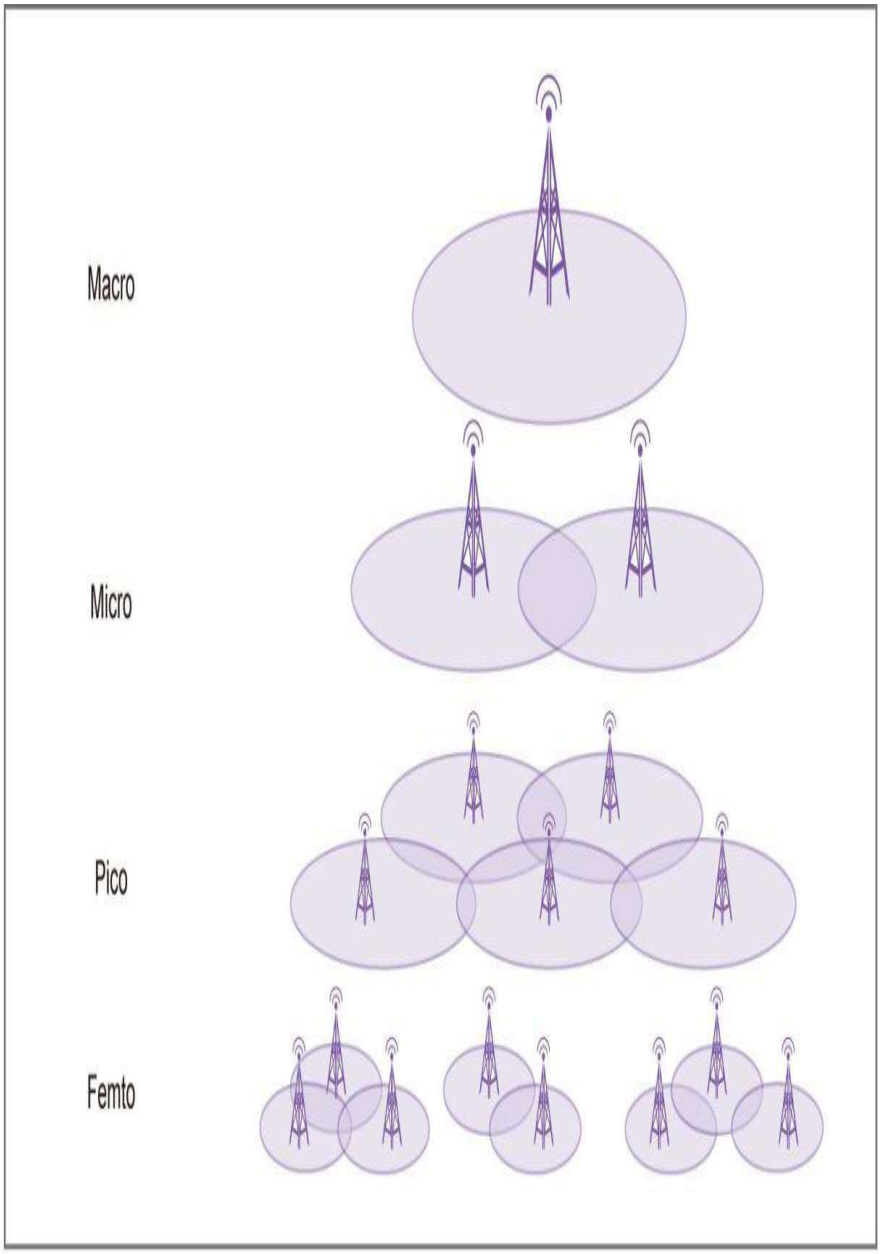


Fig. 1.10 Cellular Concept

Cell Size (Cell Types)

- **Macro Cells : (20 km in diameter)**
  - their coverage is large.
  - It is used in remote areas.
  - High-power transmitters and receivers are used.
- **Micro Cells : (400 m – 2 km in diameter)**
  - Shopping centers and airports.
- **Pico Cells : (4 – 200 m in diameter)**
  - Inside buildings
- **Femtocells : (maximum diameter is 40 m)**
  - Inside rooms (4G) High-power



Small cells → more bandwidth → more users → more base station → complex networks → leading to more interference and handovers.

Reduction in Cell size results in:

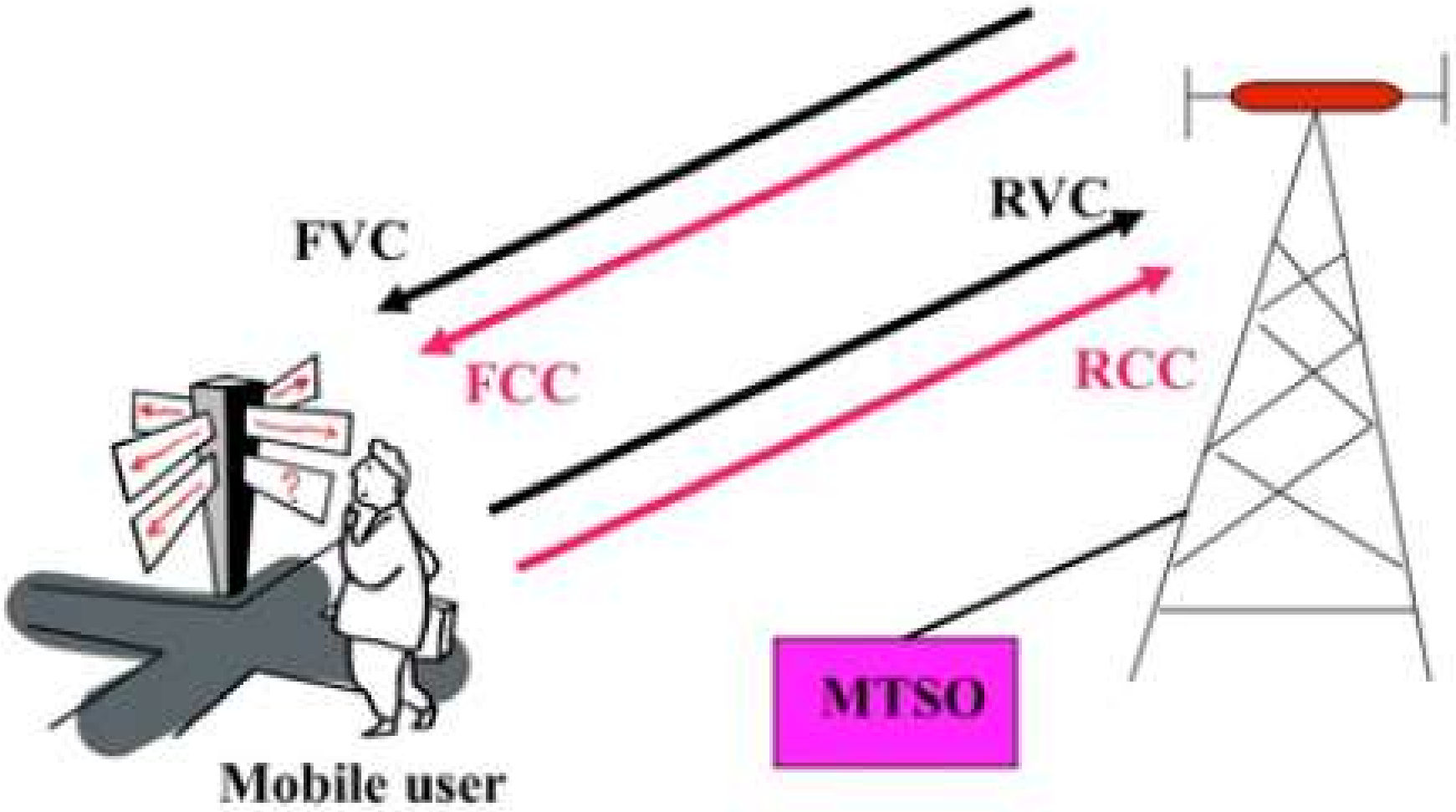
- **Increased system capacity** and **user capacity**
- **Increased number of handoffs per call** for mobile users.
- **Increased complexity** in locating the subscriber.
- **Increased interference.**
- **Lower power consumption in mobile terminals** as the transmitted signal (during the uplink) may be small due to the near distance between the MS and the BS.
- **Lower power consumption in the base station** as the transmitted signal (during the downlink) may be small due to the near distance between the BS and the MS.

Uplink and downlink, also called upload and download, refers to the two-way communication between a cell tower and your phone.

Downlink definition – signal coming from a cell tower to your cellular device. **Uplink definition** – signal leaving your cellular device and going back to a cell tower.

Channel Types in Cellular Systems

- Control Channel (Forward & Reverse).
- Voice Channel (Forward & Reverse).



Common Air Interface (CAI)

- Common Air Interface: A Standard that defines Communication between a Base Station and a Mobile Station.
- Specifies Four Channels [Voice Channels and Control / Setup Channels]

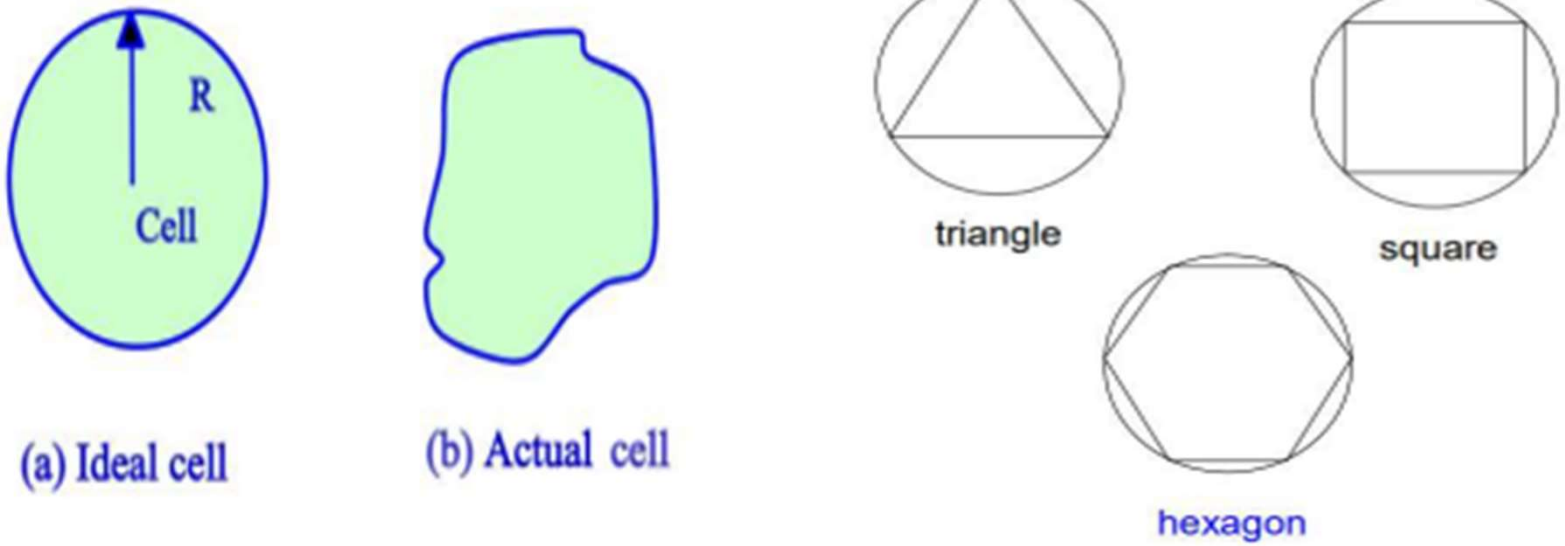
  - **FVC: Forward Voice Channel**
  - **RVC: Reverse Voice Channel**
  - **FCC: Forward Control Channel**
  - **RCC: Reverse Control Channel**

Cell Structure

Area divided into cells

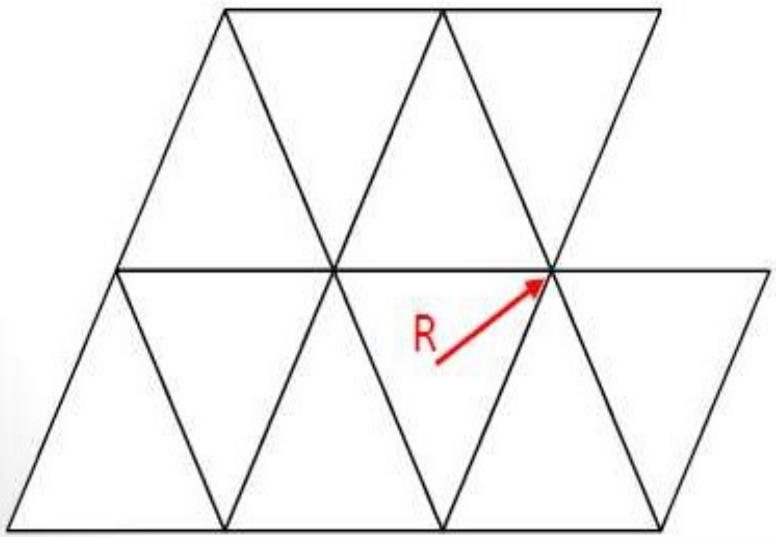
- Each with its own antenna
- Each with its own range of frequencies
- Served by base station (Transmitter, receiver, control unit)
- Mobile stations communicate only via the base station
- Adjacent cells on different frequencies to avoid crosstalk

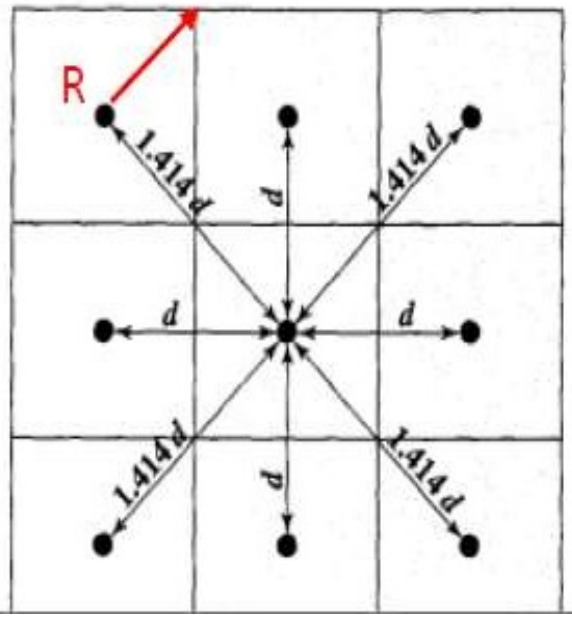
- **Cell Shape**

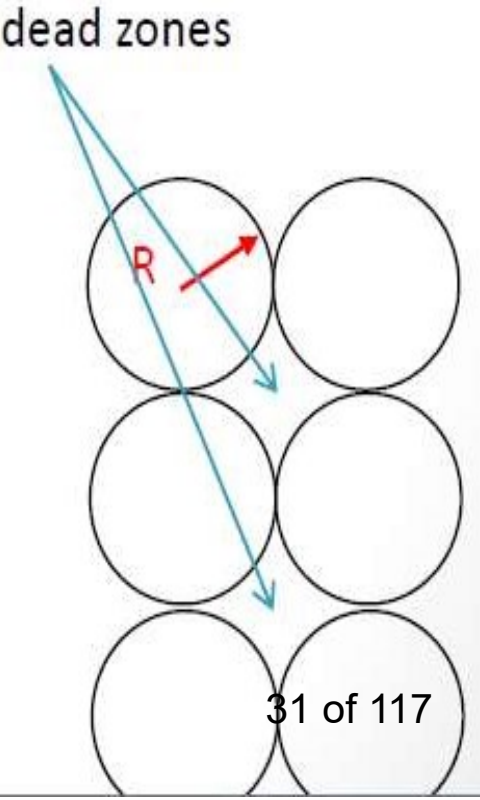


Cell Shape

- Square & Triangle Patterns
  - Not suitable (different distance from the cell's centre to different point in the perimeter)
  - Square for example :
    - Width *d* cell has four neighbors at distance *d* and four at distance 1.414 *d*
    - Better if all adjacent antennas equidistant
      - Simplifies choosing and switching to new antenna
- Circular Patterns
  - Ideal shape, but has dead zones

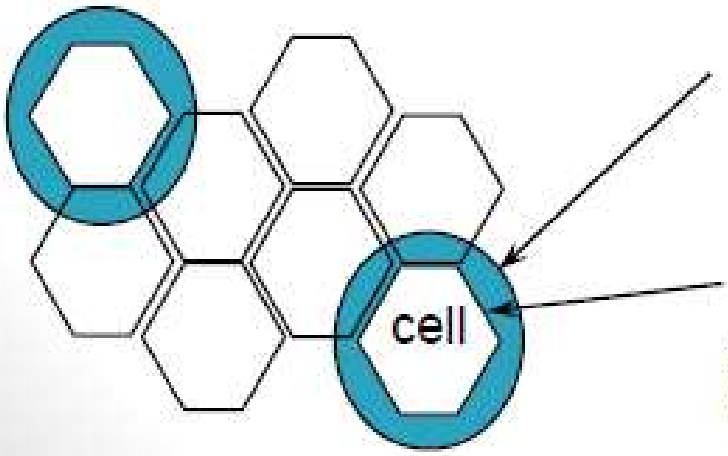






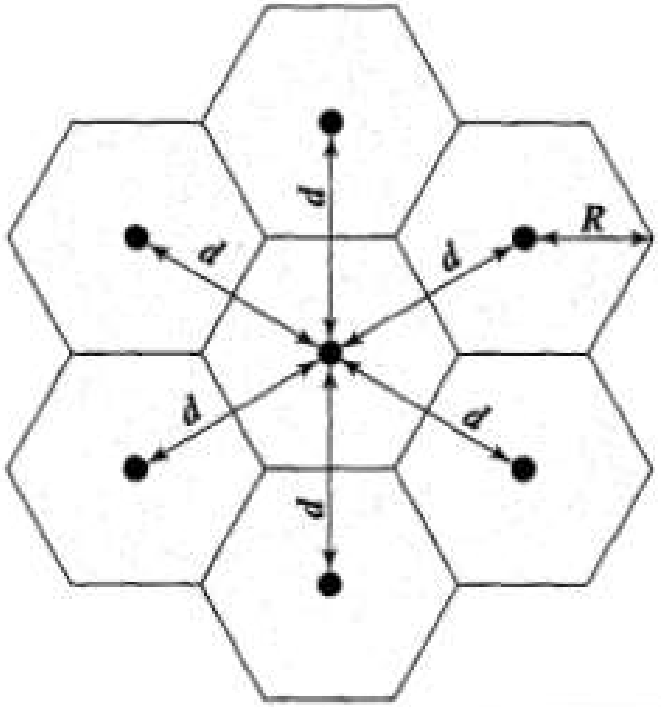
Cell Shape…….

- Hexagon
  - Provides equidistant antennas
  - Radius defined as radius of circum-circle
    - Distance from center to vertex equals length of side
  - Distance between centers of cells radius *R* is 1.7 *R*
  - Not always precise hexagons in reality
    - Topographical limitations
    - Local signal propagation conditions
    - Location of antennas



possible radio coverage of the cell

idealized shape of the cell



1.2 Cellular System Concepts (Continued)

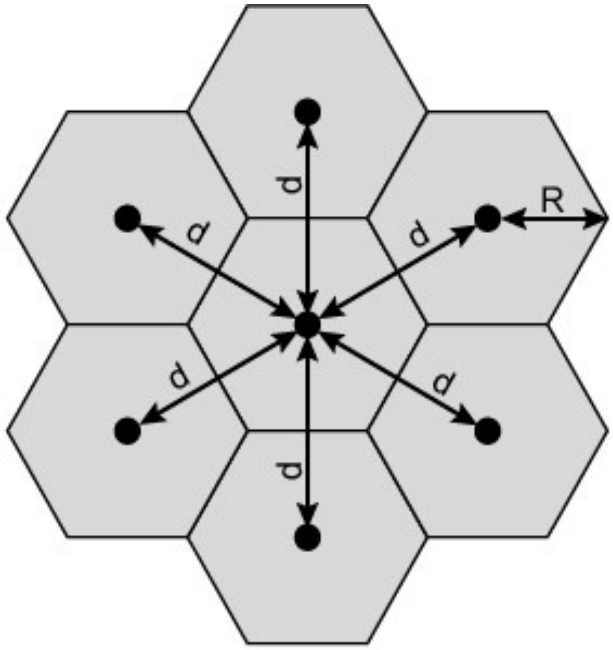
1.2.1 Cellular Geometries

- While it might seem natural to choose a circle to represent the coverage area of a base station, adjacent circles cannot be overlaid upon a map without leaving gaps or creating overlapping regions.
- by using the hexagon geometry, the fewest number of cells can cover a geographic region as shown in Fig. 1.11 and R is the radius of a cell.
- The hexagonal area is equal to six triangle and expressed as:

$$Hexagonal\ Area = 6x\Delta = 6x0.5R^2 \sin 60^0 = 2.6R^2 \quad (1.1)$$

- The distance between two adjacent hexagon, d is:

$$d = \sqrt{3}R \quad (1.2)$$





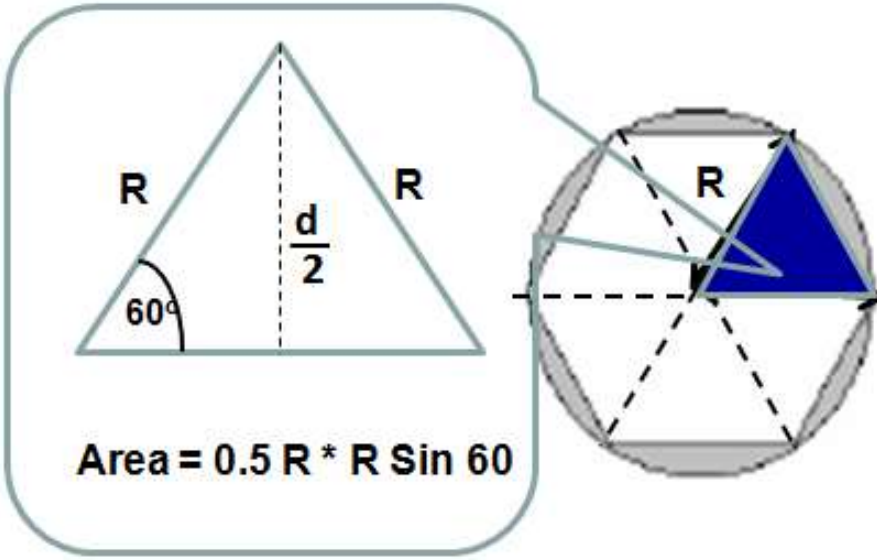


Fig. 1.11 Hexagonal shape cell

1.2 Cellular System Concepts (Continued)

1.2.1 Cellular Geometries

- Hexagonal shape covers the entire area without any overlap and gaps. Actual radio coverage of a cell is known as footprint.
- It permits easy and manageable analysis of a cellular system.
- Base Station (BS) transmitters are placed either at the center (center-excited cells) or on 3 of the 6 hexagon vertices (edge-excited cells).
- For center-excited cells, normally omnidirectional antennas are used. For edge-excited cells we use sectored directional antennas.

Sectorization

- For more efficient use of available spectrum and hence enhancing the # of users in the system, each cell is divided into three sectors of 120° as shown in Fig. 1.12.
- In each sector a directional antenna is used whose narrow beams allow reusing the channels more often.
- Sectorization is suitable to use in dense urban areas



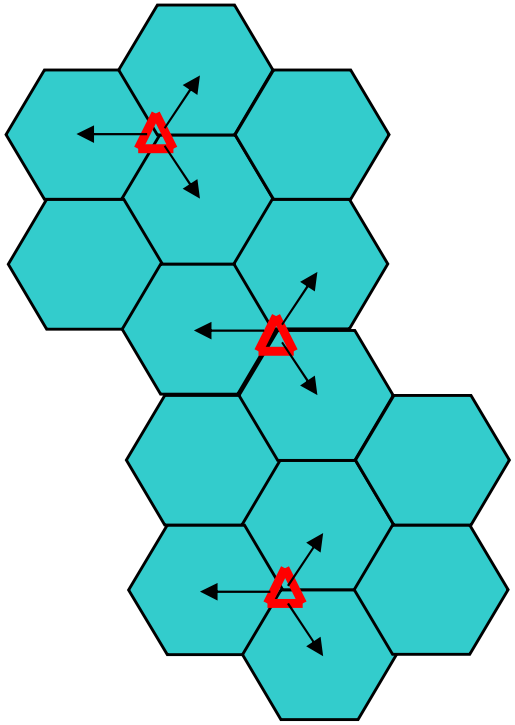
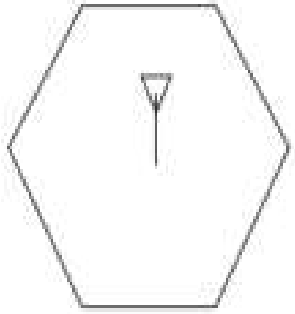


Fig. 1.12 Sectorization

Cell Excitation Modes

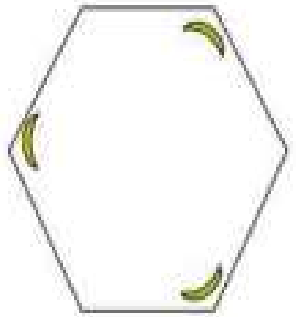
Center-excited cell

- Base station transmitter is in the center of the cell.
- Omni-directional antennas are used.



Edge-excited cell

- Base station transmitters are on three of the six cell vertices.
- Sectored directional antennas are used.



1.2 Cellular System Concepts (Continued)

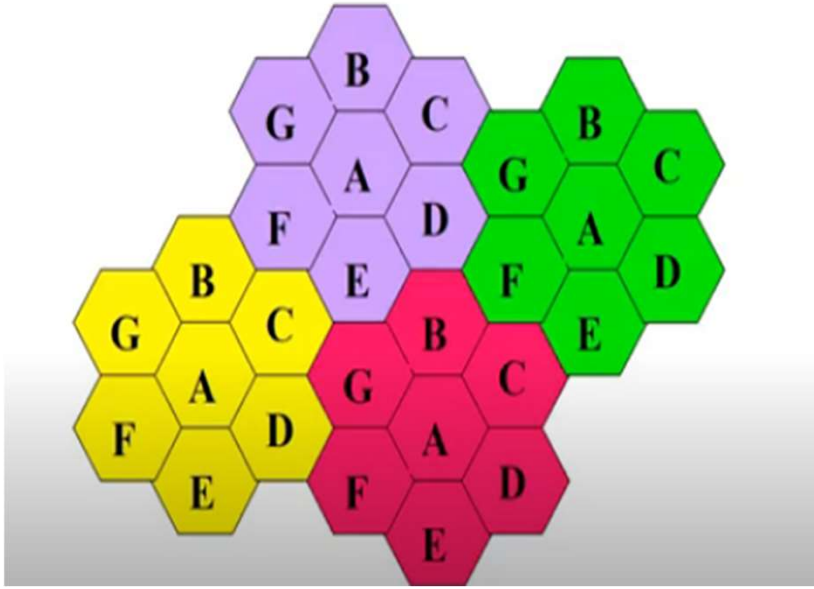
1.2.3 The Cellular Frequency Reuse

- Each BS is allocated a group of channels for use within a small geographic area called a cell.
- Cells are grouped into clusters as shown in Fig. 1.13 where N is the number of cells in a cluster. Available band is distributed among the cells of the cluster.
- The frequency reuse factor of a cellular system = 1/N, and represents the share of each cell of the total number of the duplex channels of a cluster.
- BSs in adjacent cells are assigned channel groups which contain completely different channels than neighboring cells.
- Hence, the same group of channels may be used to cover different cells that are separated from one another by distances large enough to keep interference levels within tolerable limits.
- Selecting or allocating channel groups to BSs is called **frequency reuse** or **frequency planning**.

- **<u>High capacity is achieved</u>**:
  - by limiting the coverage of each base station

  Transmitter to a small geographical area called a *cell*
  - by reusing the same radio channels in another

  base station located some distance away – *Frequency reuse*.



1.2 Cellular System Concepts (Continued)

1.2.3 The Cellular Frequency Reuse

- For a cluster number of cells, N =1, the frequency reuse factor = 1/N=1 is as shown in Fig. 1.13.
- This means that the whole band of frequency is used in a cell and reused in each of the adjacent cells. Because the same frequency is used in all cells, high interference occurs in this system, this problem is solved in 3G mobile system.

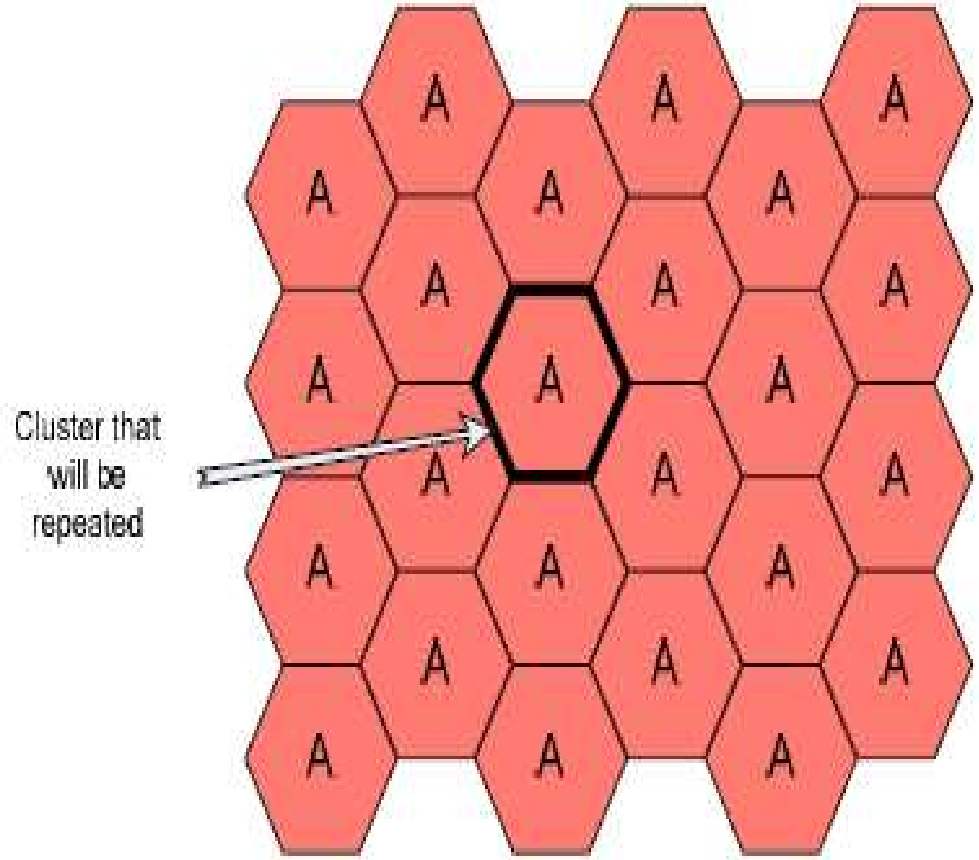


Fig. 1.13 Clusters with N =1 and reuse factor of 1.

1.2 Cellular System Concepts (Continued)

1.2.3 The Cellular Frequency Reuse

- For a cluster number of cells, N =3, the frequency reuse factor = 1/N= 1/3 is shown in Fig. 1.14(a).
- For a cluster number of cells, N =4, the frequency reuse factor = 1/N= 1/4 is shown in Fig. 1.14(b).

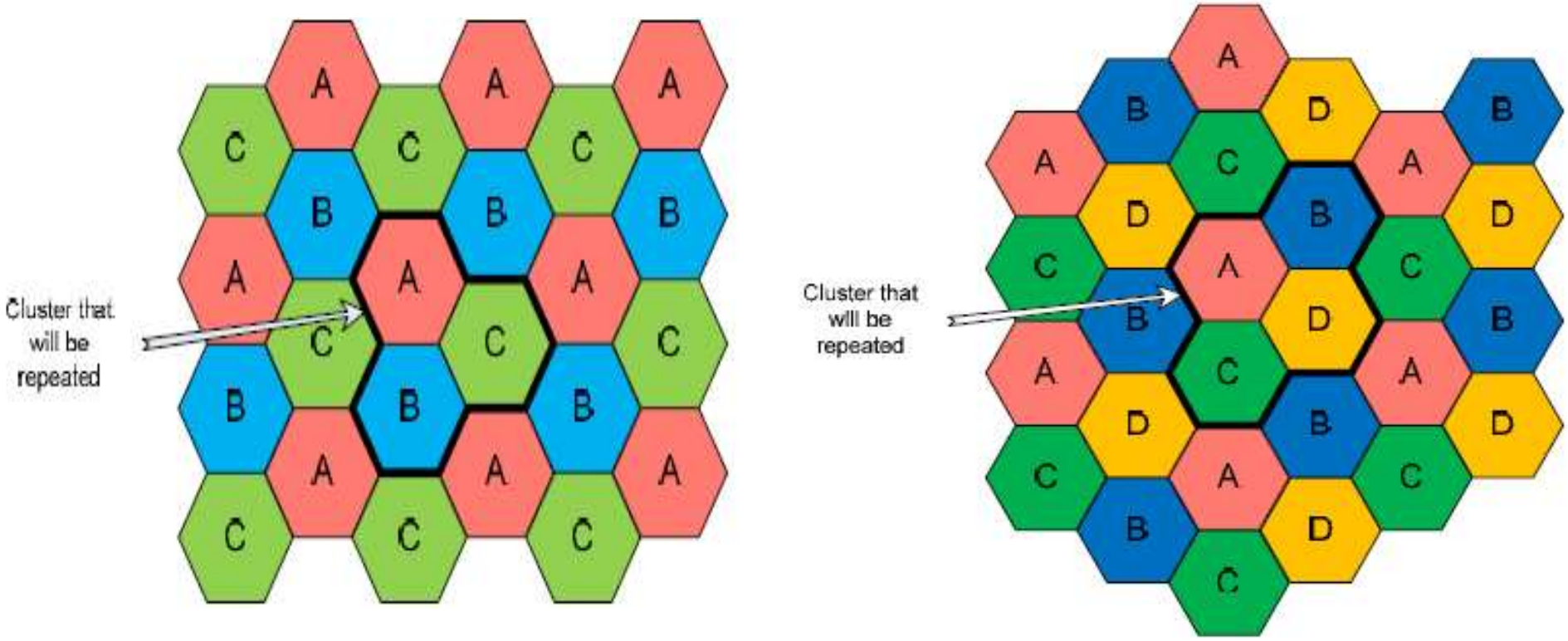
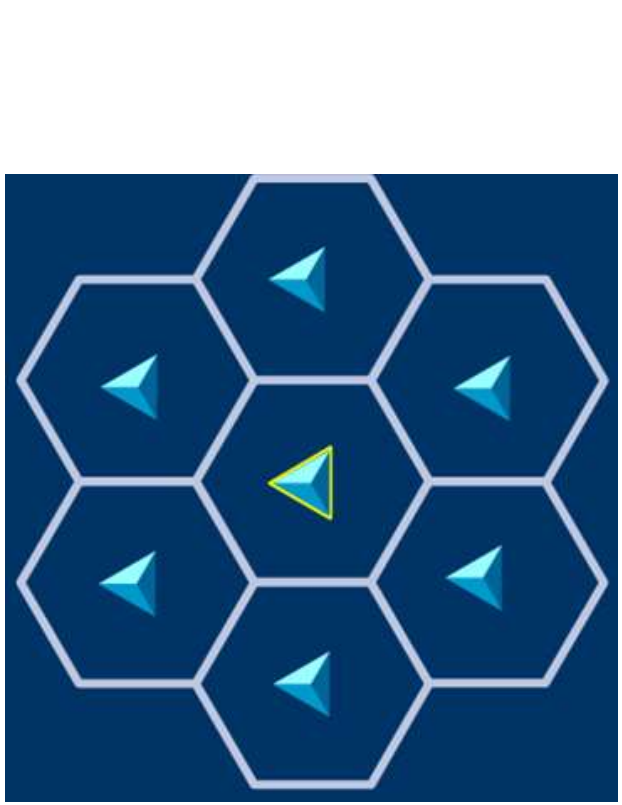


Fig. 1.14 (a) Frequency Reuse Factor, 1/N = 1/3 (b) Frequency Reuse Factor, 1/N = 1/4

1.2 Cellular System Concepts (Continued)

1.2.3 The Cellular Frequency Reuse

- For a cluster number of cells, N =7, the frequency reuse factor = 1/N= 1/7 is shown in Fig. 1.14(c).



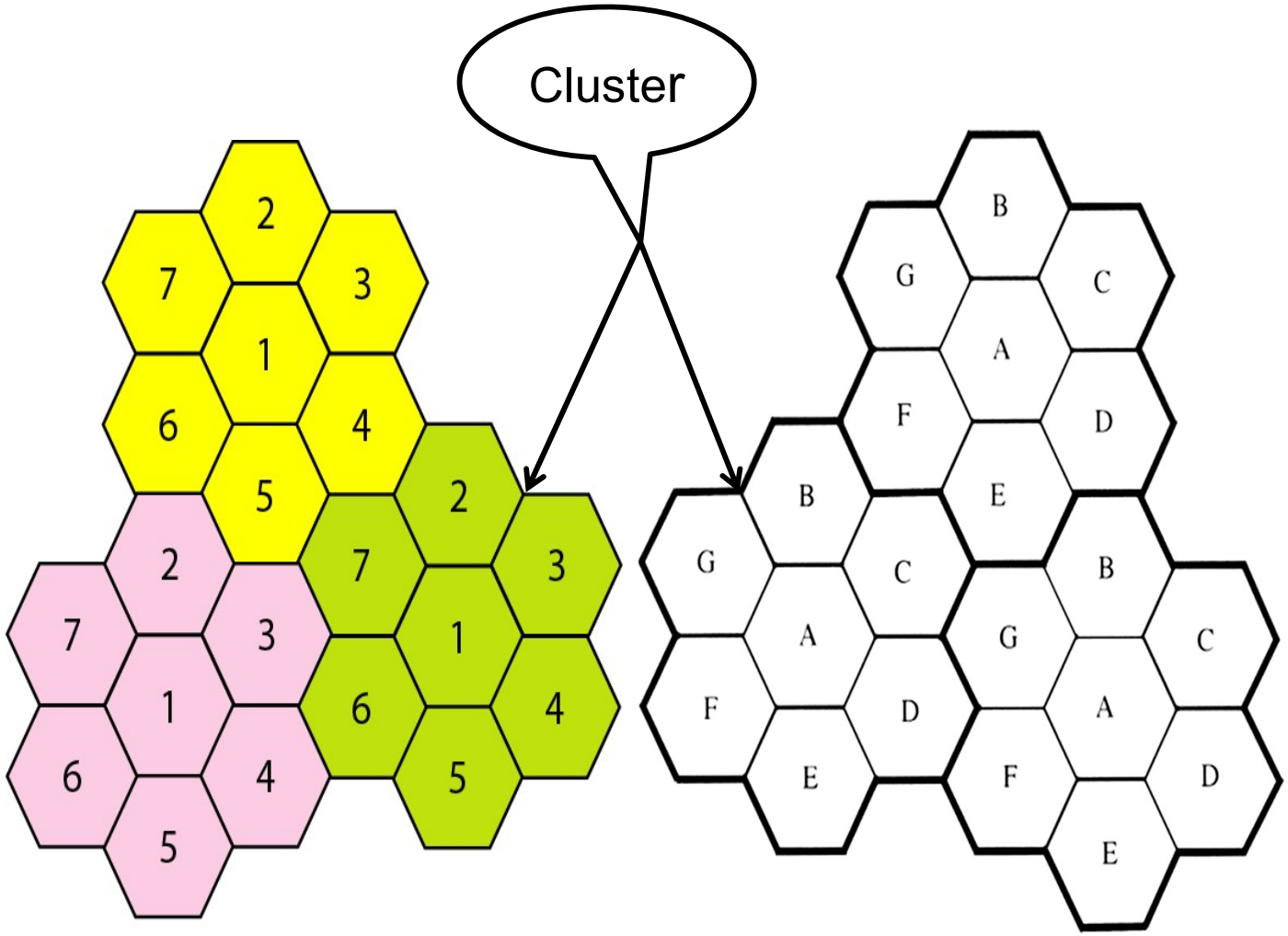


Fig. 1.14(c) Clusters with N =7 and reuse factor of 1/7.

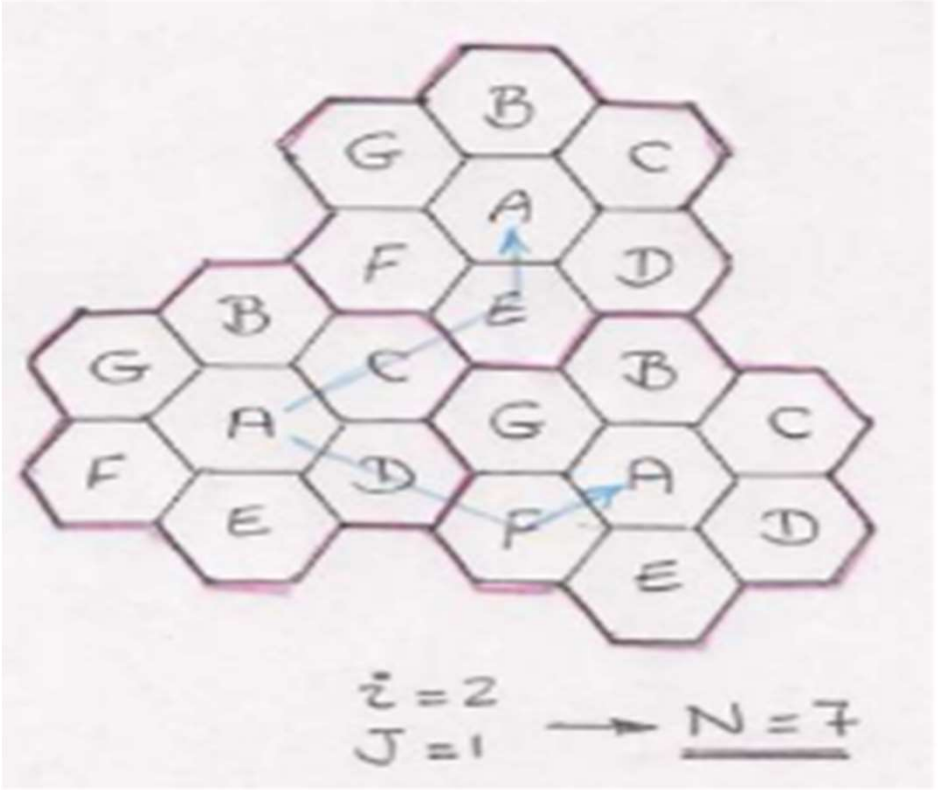
- The **Frequency Reuse factor** is given as 1/N & and each cell is assigned 1/N of total channels.

Location of Co-channel cells

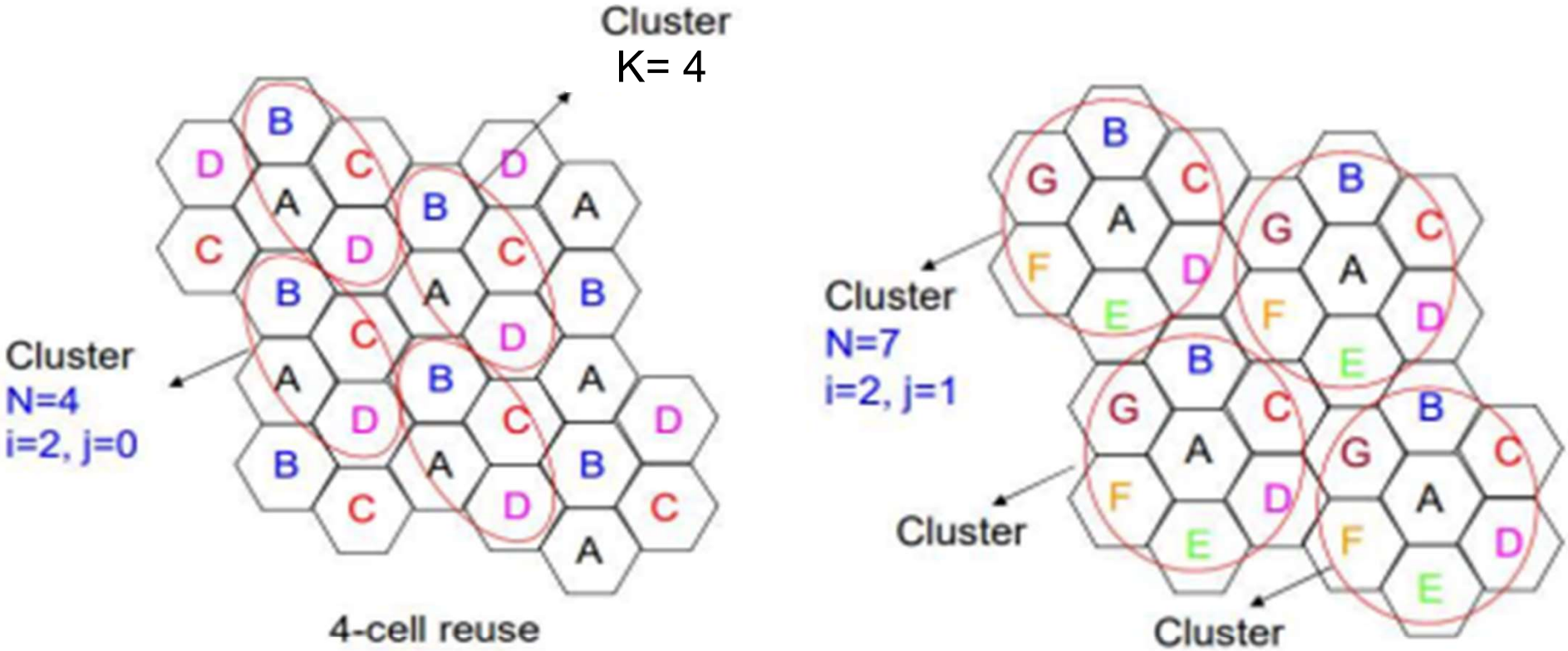
- The geometry of the hexagon is such that no. of cells per cluster is N (cluster size), and can only have values that satisfy the equation

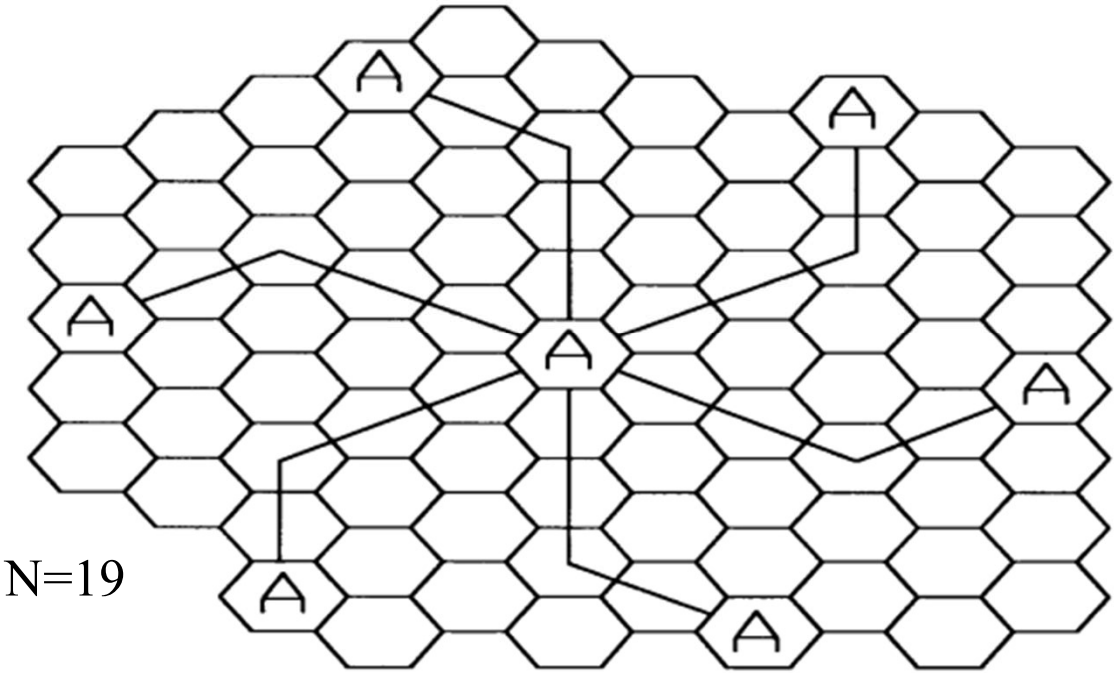
$$N = i^2 + ij + j^2$$

- The nearest co-channel neighbors of a particular cell can be found by doing what follows:
  - i and j are **non-negative** integers
  - i and j are **shift parameters**
  - Move i cells along any chain of hexagons.
  - Turn 60 degrees counter-clockwise.
  - Then move j cells.



- N, the cluster size is typically 4, 7, or 12.
- In GSM normally N =7 is used.





for i=3 and j=2, N=19

1.2 Cellular System Concepts (Continued)

1.2.4 Frequency Planning of Cellular System

- Let D be the frequency reuse distance; the distance between centers of the nearest co-channel cells (the cells that use the same band of frequencies).
- Refer to Fig. 1.15, if R is the radius of a cell, the distance between centers of adjacent cells, d and the step size → (i or j =1) where i and j are non-negative integers. They represent the number of steps in the U or V axis and N is the number of cells in a cluster, so we can calculate:

$$d = \sqrt{3}R \quad \text{meter} \qquad (1.2)$$

$$D = d\sqrt{i^2 + j^2 - 2ij\cos(\alpha)} \ \text{meter} \qquad (1.3a)$$

- Because $\alpha = 120^\circ$, $\therefore \cos\alpha = -0.5$
- Because the step size $d = \sqrt{3}R$ meter

- Then the frequency reuse distance, D is:

$$D = \sqrt{3}R\sqrt{i^2 + j^2 + ij} \quad \text{meter} \qquad (1.3b)$$

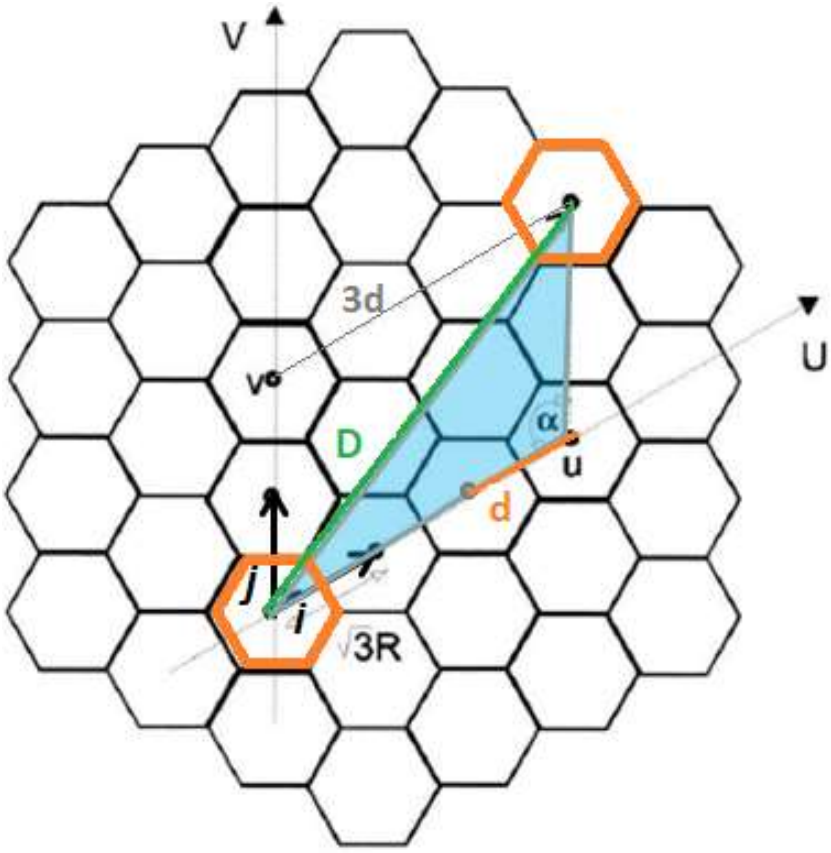
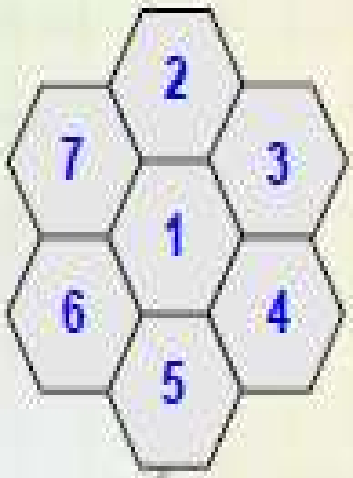
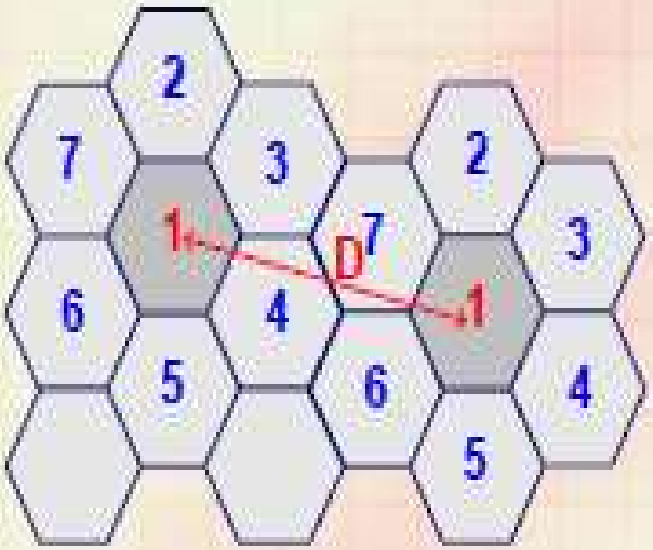
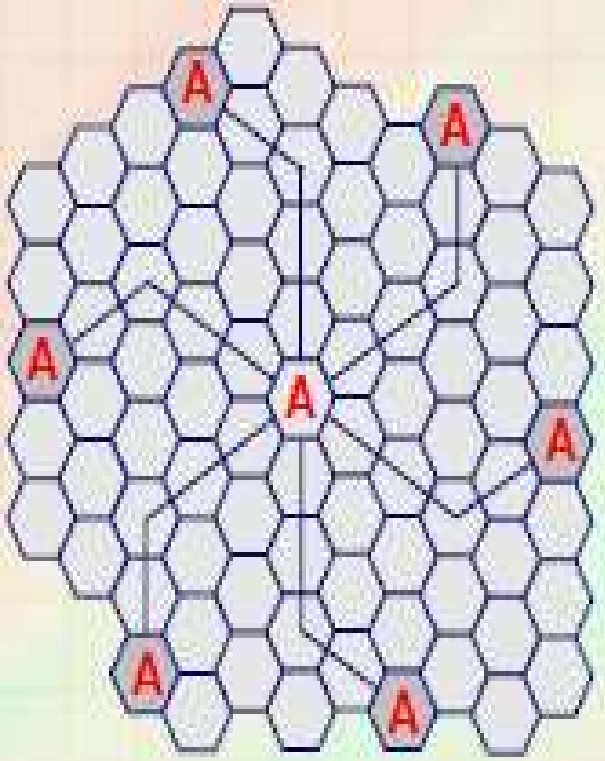


Fig. 1.15 Hexagonal shape cell

Frequency reuse







$$\frac{D}{R} = \sqrt{3N}$$

1.2 Cellular System Concepts (Continued)

1.2.4 Frequency Planning of Cellular System

- To find the nearest co-channel neighbors of a particular cell, one must do the following:

1. move i cells along any chain of hexagons and then
2. turn 60 degrees counter-clockwise and move j cells.

- This is illustrated in Fig. 1.16 for i = 1 and j = 2 (example, N = 7).
- Each frequency is reused after the same distance, D and is calculated as:

	Area	# of cells
Cell	2.6 R^2	1
Large hexagonal	2.6 D^2	3 N

$$\frac{Area(\text{large})}{\text{Area(cell)}} = \frac{2.6D^2}{2.6R^2} = \frac{3N}{1} \qquad (1.4)$$

$$D^2 = 3R^2N \Rightarrow D = \sqrt{3}R\sqrt{N} \qquad (1.5)$$

Compare (1.3b) with (1.5), we get:

$$N = \left(i^2 + j^2 + ij\right) \quad (1.6) \qquad D = \sqrt{3N}R \quad (1.7)$$

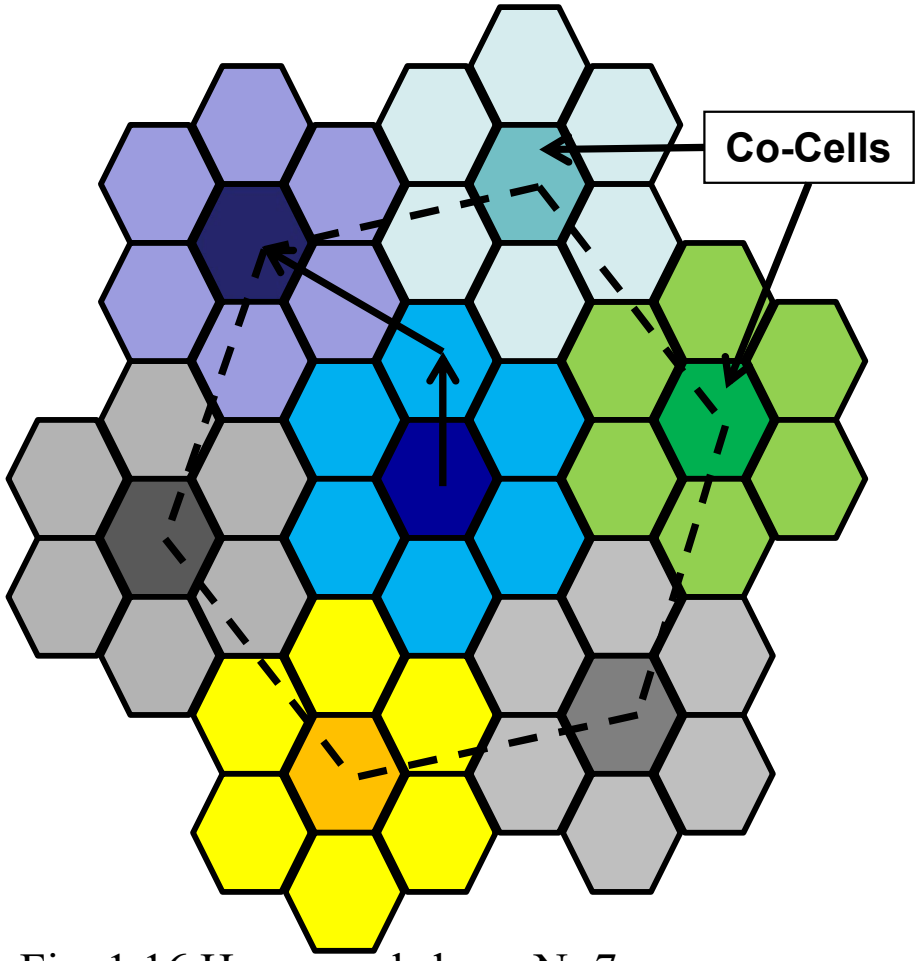


Fig. 1.16 Hexagonal shape N=7

1.2 Cellular System Concepts (Continued)

1.2.4 Frequency Planning of Cellular System

- Define the co-channel reuse ratio (interference rejection ratio), Q = D/R. From (1.7) Q is given by:

$$Q = D / R = \sqrt{3N} \qquad (1.7)$$

- To find the nearest co-channel neighbors of a particular cell, one must do the following:
  1. move i cells along any chain of hexagons and then
  2. turn 60 degrees counter-clockwise and move j cells.
- This is illustrated in Fig. 1.17, for i = 2 and j = 2 (then N = 12).

$$N = \left(i^2 + j^2 + ij\right) \qquad (1.6)$$

$N = 1 \quad i = 1, j = 0$
$N = 3 \quad i = 1, j = 1$
$N = 4 \quad i = 2, j = 0$
$N = 7 \quad i = 2, j = 1$
$N = 9 \quad i = 3, j = 0$
$N = 12 \quad i = 2, j = 2$

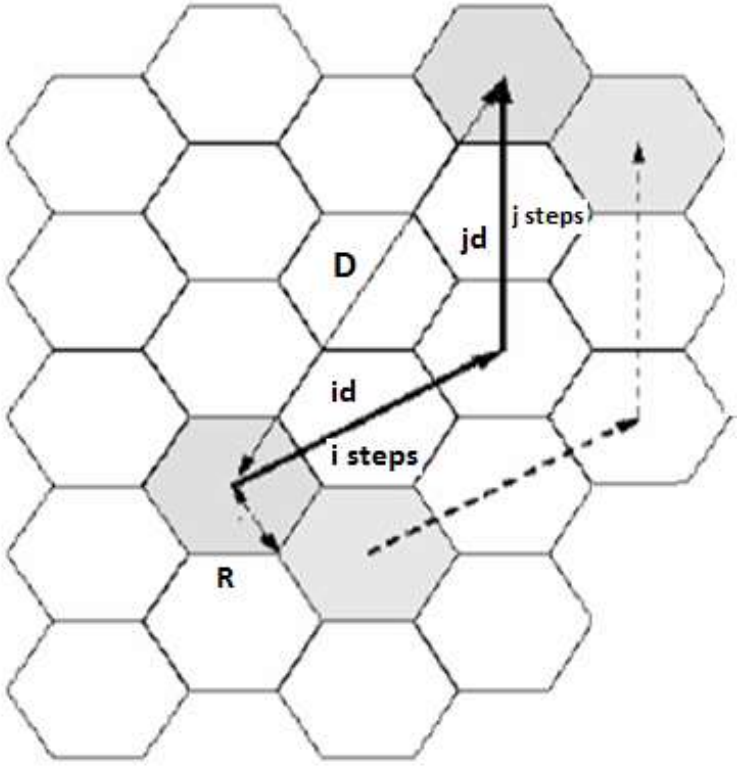


Fig. 1.17 Hexagonal shape N=12.

1.2 Cellular System Concepts (Continued)

1.2.4 Frequency Planning of Cellular System

- The following table illustrates N for different i and j and N is given Eq. (1.6) as:

$$N = \left(i^2 + j^2 + ij \right) \qquad (1.6)$$

N		j												
		0	1	2	3	4	5	6	7	8	9	10	11	12
i	0	0	1	4	9	16	25	36	49	64	81	100	121	144
	1		3	7	13	21	31	43	57	73	91	111	133	157
	2			12	19	28	39	52	67	84	103	124	147	172
	3				27	37	49	63	79	97	117	139	163	189
	4					48	61	76	93	112	133	156	181	208
	5						75	91	109	129	151	175	201	229
	6							108	127	148	171	196	223	252
	7								147	169	193	219	247	277
	8									192	217	244	273	304
	9										243	271	301	333
	10											300	331	364
	11												363	397
	12													432

- A small value of Q provides a larger capacity since the cluster size N is small, whereas a large value of Q improves the transmission quality, due to a smaller level of co-channel interference.
- The parameter Q is called the (frequency reuse ratio) or (interference rejection ratio) and is related to the cluster size, for hexagonal geometry:

From the above equation:

- Small (Q) means the small value of cluster size (N) and increase in cellular capacity and increased interference.
- Large (Q) leads to a decrease in system capacity but an increase in transmission quality and a decrease in interference.
- Choosing the options is very careful for the selection of (N).

- A cellular system having 'S' duplex channels, each cell is allocated 'K' channels (***K*** < ***S)*** channels are allocated to N cells into unique and disjoint channels, the total no of available channels is

$$S = NK$$

- N cells collectively using all the channels is called a cluster (a group of adjacent cells).
- If the cluster is repeated M times, the capacity C of the system is given as

$$C = MS = MkN$$

C: Total Capacity of the system

M: Number of replications of the cluster in the systems

S: Total Number of available radio channels/cluster

K: Number of channels per cell

N: Number of cells per cluster (cluster size)

- The capacity of the system is directly proportional to the no. of times the cluster is repeated.
- Reducing the cluster size N while keeping the cell size constant, more clusters are required to cover the given area and hence more capacity.
- Co-channel interference is dependent on cluster size; a large cluster size has less interference and vice versa.

Given a cell radius R and a cluster size N, the geographic area covered by a cluster is:

$$A_{cluster} = N\,A_{cell} = N3\sqrt{3}R^2/2.$$

If the total serviced area is A_{total}, then the number of clusters M that could be accommodated is given by:

$$M = A_{total}/A_{cluster} = A_{total}/(N3\sqrt{3}R^2/2).$$

Ex (1)

Out of the numbers 3, 8, 15, 21, 23, 30, and 47, what values are possible for a cluster size in a regular hexagonal geometry of cellular architecture? Justify the same using the relationship between the cluster size N and shift parameters i and j.

Ans:

Cluster size, $N = i^2 + j^2 + i \times j$, where i and j are shift parameters of regular hexagonal geometry and can have any integer values such as 0, 1, 2, 3, …. so on.

For $i = 1, j = 1$; $N = i^2 + j^2 + i \times j$; $N = 3$ is possible.

For $i = 4, j = 1$; $N = i^2 + j^2 + i \times j$; $N = 21$ is possible.

Ex (2)

A large city with an area of 1000 km^2 is required to be covered by finite number of cells with a radius of 2 km each. How many cell sites would be required, assuming regular hexagonal shaped cells?

Ans:

Total service area, $A_{service}$ = 1000 km^2 (Given)

Radius of a cell, R = 2 km (Given)

Area of a regular hexagonal cell, $\mathbf{A_{hexagon\ cell} = (3\sqrt{3} / 2) \times R^2}$

Or, $A_{hexagon\ cell}$ = 10.4 km^2

Number of cell sites = $A_{service} / A_{hexagon\ cell}$ = 1000 / 10.4 = **96 cell sites** **Ans.**

Ex (3)

A new cellular service provider decides to employ a cluster of 19 cells for frequency reuse.

(a) Draw one such cluster structure?

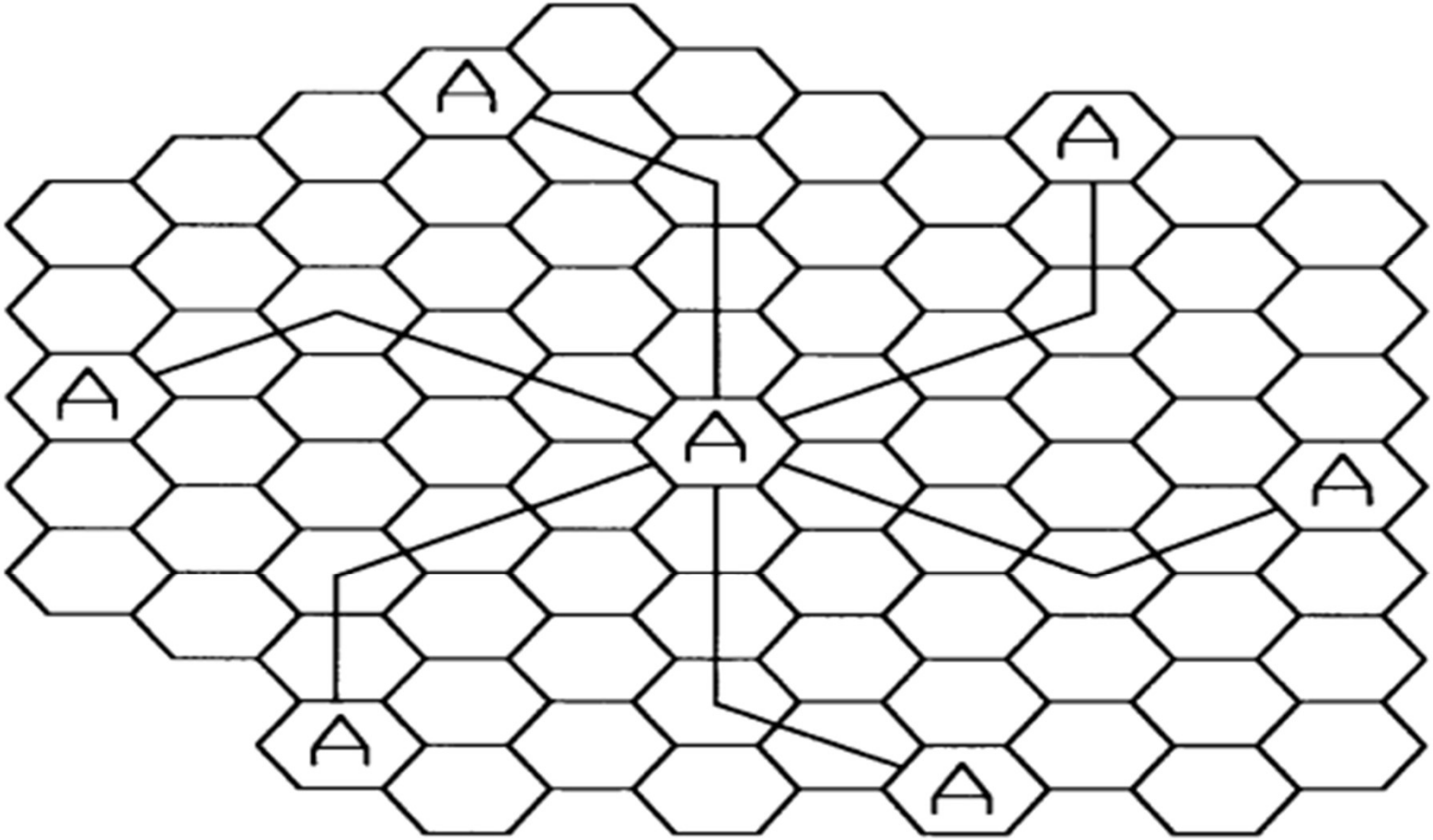
(b) Is there an alternative cluster of 19 cells available for part (a)? If yes, how?

(c) Determine the reuse distance for part (a), if the radius of a cell is 2 Kms.

Ans:

We know that for $N = 19$; $i = 3$ and $j = 2$

Yes, there is an alternative way of representing cluster of 19 cells. One way is to turn 60 degrees clockwise (instead of counterclockwise) and move j number of cells, after moving i number of cells along any chain of hexagons. Other alternative ways are to draw cluster structure of $N = 19$ with $i = 2$ and $j = 3$.



for i=3 and j=2, N=19

(c) Radius of a cell, R = 2 km (Given)

Cluster size, N = 19 (Given)

$D / R = \sqrt{3N}$ or $D = R \times \sqrt{3N} =$ **15 Km**

Ex (4)

Calculate the number of channels per cell for a cluster size of 7 in a cellular system which has 1001 radio channels available for handling traffic. The serving area of the complete system is 2100 Km^2 and the area of the cell is 6 Km^2.

Ans:

Service area of the cellular system, A_{sys} = 2100 km^2 (Given)

Service area of a cell, A_{cell} = 6 km^2 (Given)

Total number of available channels, **S** = 1001 (Given)

Cluster size, ***N*** = 7 (Given)

The coverage area of a cluster, $A_{cluster}$ = ***N*** x A_{cell}

Therefore, $A_{cluster}$ = 7 x 6 km^2 = 42 km^2

The number of times that the cluster has to be replicated to cover the entire service area of cellular system = $A_{sys} / A_{cluster}$

Or, number of clusters, M = 2100 / 42

Hence, number of clusters, M = 50 clusters

Since all available radio channels are allocated to one cluster, and then cluster is repeated many times to cover the service area. Therefore,

The number of channels per cell, K = **S / N**

Or, cell capacity, K = 1001 / 7

Hence, cell capacity, K = **143 channels/cell** **Ans.**

The system capacity, C = **S** x M

Or, system capacity, C = 1001 x 50

Hence, the system capacity, C = **50,050 channels** **Ans.**

Ex (5)

A cellular service provider is allocated RF spectrum of 30 MHz. The cellular system is configured with a cluster size of 12. It has simplex channel bandwidth of 25 KHz and 10 channels per cell are reserved as control channels.

(a) Find the reuse distance if the radius of each cell is 5 Kms.

(b) How many channels per cell are available for traffic purpose?

(c) How many calls can be simultaneously processed by each cell if 8 users can share each channel?

Ans:

(a) Radius of a cell, R = 5 km (Given)

Cluster size, **N** = 12 (Given)

We know that **(Refer Eq. 4.15)**

$D / R = \sqrt{3N}$ or $D = R \times \sqrt{3N} = 30$ **Km** **Ans.**

(b) Total allocated RF bandwidth = 30 MHz (Given)

Simplex channel bandwidth = 25 KHz (Given)

Duplex channel bandwidth = 2 x 25 KHz = 50 KHz

Total number of channels = 30 MHz / 50 KHz = 600

Since all available channels are allocated to one cluster, and then cluster is repeated many times to cover the service area. Therefore,

Number of channels per cluster = 600

Number of cells per cluster = 12 (Given)

Therefore, total number of channels per cell = 600 / 12 = 50

Number of control channels per cell = 10 (Given)

Therefore, number of traffic channels per cell = 50-10 = **40** **Ans.**

(c) Number of users per traffic channel = 8 users (Given)

Therefore, number of simultaneous calls per cell = 40 x 8 = **320** **Ans.**

Ex (6)

A cellular system has a total RF bandwidth of 12.5 MHz and Duplex channel spacing of 30 KHz. The system contains 20 control channels. The system is required to cover a total area of 3600 Km^2.

(a) Calculate the number of traffic channels/cell, if the cluster size is 9.

(b) Find the reuse distance for part (a), if the area of each cell is 8 Km^2.

(c) Determine the total number of cells needed to cover the entire area.

(d) How many calls can be simultaneously processed by each cell if 8 users can share each channel?

Ans:

(a) Total allocated RF bandwidth = 12.5 MHz (Given)

Duplex channel bandwidth = 30 KHz (Given)

Total number of channels in system = 12.5 MHz / 30 KHz = 416

Number of control channels in the system = 20 (Given)

Therefore, number of traffic channels in the system = 416-20 = 396

Since all available traffic channels are allocated uniformly among cells of a cluster, and then cluster is repeated many times to cover the service area.

Therefore, number of traffic channels per cluster = 396

Number of cells per cluster (or cluster size), **N** = 9 (Given)

Therefore, number of traffic channels per cell = 396 / 9 = **44** **Ans.**

(b) Area of a cell, A_{cell} $= 8\ Km^2$

We know that A_{cell} $= 2.6\ R^2$ (Assuming regular hexagonal cell)

Therefore, radius of a cell, R $= 1.75$ km

We know that **(Refer Eq. 4.15)**

$D / R = \sqrt{3N}$ or $D = R \times \sqrt{3N} =$ **9 Km** **Ans.**

(c) Service area of the cellular system, A_{sys} $= 3600\ km^2$ (Given)

Service area of a cell, A_{cell} $= 8\ km^2$ (Given)

Therefore, number of cells $= A_{sys} / A_{cell} =$ **450 cells** **Ans.**

(d) Number of users per traffic channel $= 8$ users (Given)

Therefore, number of simultaneous calls per cell $= 44 \times 8 =$ **352** **Ans.**

1.3 Channel Assignment Strategies (CAS)

- A scheme for *increasing capacity* and *minimizing interference* is required for effective utilization of radio spectrum.
- Channel Assignment strategies means to allocate the available channels to the cells in a cellular system whenever a user wants to make a call request then by using channel assignment strategies their requests are fulfilled. Channel Assignment Strategies are designed in such a way that there is efficient use of frequencies, time slots and bandwidth.

- Choice of CAS impacts the performance of the system for call management.
- CAS can be classified as either *fixed* or *dynamic* or *Hybrid Channel Assignment*

1.Fixed Channel Assignment (FCA)

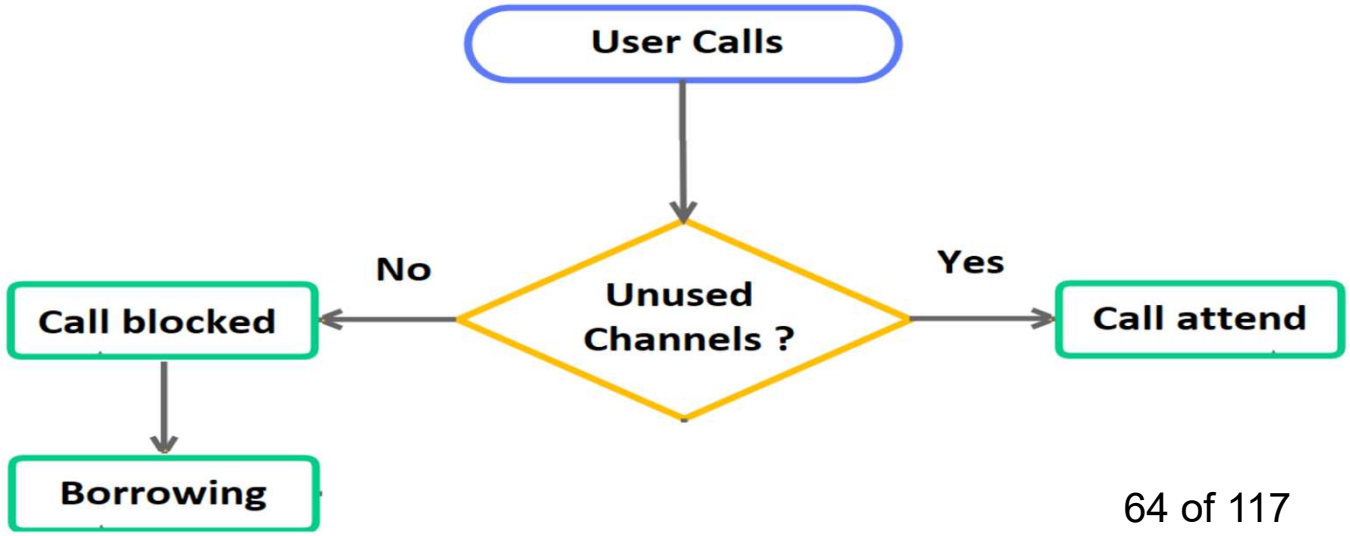
2.Dynamic Channel Assignment (DCA)

3.Hybrid Channel Assignment (HCA)

1. Fixed Assignment Strategy (FAS)

It is a strategy in which fixed number of channels or voice channels are allocated to the cells. Once the channels are allocated to the specific cells then they cannot be changed.

- Fixed channel assignment are **very simple**
- Requires **least amount of processing**.
- Any call attempt within the cell can only be served by the unused channel in that particular cell
- If all the channels in the cell are occupied, the call is *blocked* (**disadvantage**) and the user does not get service.
- In a variation of FCA – called *borrowing strategy*, a cell can borrow channels from its neighboring cell if its own channels are full.
- Cells in this strategy are allowed to **borrow** channels from adjacent cells if their channels are fully occupied while adjacent cells have free channels.
- **Ensure that no interference** occurs by moving the channel from one cell to another.
- **MSC monitors** the borrowing process to ensure that there is no interference with any of the calls in the donor cell



2. Dynamic Channel Assignment (DCA)

• Voice channels are not allocated to different cells permanently.

• Each time a call request is made, the **BS requests** a channel from the **MSC**.

• MSC allocates a channel to the requesting cell using an algorithm that takes into account :

– the likelihood of future blocking

– The reuse distance of the channel (should not cause interference)

– MSC collect **real time data** on:

- channel occupancy
- traffic distribution
- radio signal strength indication (RSSI) of all channels on continuous basis

– This increases the storage and computational load on MSC.

– Provides the advantage of **increased channel utilization** and **capacity** and **decreased probability of a blocked call**.

• To ensure min. QoS, MSC only allocates a given frequency, not currently in use in the cell or any other cell which falls within the limiting reuse distance.

• On the system channel assignment based on the real traffic:

1) Low traffic ⟶ Fewer channels.

2) High traffic ⟶ More channels.

3. Hybrid Channel Assignment (HCA)

- Hybrid Channel Allocation is a combination of both Fixed Channel Allocation (FCA) and Dynamic Channel Allocation (DCA). In this total number of channels or voice channels are divided into fixed and dynamic sets. If a user make a call, then first fixed set of channels are utilized but if all the fixed sets are busy then dynamic sets are used.

- The main purpose of HCA is to work efficiently under heavy traffic

1.4 Interference in Cellular System

- Interference is the major limiting factor in the performance of cellular radio systems.
- Interference on **voice channels** causes **cross talk**, where the subscriber hears interference in the background due to an undesired transmission.
- On **control channels**, interference leads to **missed and blocked calls** due to errors in the digital signaling.
- Interference is more severe in urban areas, due to the greater RF noise floor and the large number of base stations and mobiles.

Three major cellular interference:

1. Near end far end interference which is caused by another mobile in the same cell.
2. Adjacent channel interference (ACI) is caused by signals from neighbouring frequencies.
3. Co-Channel Interference (CCI) is caused by signals at the same frequency either from another BTS operating in the same frequency band or other non-cellular system operating in the same band.

1.4. Interference in Cellular System (Continued)

1.4.1 Near End Far End Interference

- The purpose of Power Control (PC) is to ensure that each user receives and transmits just enough energy to have service but to prevent:

  - Blocking of distant users (near-far-effect)
  - Exceeding reasonable interference levels

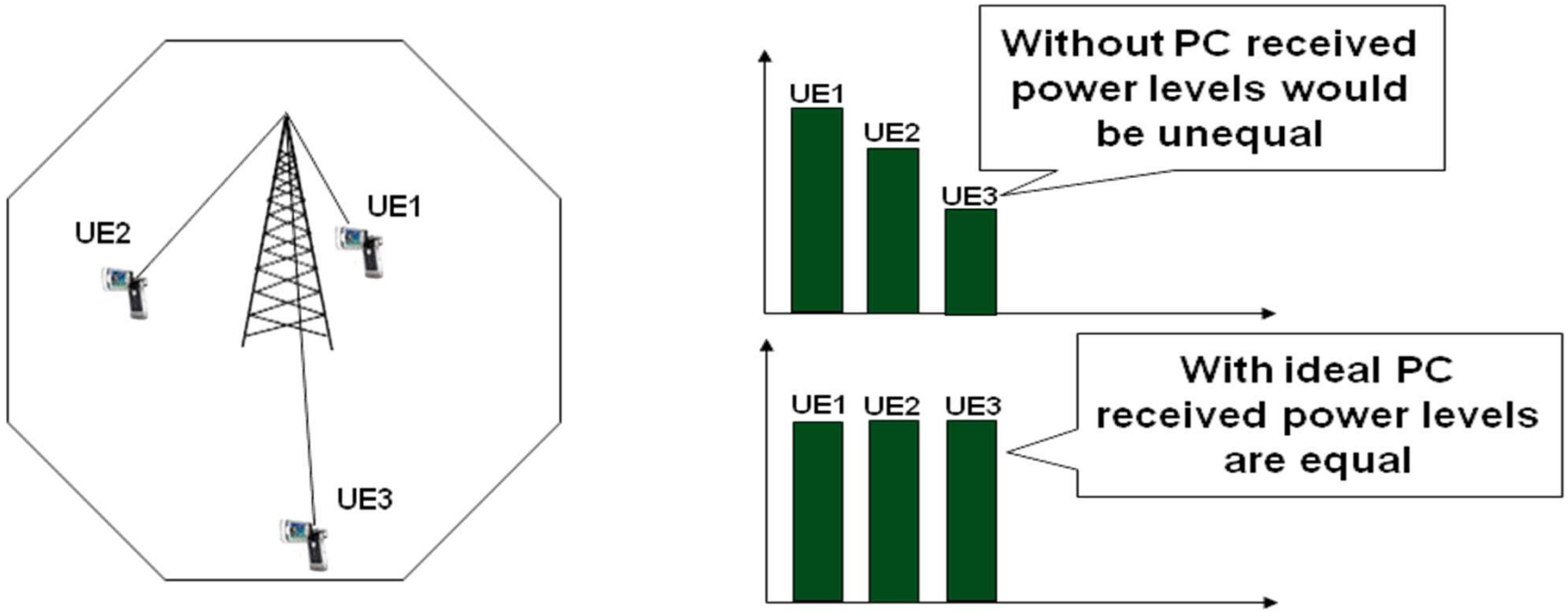


Fig. 1.22 Near end far end interference .

1.4. Interference in Cellular System (Continued)

1.4.2 Adjacent Channel Interference

- It is the interference from adjacent in frequency to the desired signal due to:
  - Imperfect receiver filters allow nearby frequencies to leak into the passband
  - Performance degrade seriously due to near-far effect.
- Adjacent channel interference can be minimized through careful filtering.
- Keep the frequency separation between each channel in a given cell as large as possible by using channel assignment strategy.

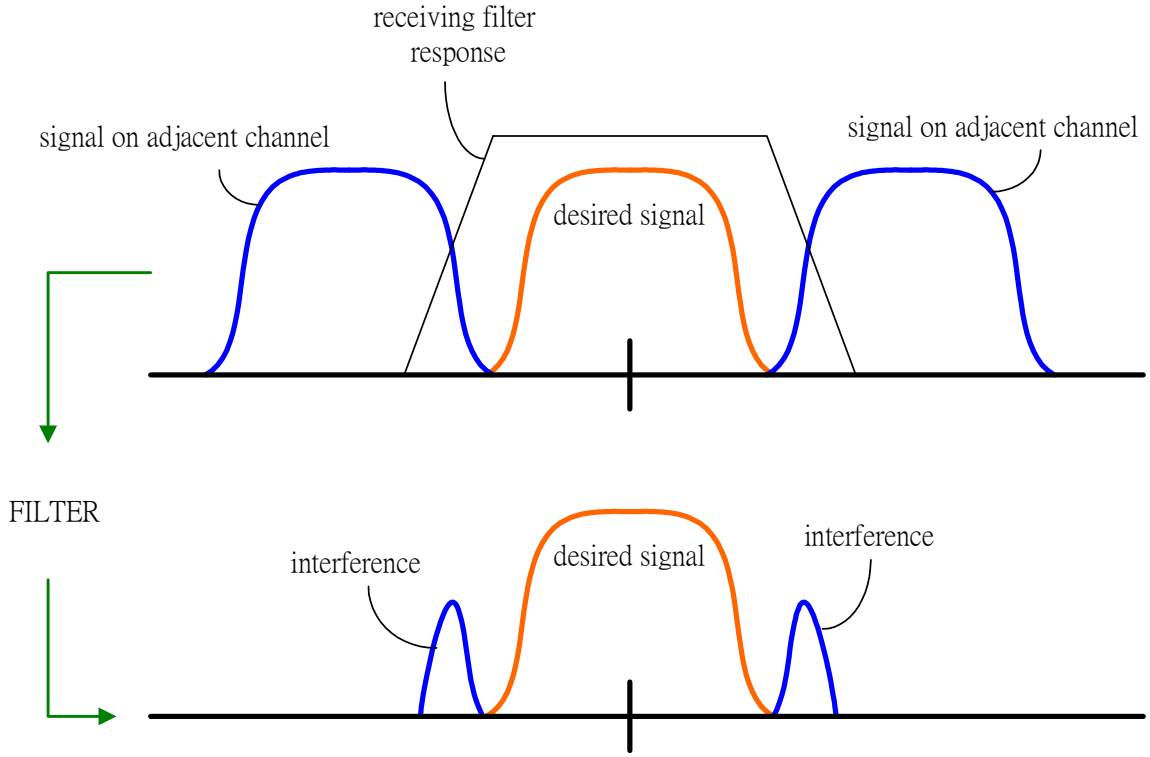


Fig. 1.23 Adjacent Channel Interference.

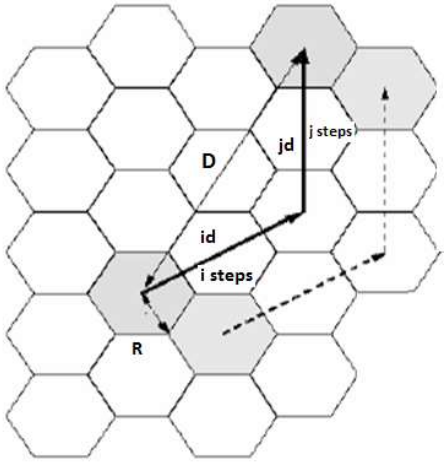
1.4. Interference in Cellular System (Continued)

1.4.3 Co-channel Interference

- Cells which use same frequency (frequency re-use) are called co-channel cells.
- Interference between their signals is called co-channel interference.
- This type cannot be solved by raising SNR; it actually increases by raising transmission power.
- It can be reduced by separating co-channel cells physically by a minimum distance to provide sufficient isolation due to interference.
- For a hexagonal geometry, it was shown that the co-channel reuse ratio is given by Eq. (2.1) as:

- Where N is the cluster size. As Q increases, spatial separation between co-channels increases ➔ co-channel interference is reduced.
- So as Q ↑➔ transmission quality increases – whereas Q ↓➔ system capacity ↑
- If n_0 is the number of the co-channel interfering cells, then the signal-to-Interference ratio (S/I or SIR) for a mobile receiver that is monitoring a forward channel can be given as:

$$Q = D / R = \sqrt{3N} \qquad (2.1)$$



- where I_i is the mean interference power originating from the ith cell.

$$S / I = \frac{S}{\sum_{i=1}^{n_0} I_i} \qquad (2.2)$$

1.4 Interference in Cellular System (Continued)

1.4.3 Co-channel Interference

- Propagation measurements in a mobile radio channel show that the average received signal strength at any point decays as a power law of the distance of separation between a transmitter and receiver
- The average received power, P_r, at a distance d from the transmitting tower is given by:

$$P_r(d) = P_r(d_0)(d/d_0)^{-n} \qquad (2.2a)$$

- where $P_r(d_0)$ is the power at a point that is at a small distance d_0 from the transmitting antenna at a distance, d and n is the path loss exponent (n = 2 → 6). In dBm, Eq. (2.2a) can be written in the form:

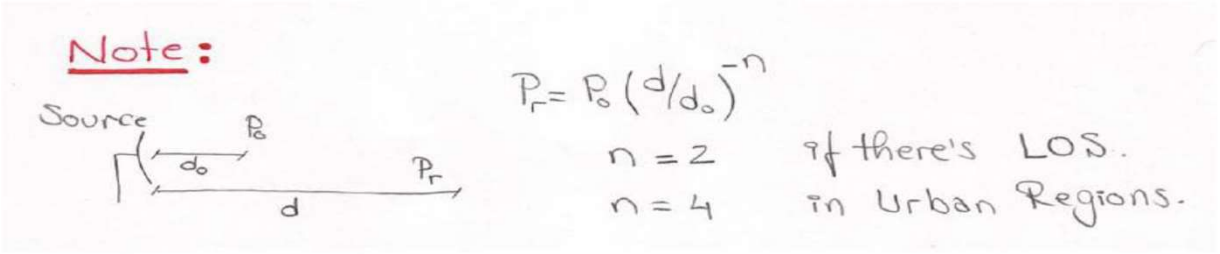
$$P_r(d)dBm = P_r(d_0)dBm - 10n\log(d/d_0) \qquad (2.2b)$$

- If we consider the forward link from the serving BTS, D_i is the distance of the ith interferer from the mobile. The received power at a given mobile due to the ith interferer is proportional to $(D_i)^{-n}$.
- When transmit power for each BTS is equal and the path loss exponent is same for all area, the S/I for a mobile can be approximated as:

$$S/I = \frac{S}{\sum_{i=1}^{n_0} I_i} = \frac{P_0(R/d_0)^{-n}}{\sum_{i=1}^{n_0} P_0(D/d_0)^{-n}} = \frac{R^{-n}}{\sum_{i=1}^{n_0} D_i^{-n}} \qquad (2.3)$$

$$S = P_0(R/d_0)^{-n}$$

$$I = P_0(D/d_0)^{-n}$$



1.4 Interference in Cellular System (Continued)

- Considering only first tier of interfering cells, if all interfering BSs are at a distance, D, from the desired BS, then S/I can be simplified as:

$$S/I = (D/R)^n / n_0 = \left(\sqrt{3N}\right)^n / n_0 \quad (2.4)$$

- **The received power**

$$P_r \approx \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d^2}$$

The received power

The transmit power

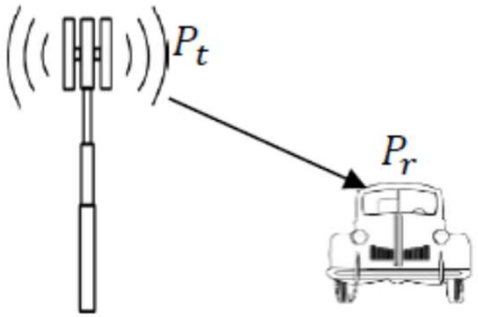
The transmit antenna gain

The receive antenna gain

Distance between TX and RX

The Wave-length

(free space) Model



Example 7

Consider the AMPS system in which Signal to Interference ratio, S/I =18 dB is required for the accepted voice quality. Assume that the number of the co-channel interfering cells in the first tier is $n_0 = 6$ and the path loss exponent is n = 4 calculate:

a) The minimum cluster size, N to meet this system requirement.
b) The minimum cluster size of GSM system in which S/I = 12 dB is required.

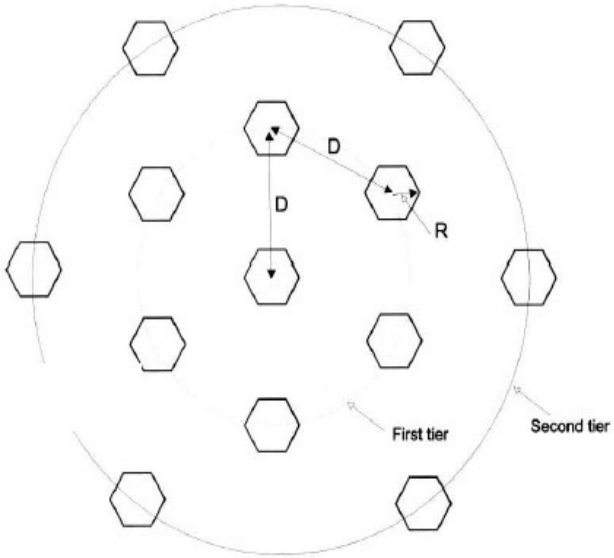


Fig. 1.24 Space distribution of interfacing cells.

1.4 Interference in Cellular System (Continued)

1.4.3 Co-channel Interference

Example 7 Solution

$N = 1$	$i = 1, j = 0$
$N = 3$	$i = 1, j = 1$
$N = 4$	$i = 2, j = 0$
$N = 7$	$i = 2, j = 1$
$N = 9$	$i = 3, j = 0$
$N = 12$	$i = 2, j = 2$

For AMPS

$$\because S/I\Big|_{dB} = 10\log\frac{S}{I} = 18$$

$$\therefore \frac{S}{I} = 10^{1.8} = 63.1$$

$$\because S/I = \left(\sqrt{3N}\right)^n / n_0$$

$$63.1 = \left(\sqrt{3N}\right)^4 / 6 \qquad \frac{9N^2}{6} = 63.1$$

$$N = \sqrt{\frac{63.1x6}{9}} = 6.49$$

$so\ N$ should be at least 6.49

$$\therefore N\Big|_{\min} = 7$$

For GSM

$$\because S/I\Big|_{dB} = 10\log\frac{S}{I} = 12$$

$$\therefore \frac{S}{I} = 10^{1.2} = 15.85$$

$$\because S/I = \left(\sqrt{3N}\right)^n / n_0$$

$$15.85 = \left(\sqrt{3N}\right)^4 / 6$$

$$\frac{9N^2}{6} = 15.85$$

$$N = \sqrt{\frac{15.85x6}{9}} = 3.25$$

$so\ N$ should be at least 3.25

$$\therefore N\Big|_{\min} = 4$$

1.4. Interference in Cellular System (Continued)

1.4.3 Co-channel Interference

Example 8

If a signal-to-interference ratio, S/I = 15 dB is required for satisfactory forward channel performance of a cellular system. Calculate the frequency reuse factor and cluster size, N that should be used for maximum capacity if the path loss exponent is a) n = 4, b) n = 3. Assume that the number of the co-channel interfering cells in the first tier is $n_0 = 6$, and all of them are at the same distance, D from the mobile. Consider the possible cluster size values to be, N = 3, 7, 12 or 13.

Solution

a) n = 4

- First, let us consider a seven-cell reuse pattern, N = 7. Using Eq. (1.16), the co-channel reuse ratio:

$$Q = D/R = \sqrt{3N} = \sqrt{3x7} = 4.583$$

- Using Eq. (2.4), the signal-to-interference ratio is given by

$$S/I = (D/R)^n / n_0 = (\sqrt{3N})^n / n_0 = (\sqrt{3x7})^4 / 6 = 75.3 \Rightarrow 18.66 \ \ dB$$

- Since this is greater than the minimum required S/I, N = 7 can be used.

1.4. Interference in Cellular System (Continued)

1.4.3 Co-channel Interference

Example 8 Solution

(a) n = 3

- First, let us consider a seven-cell reuse pattern N = 7. Using Eq. (2.4), the signal-to-interference ratio is given by

$$S/I = (D/R)^n / n_0 = (\sqrt{3N})^n / n_0 = (\sqrt{3x7})^3 / 6 = 16.04 \Rightarrow 12.05 \ \ dB$$

Since this is less than the minimum required S/I, we need to use a larger N.

- The next possible value of N is 12 (given).

The corresponding co-channel ratio is given by Eq. (2.1) as:

$$Q = D/R = \sqrt{3N} = \sqrt{3x12} = 6$$

- Using Eq. (2.4), the signal-to-interference ratio is given by

$$S/I = (D/R)^n / n_0 = (\sqrt{3N})^n / n_0 = (\sqrt{3x12})^3 / 6 = 36 \Rightarrow 15.56 \ \ dB$$

- Since this is greater than the minimum required S/I, N = 12 is used.

1.4. Interference in Cellular System (Continued)

1.4.3 Co-channel Interference

Example 9

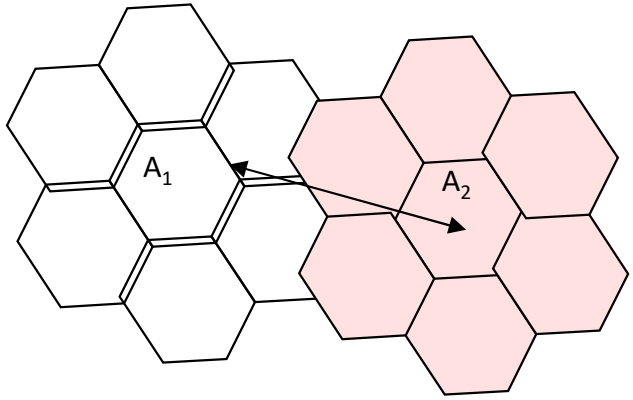
In a GSM network, cell A_1 is the desired cell and A_2 is a co-channel cell as shown in the following figure with cluster size, N = 7. For a mobile located in cell A_1, find the minimum cell radius R to give a forward link signal-to-interference ratio, S/I ≥ 26 dB. Assume the following:

- Co-channel interference is due to base A_2 only.
- The carrier frequency, f_c = 890 MHz.
- Reference distance, d_o = 1 km (assume free space propagation from the transmitter to d_o).
- Assume Omni directional antennas for both transmitter and receiver, where G_{base} = 6 dBi and G_{mobile} = 3 dBi.
- Transmitter power, P_t = 10 W (assume equal power for all base stations).
- Assume log-normal path loss model is applied between the mobile and base A_1, with loss exponent, n_1 = 2.5 and A_2 with loss exponent, n_2 = 4.

Solution

Given P_t = 10 W , $G_{base} = G_t$ = 6 dBi = 4,
$G_{mobile} = G_r$ = 3 dBi = 2 and d_o = 1 km.

$$\lambda = c / f_c = 3x10^8 /(890x10^6) = 0.337 \ \ m$$



1.4. Interference in Cellular System (Continued)

1.4.3 Co-channel Interference

Example 9 Solution For free space propagation the received power at d_o, $P_r(d_o)$ is given by Eq.(1.5):

$$P_r(d_0) = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d_0^2} = \frac{10x4x2x(0.337)^2}{(4\pi)^2 (1000)^2} = 5.76x10^{-8}\ W$$

$$P_r(d_0) = 10\ \log\ (5.76x10^{-8}) = -42.4\ dBm$$

- At the boundary of the cell A_1, the received power due to the transmitter A_1 is given by Eq. (1.9) as:

$$P_r(R) = P_0(R/d_0)^{-n} \quad in\ dBm \quad P_r(R)dBm = P_r(d_0) - 10n_1 \log(R/d_0)$$

- so $P_r(R)dBm = P_r(d_0) - 10n_1 \log(R/d_0) = -42.4 - 25\log(R/d_0)$
- The distance between the cell A_1 and Cell A_2, is given by Eq. (1.5) as:

$$D = \sqrt{3}R\sqrt{i^2 + j^2 + ij} = \sqrt{3N}R = \sqrt{3x7}R = 4.6R$$

- At the boundary of the cell A_1, the received power due to the transmitter A_2 is:

$$P_r(d_1)dBm = P_r(d_0) - 10n_2 \log(d_1/d_0) = -42.4 - 40\log(d_1/d_0)$$

where $d_1 = D - R = 4.6R - R = 3.6R$

- The signal-to-interference ratio, S/I is given by:

$$S/I = P_r(R) - P_r(d_1) = -42.4 - 25\log(R/d_0) - [-42.4 - 40\log(R/d_0) - 22.25]$$

$$= 22.24 + 15\log(R/d_o)$$

- C/I =26 dB= 22.24 + 15 log (R/ d_o) So, the cell radius , R =1.65 km.

1.5 Handover or Handoff

1.5.1 Handover or Handoff Definition

- If an MS approaches a cell boundary, the signal strength decreases, and the MS is passed to a transmitter of the next (closest BTS) cell, then MS is assigned a new idle channel belongs to the new cell.
- In other words, if a mobile station moves during the call from one cell to another, the signal may become weak. To solve this, the Base Station Controller (BSC) monitors the signal until it diminishes and if so, the MSC seeks a new cell that is better for the call and assigns a new voice channel.

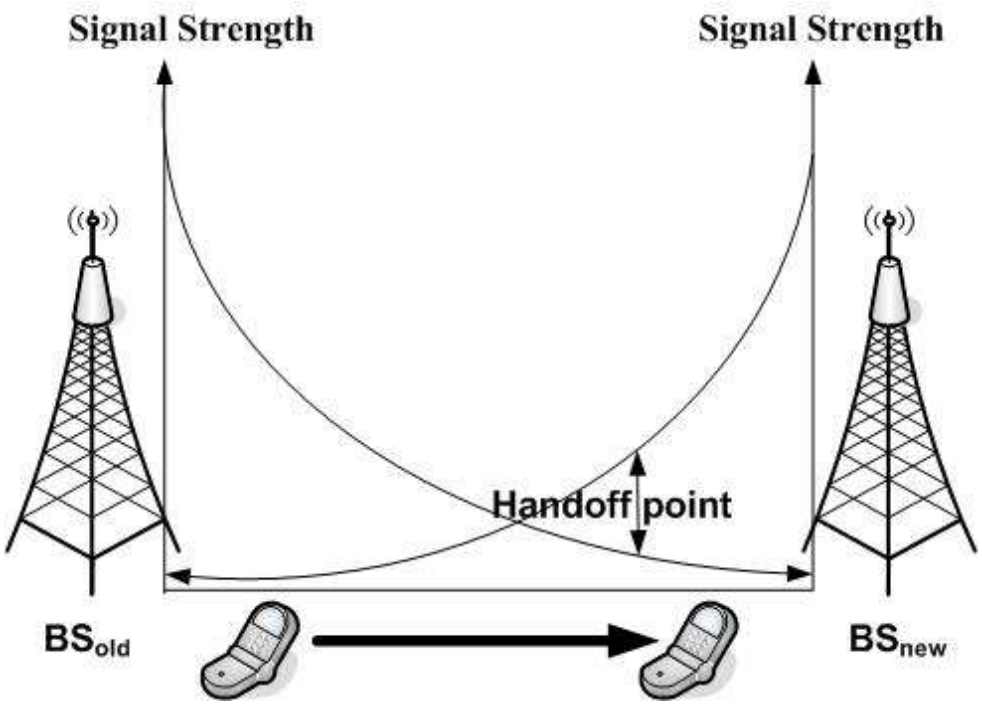


Fig. 1.20(a) Handover or handoff action.

1.5 Handover or Handoff (Continued)

1.5.2 Handover or Handoff classification

1. **Hard Handoff:** A hard handoff is essentially a "***break before make***" connection. Here the link to the prior base station is terminated before or as the user is transferred to the new cell's base station. This means that the mobile is linked to no more than one base station at a given time. A hard handoff occurs when users experience an interruption during the handover process caused by frequency shifting.

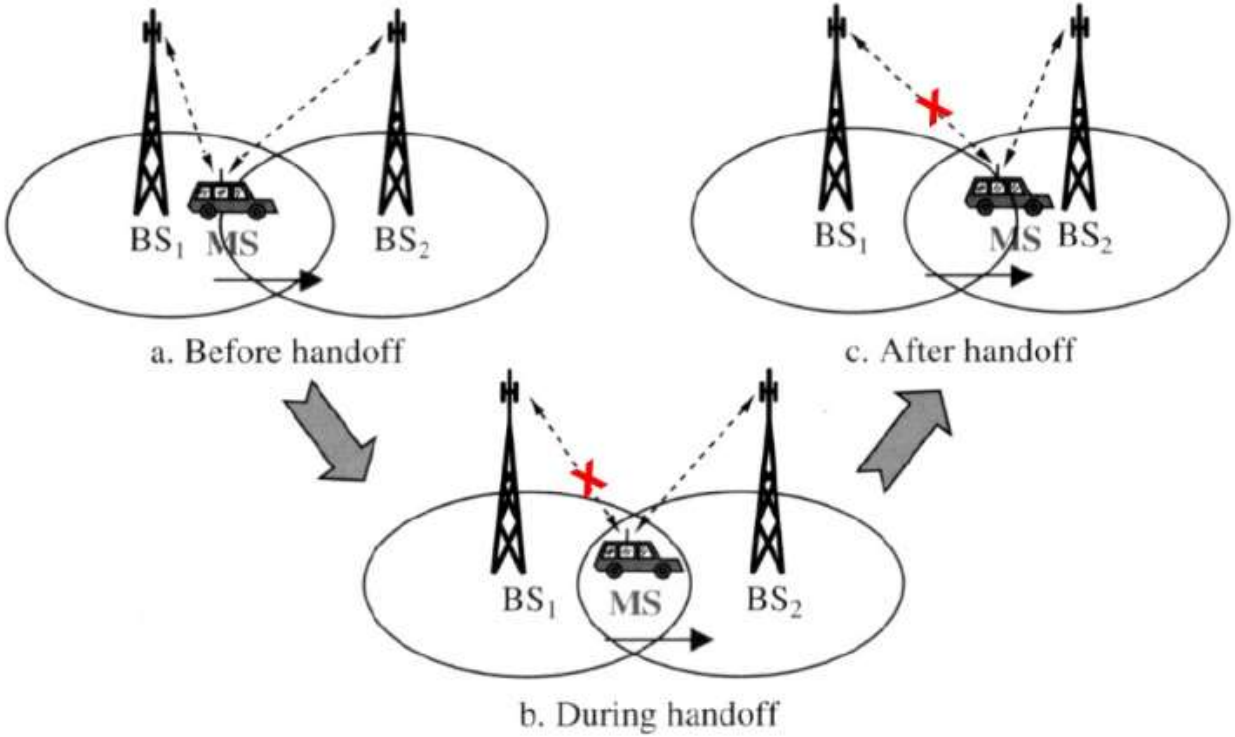


Fig. 1.20(b) Hard handoff concept.

1.5 Handover or Handoff (Continued)

1.5.2 Handover or Handoff classification

2. Soft Handoff: It is the ability to select between the instantaneous received signals from different base stations. Here the channel in the source cell is retained and used for a while in parallel with the channel in the target cell. In this approach, the connection to the target is established before the connection to the source is broken, hence this is called *"**make-before-break**"*.

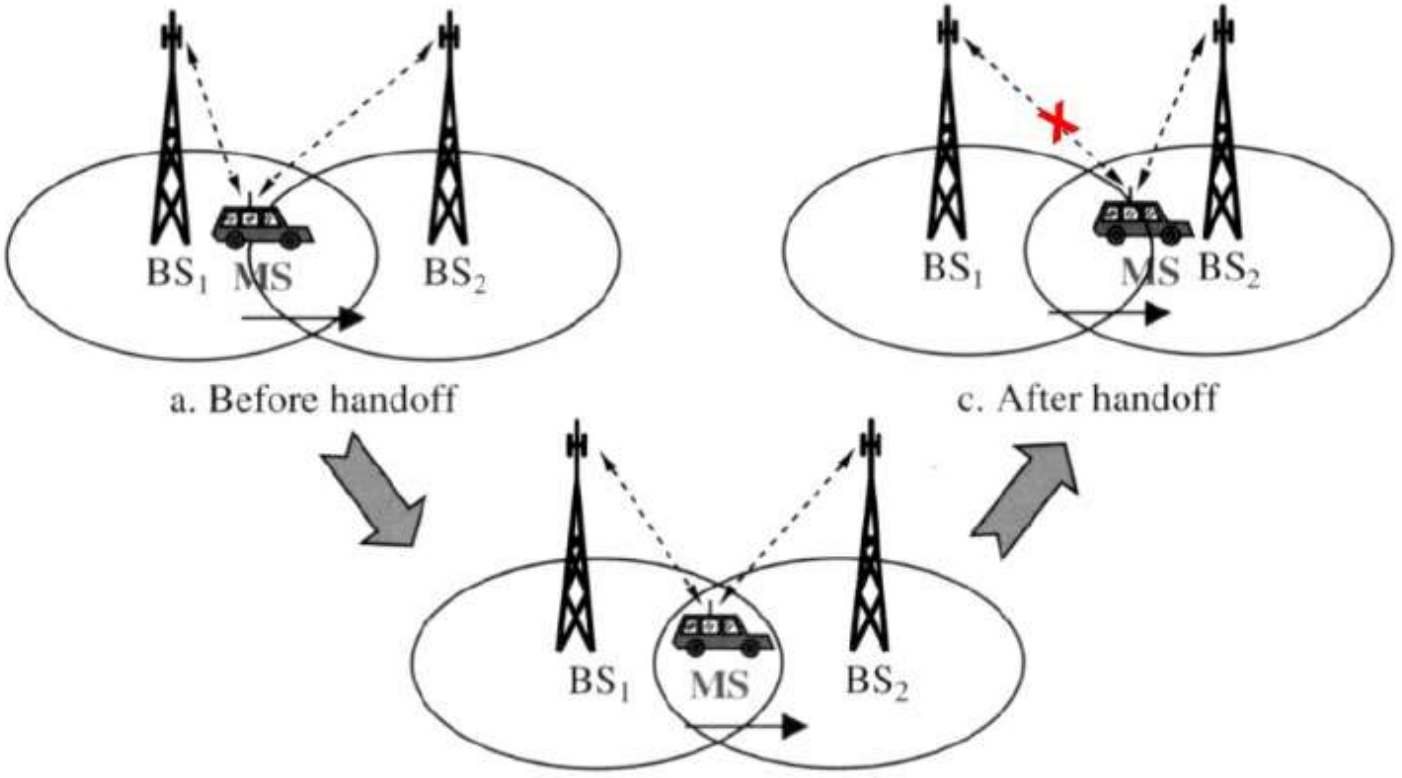


Fig. 1.20(c) Soft handoff concept.

1.5 Handover or Handoff (Continued)

1.5.2 Handover or Handoff classification

2. Soft Handoff:

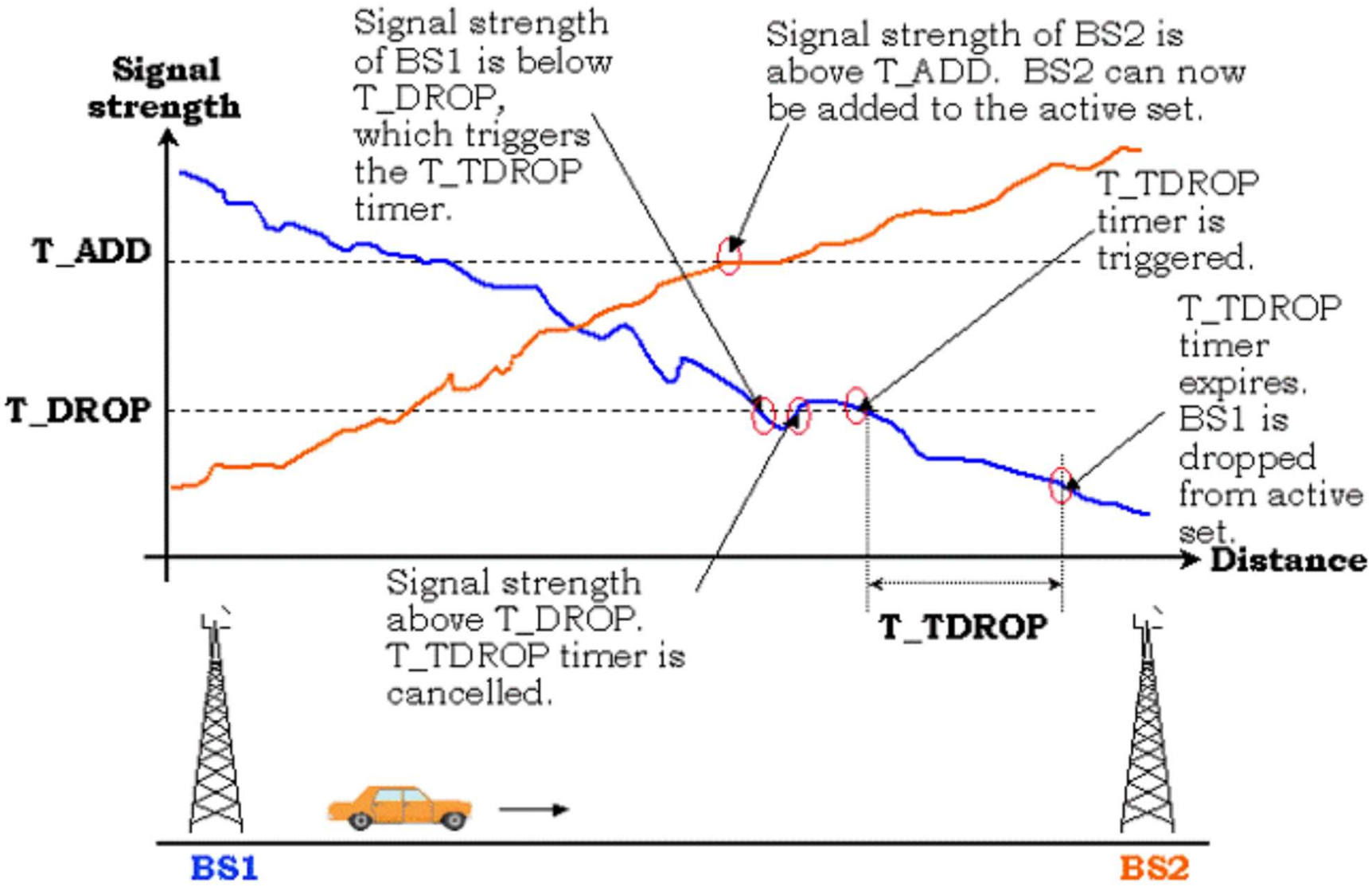


Fig. 1.20(d) Soft handoff concept.

1.5 Handover or Handoff (Continued)

1.5.3 Handoff Strategies

- System designers must specify an optimum signal level at which a handoff is initiated.
- Once a particular signal level is specified as the **minimum usable signal for acceptable voice** quality, P_{rmin} at the BS receiver, a slightly stronger signal level is used as a threshold, P_{rHO} at which a handoff is made.
- This margin, given by the difference:

$$\Delta P = P_{rHO} - P_{rmin}$$

- ΔP cannot be too large or too small.
  - If ΔP is **too large**, **unnecessary handoffs** which burden the Mobile Switching Center (MSC) may occur.
  - and if ΔP is **too small**, there may be **insufficient time to complete a handoff** before a call is lost due to weak signal conditions.
- In deciding when to handoff, it is important to ensure that the drop in the measured signal level is not due to **momentary fading** and that **the mobile is actually moving away from the serving BS**. This is done by monitoring the channel for a certain period before handoff is initiated.
- Handoff scenario at cell boundary is illustrated in Fig. 1.20(d).

1.5 Handover or Handoff (Continued)

1.5.3 Handoff Strategies

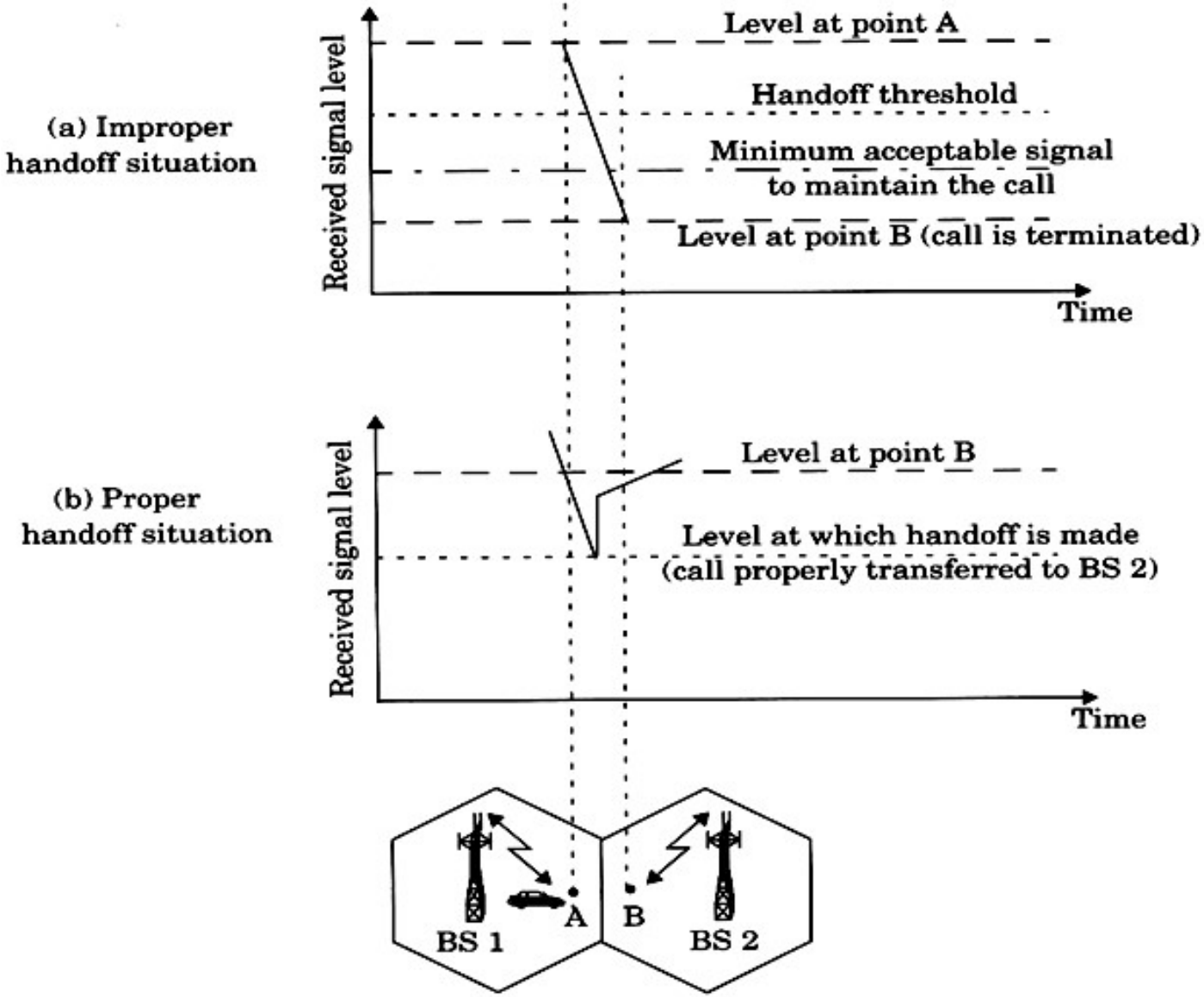


Fig. 1.21(a) Illustration of handoff scenario at cell boundary.

1.5 Handover or Handoff (Continued)

Example 10

Suppose that a mobile station is moving along a straight line from base stations BS1 and BS2 as in Fig. 1.21(b). If the distance between them is D = 6 km, P_o = 0 dB_m, d_o = 1 m and n = 2.5. Assume the minimum usable signal level for acceptable voice quality at the base station receiver is P_{rmin} = - 88 dBm, mobile average speed is v_{mob} = 80 km/h and the threshold level used by the switch for handoff initiation is P_{rHO} = -80 dBm.

a) Determine the minimum distance, d_{min} for acceptable signal level.

b) Find the maximum time required for processing hand off required by MSC (Δt).

c) If user mobility decrease to v_{mob} = 10 m/sec, the handoff margin, $\Delta P = P_{rHO} - P_{rmin}$ will decrease or increase?

Solution D = 6 km

$P_r(d_0)$ = 0 dBm at d_0 = 1 m n= 2.5

P_{rmin} = -88 dBm P_{rHO} = -80 dBm

v = 80 km/h = 22 m/sec

a) $P_r(d) = P_r(d_0) - 10*n*\log_{10}(d/d_o)$

P_{rmin} = -88 dBm at d_{max}

$-88 = 0 - 10(2.5)\log_{10}(d_{max}/1)$

$d_{max} = 10^{88/25} = 3311$ m = 3.311 km

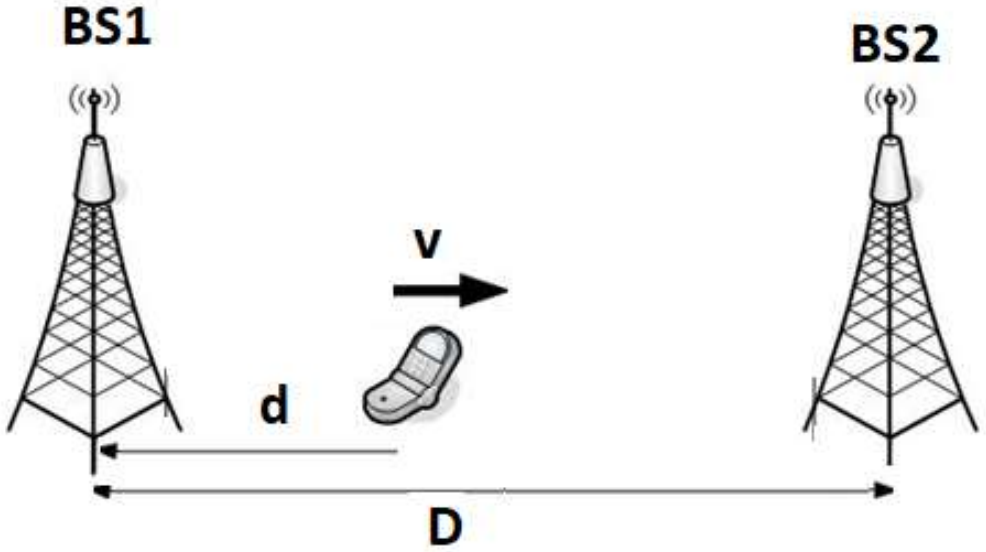


Fig. 1.21(b) Handoff of MS.

1.5 Handover or Handoff (Continued)

b) $P_{rHO} = -80$ dBm.

$-80 = 0 - 10(2.5)\log_{10}(d_{Ho}/1)$

$d_{Ho} = 10^{80/25} = 1584$ m $= 1.548$ km

The processing distance required by MSC to perform hand off, Δd is given by:

$\Delta d = d_{max} - d_{Ho} = 3.311 - 1.584 = 1.727$ km $= 1.727$ km

$$\Delta t = \frac{\Delta d}{v_{mob}} = \frac{3.311 - 1.584}{80} = \frac{3311 - 1584}{22} = 77.7 \text{ sec}$$

c) If the user mobility changes, we have the following situations:

$$\Delta t = \frac{\Delta d}{v_{mob}} \quad if\ \Delta t = cons\tan t$$

When increasing the mobile velocity, v_{mob} and processing hand off required by MSC, Δt = constant, then Δd increases, therefore, $\Delta P = P_{rHO} - P_{rmin}$ increases.

When decreasing the mobile velocity, v_{mob} and processing hand off required by MSC, Δt = constant, then Δd decreases, therefore, $\Delta P = P_{rHO} - P_{rmin}$ decreases.

So, If user mobility decrease to $v_{mob} = 10$ m/sec, the handoff margin, $\Delta P = P_{rHO} - P_{rmin}$ will decrease.

1.6 Improving Coverage & Capacity in Cellular Systems

• As the demand for wireless service increases, the number of channels assigned to a cell becomes insufficient to support the required number of users.

• Hence cellular design techniques are needed to provide more channels per unit coverage area.

• Techniques such as **cell splitting**, **sectoring**, and **zone microcell** approaches are used to expand the capacity of cellular systems.

• **Cell splitting** allows an orderly growth of the cellular system.

• **Sectoring** uses directional antennas to control the interference and frequency reuse of channels.

Microcell zone concept distributes the coverage of a cell and extends the cell boundary to hard-to-reach places.

Introduction (contd)

- While cell splitting increases the number of base stations to increase capacity, sectoring and zone microcells depend on base station antenna placements to improve capacity by reducing co-channel interference.

- Cell splitting and zone microcell techniques do not suffer trunking inefficiencies experienced by sectored cells.

- They enable the base station to oversee all handoff chores related to the microcells, thus reducing the computational load at the MSC.

- These three capacity improvement techniques will be explained in detail.

Introduction to Cell Splitting (number of base stations increased)

• Cell splitting is the process of subdividing a congested cell into smaller cells with
  – their own BS
  – a corresponding reduction in antenna height
  – a corresponding reduction in transmit power

Advantages

- achieves capacity improvement by rescaling the system
- increases the number of channels per unit area
- Splitting the cell reduces the cell size
- more number of cells to be used.
- The idea is to keep the co-channel reuse ratio (interference rejection ratio) Q = D/R constant while decreasing R.

Disadvantages

- Splitting the cell reduces the cell size
- more number of cells to be used.
- channel Assignments become more complicated

More number of cells ► more number of clusters ► more channels ►high capacity

Example for Cell Splitting

- An example of cell splitting is shown in Figure 1.
- The base stations are placed at corners of the cells & area served by base station A is assumed to be saturated with traffic.
- New base stations are needed in the region to increase the number of channels in the area and to reduce the area served by the single base station.
- In Figure 1, the original base station A has been surrounded by six new microcells.
- The smaller cells were added in such a way as to preserve the frequency reuse plan of the system.
- For example, the microcell base station labeled G was placed halfway between two larger stations utilizing the same channel set G.
- Hence cell splitting merely scales the geometry of the cluster.
- In this case, the radius of each new microcell is half that of the original cell.

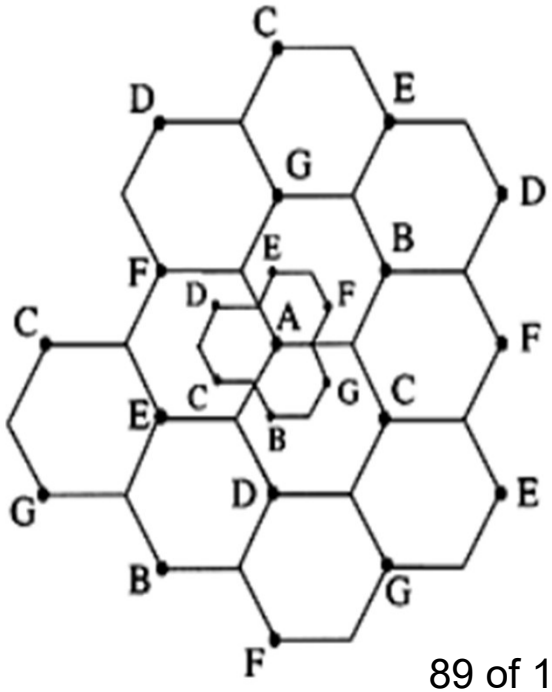
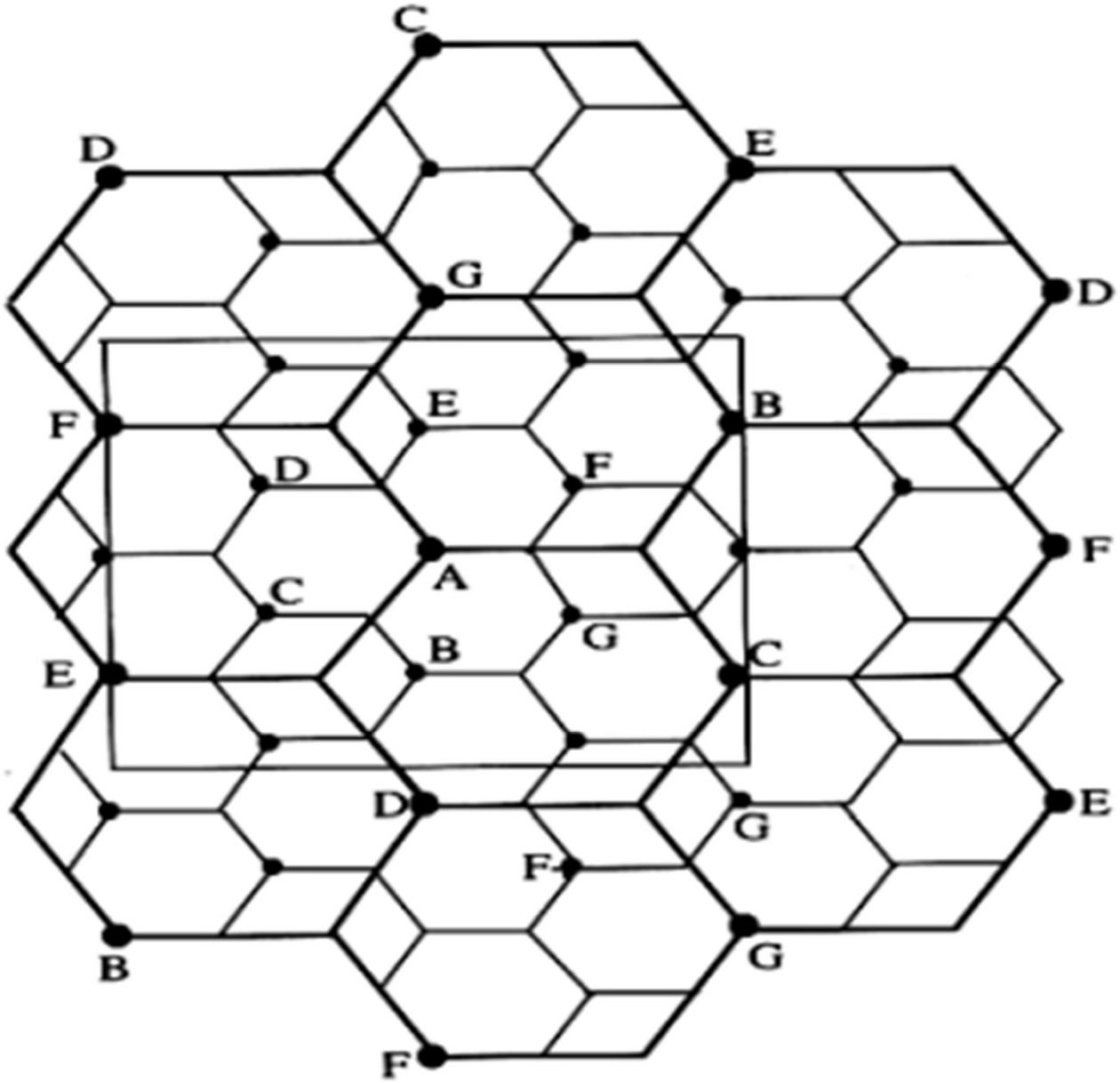


Illustration of cell splitting in 3x3 square centered around base station A

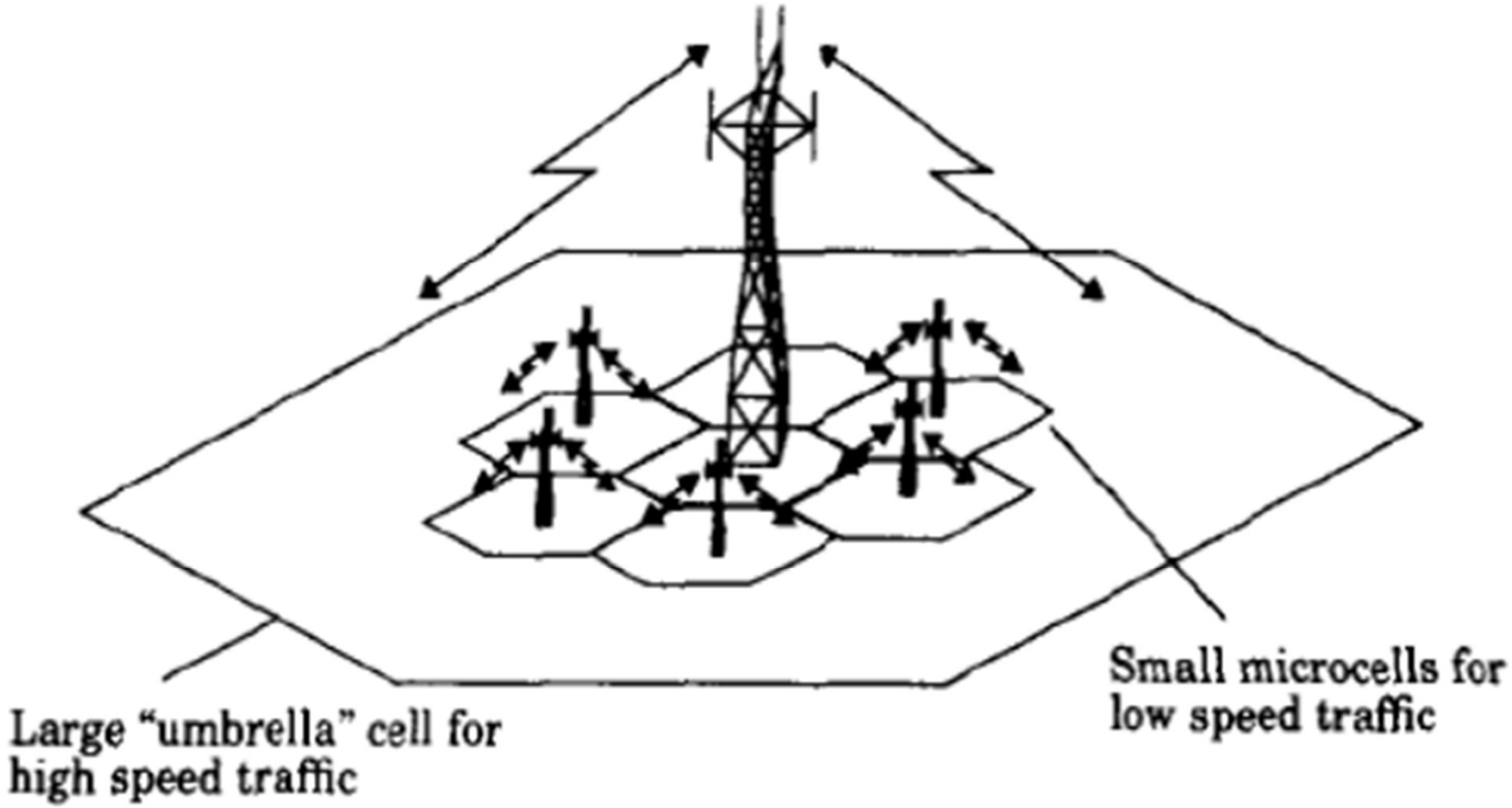


Overcome – Handoff

- In practice not all the cells are split at the same time hence different-sized cells will exist simultaneously.
- In such situations, special care needs to be taken to keep the distance between co-channel cells at a minimum and hence channel Assignments become more complicated.
- To overcome the handoff problem:
  - Channels in the old cell must be broken down into two channel groups - one for smaller cells & other for larger cell
  - The larger cell is dedicated to high-speed traffic so that handoffs occur less frequently
  - The smaller cell is dedicated to low-speed traffic so that handoffs occur more frequently
  - Initially, the small power group has fewer channels & large power group has a large no of channels but at the maturity of the system large power group does not have any channel

Umbrella Cells

Umbrella cell approach is possible by using different antenna heights and different power levels. By using this approach, it is possible to provide large and small cells which are co-located at a single location.



Sectoring (use of directional antenna)

- Another way to increase capacity is to keep the cell radius unchanged and to decrease the D/R ratio.
- The co-channel interference may be decreased by replacing a single **omnidirectional** antenna at the base station with several **directional** antennas.
- The factor by which co-channel interference is reduced depends on the amount of sectoring used.

Advantages

- Sectoring increases SIR increases the number of channels per unit area.
- capacity improvement by reducing the number of cells in a cluster.
- reduce the relative interference without decreasing the transmit power.

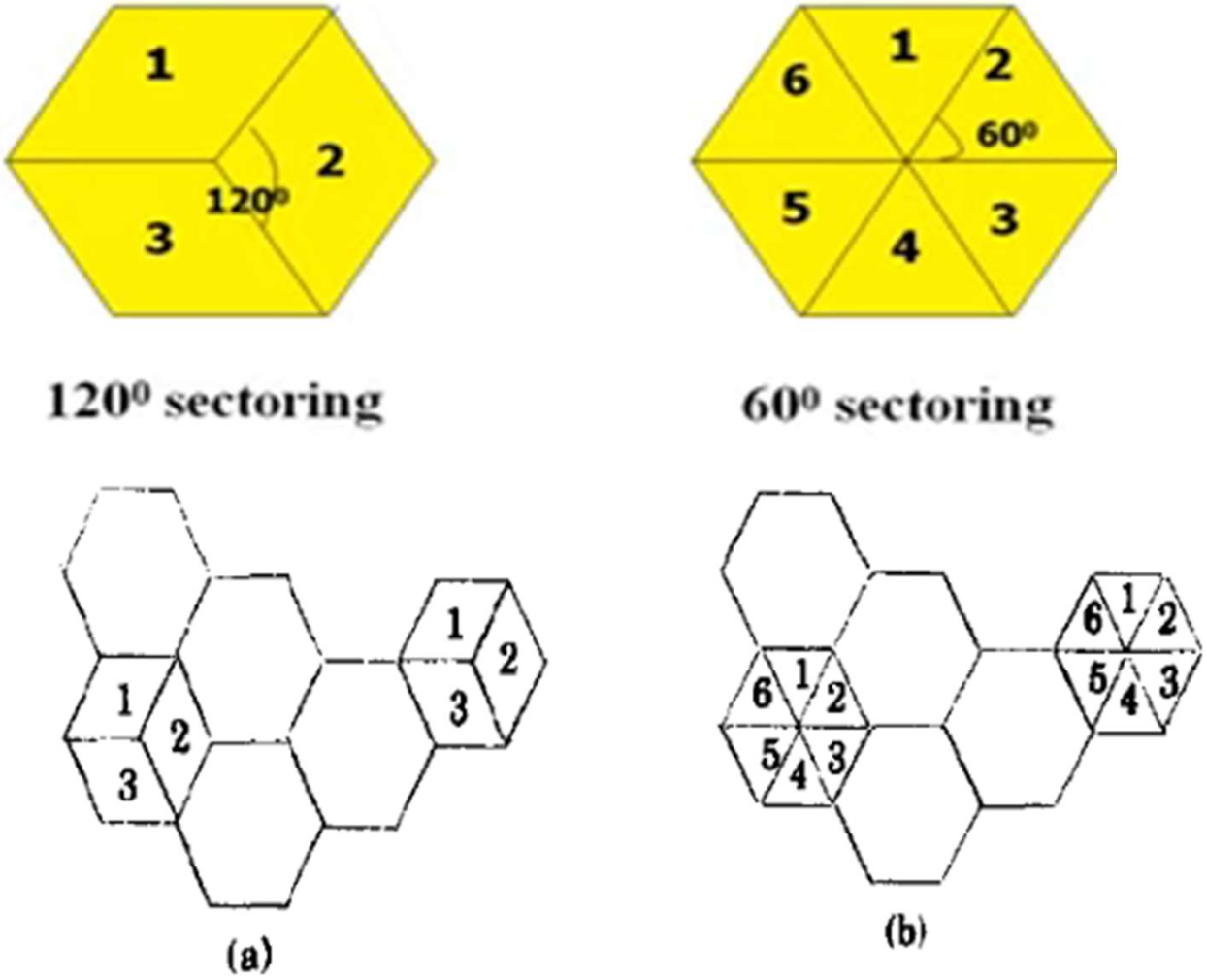
Disadvantages

- increased number of handoffs required
- increased load on the switching and control link elements of the mobile system.
- increased number of antennas at each base station
- decrease in trunking efficiency due to channel sectoring at the base station

Reducing Co-channel Interference

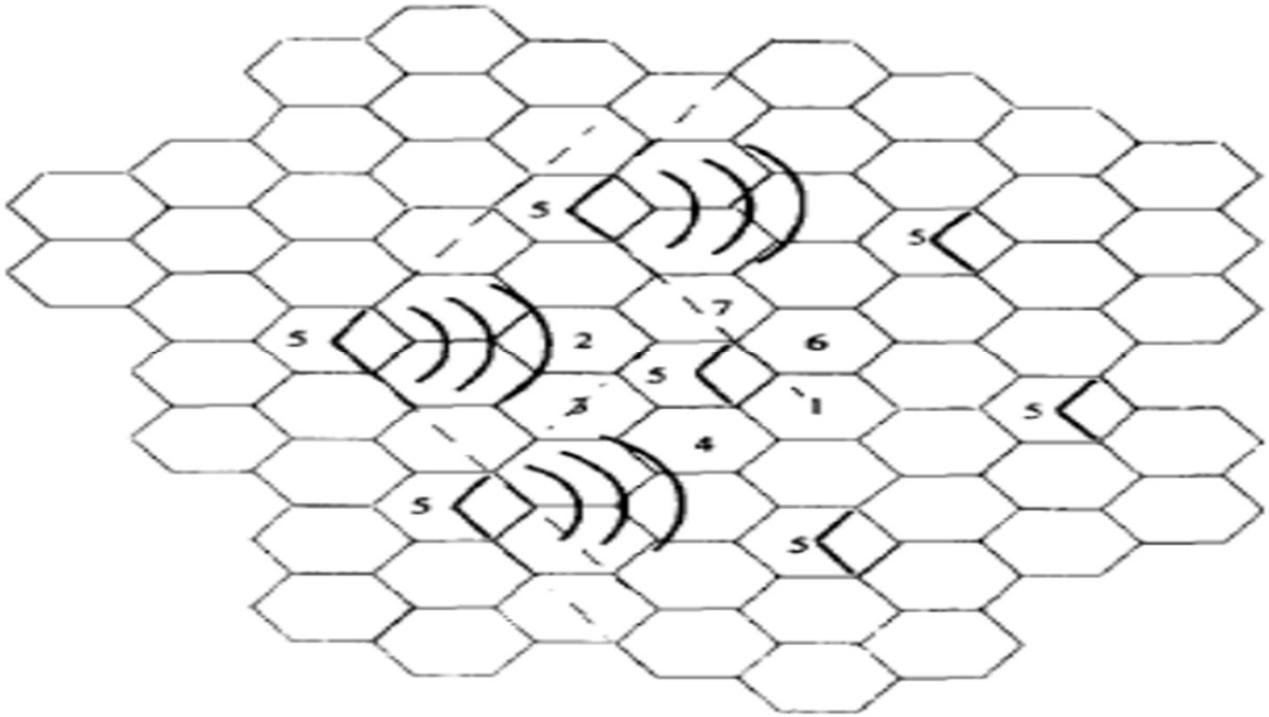
- A cell is normally partitioned into three 120° sectors or six 60° sectors as shown in Figure 1(a) and 1 (b).

Figure 1 (120 & 60) degrees Sectoring



120-degree Sectoring reducing Interference

- In Figure 2, consider the interference experienced by a mobile located in the right-most sector in the center cell labeled “5”.
- There are three co-channel cell sectors labeled “5” to the right of the center cell & three to the left of the center cell.
- Out of these six co-channel cells, only two cells have sectors with antenna patterns which radiate into the center cell.
- Hence a mobile in the center cell will experience interference on the forward link from only these two sectors.
- This S/I improvement allows to decrease the cluster size N in order to improve the frequency reuse & the system capacity



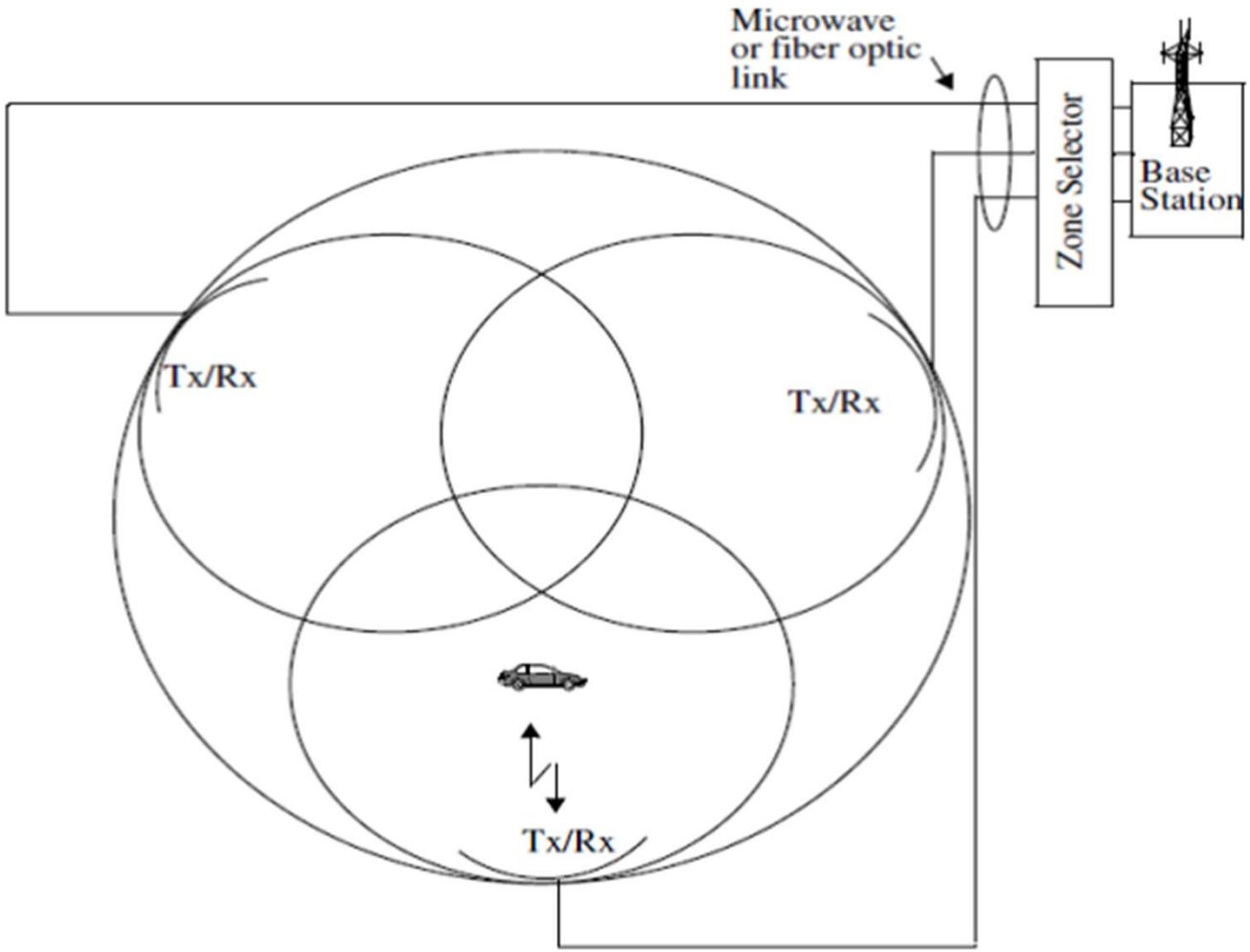
Microcell Zone Concept

- A solution to the problems in sectoring, a microcell concept is introduced.
- As a mobile travels from one zone to another within the cell, it retains the same channel. This technique is particularly useful along highways or along urban traffic corridors.
- In this scheme, each of the three zone sites represented as Tx/Rx in Figure 2 are connected to a single base station and share the same radio equipment.
- Thus, a given channel is active only in the particular zone in which the mobile is traveling & hence the base station radiation is localized and interference is reduced.
- The channels are distributed in time and space by all three zones & reused in co-channel cells in the normal fashion.
- The zones are connected by coaxial cable, fiber-optic cable, or microwave link to the base station. The base station simply switches the channel to a different zone site.
- Multiple zones and a single base station make up a cell.
- As a mobile travels within the cell, it is served by the zone with the strongest signal.

Advantages

- Enhance coverage for hard-to-reach areas
- a handoff is not required at the MSC when the mobile travels between zones within the cell
- A large central base station is replaced by several lower-powered transmitters
- Decreased co-channel interference improves the signal quality & increase capacity without degradation in trunking efficiency caused by sectoring.

The Micro-cell Concept



1.7 Repeaters and relay for Range Extension

- A wireless operator needs to provide dedicated coverage for hard-to-reach areas, such as within buildings, or in valleys or tunnels.
- Radio re-transmitters, known as **repeaters**, are used to provide such range extension capabilities.
- Repeaters are **bidirectional** in nature and simultaneously send signals to and receive signals from a serving base station.
- Repeaters may be installed anywhere and capable of repeating an entire cellular band.
- Upon receiving signals from a base station, the repeater **amplifies and reradiates** the base station signals to the specific coverage region.
- Unfortunately, the received noise & and interference are also reradiated by the repeater on both the forward and reverse link.

Disadvantage

- re-radiation of interference and noise

Repeaters Vs System Capacity

- Care must be taken to properly place the repeaters & and to adjust the various forward and reverse link amplifier levels and antenna patterns.
- In practice, directional antennas or distributed antenna systems (DAS) are connected to the inputs or outputs of repeaters for spot coverage, particularly in tunnels or buildings.
- By modifying the coverage of a serving cell, an operator can dedicate a certain amount of the base station's traffic to the areas covered by the repeater.
- However, the repeater does not add capacity to the system.
- Repeaters are being used to provide coverage into and around buildings, where coverage is weak.

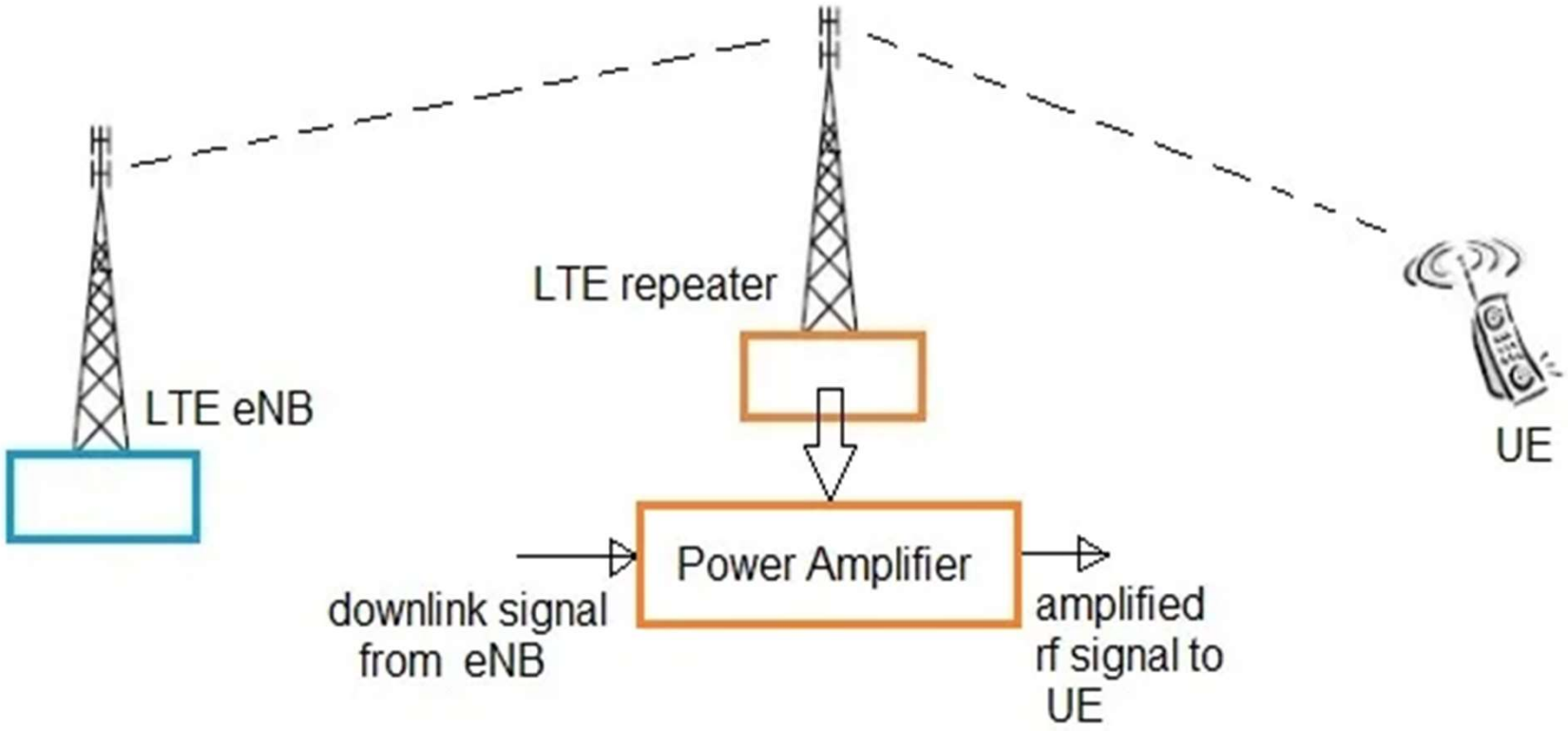
Location of Repeaters

- Determining the proper location for repeaters & and distributed antenna systems within buildings requires careful planning, to avoid re-radiation of interference into the building from the base station.
- Software products, such as Site-Planner allow engineers to determine the best placements for repeaters & and required DAS network while simultaneously computing the traffic and installation cost

Example: LTE repeater & LTE relay

LTE repeater

LTE repeater does just power amplification of the received signal from LTE eNB or LTE Base Station. After amplification, it transmits the signal back to the UEs(User Equipments) or Mobile Subscribers. Due to this received weak signal along with noise power also gets amplified. Due to this noise amplification, SNR(Signal-to-Noise Ratio) will not be improved and further degraded. LTE repeater operations have been depicted in Figure 1.



LTE relay basics and types

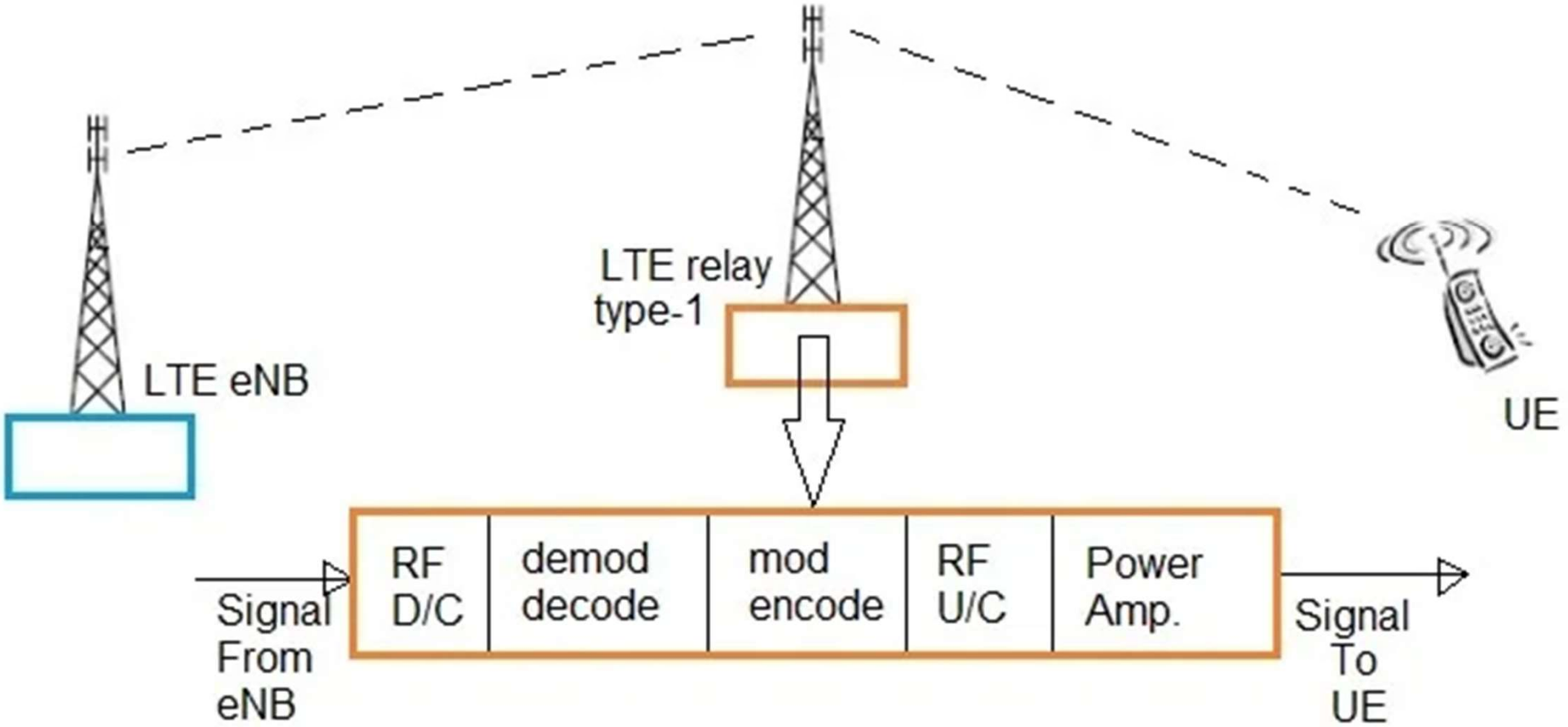
LTE relay type1

LTE relay is different than LTE repeater as it not only does amplification but also incorporates FEC functionality. Following are the steps involved in LTE relay as shown in Figure 2. This is known as LTE relay type-1.

- RF signal down conversion (RF to baseband).
- Demodulation and decoding (FEC decoder)
- modulation and encoding (FEC encoder)
- RF signal up-conversion (baseband to RF)
- RF Power Amplification

As LTE relay does amplification after decoding errors are reduced to a great extent. Hence SNR will be improved compared to LTE repeater.

Due to this LTE relay can be used even at the edge of the cell boundary where signals are weak and noisier compared to other cell areas. LTE relay supports half-duplex and full-duplex modes. It also supports inband and outband signaling modes.


LTE relay
type-1
LTE eNB
UE
RF
D/C
demod
decode
mod
encode
RF
U/C
Power
Amp.
Signal
From
eNB
Signal
To
UE

1.8 Trunking and Grade of Service

- The concept of trunking allows a large number of users to share the relatively small number of channels in a cell by providing access to each user, on demand, from a pool of available channels.
- In a trunked radio system, each user is allocated a channel on a per call basis, and upon termination of the call, the previously occupied channel is immediately returned to the pool of available channels.
- In a trunked mobile radio system, when a particular user requests service and all of the radio channels are already in use, the user is blocked, or denied access to the system.
- The fundamentals of trunking theory were developed by Erlang in the late 19th century. Today, the measure of traffic intensity bears his name.
- One Erlang represents the amount of traffic intensity carried by a channel that is completely occupied (i.e. one call-hour per hour or one call-minute per minute).
- For example, a radio channel that is occupied for thirty minutes during an hour carries 0.5 Erlangs of traffic.
- The Grade of Service (GoS) is a measure of the ability of a user to access a trunked system during the busiest hour.
- GoS is typically given as the likelihood that a call is blocked

1.8 Trunking and Grade of Service (Continued)

Unit of Traffic (Erlang)

• The fundamentals of trunking theory were developed by Erlang, a Danish mathematician, the unit bears his name.

• An Erlang is a unit of telecommunications traffic measurement.

• Erlang represents the continuous use of one voice path.

• Used to describe the total traffic volume of one hour.

• A channel kept busy for one hour is defined as having a load of one Erlang

• For example, a radio channel that is occupied for thirty minutes during an hour carries 0.5 Erlangs of traffic

• For 1 channel

– Min load = 0 Erlang (0% time utilization)

– Max load = 1 Erlang (100% time utilization)

1.8 Trunking and Grade of Service (Continued)

Key Definitions

- **Setup Time**: Time required to allocate a radio channel to a requesting user

- **Blocked Call**: Call which cannot be completed at the time of request, due to congestion

- **Holding Time**: Average duration of a typical call.

- **Request Rate (λ)**: The average number of calls requests per unit time

- **Traffic Intensity (A)**: Measure of channel time utilization measured in Erlangs. Dimensionless quantity.

- **Load**: Traffic intensity across the entire TRS (Erlangs)

1.8 Trunking and Grade of Service (Continued)

Grade of Service (contd,)

- **Grade of Service (GOS)**: Measure of the ability of a user to access a trunked system during the busiest hour.

- Also, a Measure of the congestion which is specified as a probability. ($0 \leq GOS \leq 1$)

- It is a benchmark to define the desired performance of a particular trunk system.

- **Blocked calls cleared (BCC)** or **Erlang B systems**: The probability of a call being blocked.

- **Blocked call delayed (BCD)** or **Erlang C systems**: The probability of a call being delayed beyond a certain amount of time before being granted access.

1.8 Trunking and Grade of Service (Continued)

Blocked Call Cleared Systems (BCC) (Erlang B systems)

- When a user requests service, there is a minimal call set-up time & user is given immediate access to a channel if one is available.

- If channels are already in use and no new channels are available, the call is blocked without access to the system.

- The user does not receive service but is free to try again later.

- All blocked calls are instantly returned to the user pool.

1.8 Trunking and Grade of Service (Continued)

Modeling of BCC Systems

- The Erlang B model is based on the following assumptions :
  - Calls are assumed to arrive with a Poisson distribution
  - Nearly an infinite number of users
  - Call requests are memory-less so that all users, including blocked users, may request a channel at any time
  - All free channels are fully available for servicing calls until all channels are occupied
  - The probability of a user occupying a channel is exponentially distributed.
  - Longer calls are less likely to happen
  - Finite number of channels available in the trunking pool.
  - Inter-arrival times of call requests are independent of each other

1.8 Trunking and Grade of Service (Continued)

- The traffic intensity offered by each user is:

$$A_u = \lambda H \quad \text{Erlang} \tag{2.1}$$

Where: λ is the request rate for each user.

- For a system (cell) containing N_{uscell} users and an unspecified number of channels, the total offered traffic intensity A, is given as:

$$A = N_{uscell}\ A_u \quad \text{Erlang} \tag{2.2}$$

- Furthermore, in N_{chcell} channel trunked system, if the traffic is equally distributed among the channels, then the traffic intensity per channel, A_{ch}, is given as:

$$A_{ch} = N_{uscell}\ A_u / N_{chcell} \quad \text{Erlang} \tag{2.3}$$

- When the offered traffic exceeds the maximum capacity of the system, the carried traffic becomes limited due to the limited capacity.
- The Erlang B formula determines the probability that a call is blocked and is a measure of the GoS for a trunked system which provides no queuing for blocked calls. It is given by:

$$\text{Probability of blocked calls} = P_b = \frac{\dfrac{A^{N_{chcell}}}{N_{chcell}!}}{\displaystyle\sum_{k=0}^{N_{chcell}} \frac{A^k}{k!}} = GoS \tag{2.4}$$

1.8 Trunking and Grade of Service (Continued)

- where N_{chcell} is the number of trunked channels offered by a trunked radio system or cell, and A is the total offered traffic.
- Erlang B formula provides a conservative estimate of the GoS, as the finite user results always predict a smaller likelihood of blocking. The capacity of a trunked radio system where blocked calls are lost is tabulated for various values of GoS and numbers of channels in Table 2.4.

Table 2.4 Capacity of an Erlang B system

# of Channels N_{chcell}	Capacity, A (Erlangs) for GOS = 0.01	= 0.005	= 0.002	= 0.001
2	0.153	0.105	0.065	0.046
4	0.869	0.701	0.535	0.439
5	1.36	1.13	0.900	0.762
10	4.46	3.96	3.43	3.09
20	12.0	11.1	10.1	9.41
24	15.3	14.2	13.0	12.2
30	**20.34**	**19.03**	**17.5**	**16.68**
31	**21.19**	**19.85**	**18.2**	**17.44**
40	29.0	27.3	25.7	24.5
41	**29.89**	**28.23**	**26.5**	**25.24**
70	56.1	53.7	51.0	49.2
100	84.1	80.9	77.4	75.2

1.8 Trunking and Grade of Service (Continued)

Example 11

Assume each user generates traffic intensity, $A_u = 0.1$ Erlangs, How many users per cell, N_{uscell} can be supported for blocking probability, P_b = GoS = 0.5% for the following number of trunked channels, N_{chcell} in a blocked calls cleared system?

a) $N_{chcell} = 5$ b) $N_{chcell} = 10$ c) $N_{chcell} = 20$ d) $N_{chcell} = 100$.

Solution

From Table 2.4, we can find the total capacity in Erlangs for the 0.5% GoS for different numbers of channels. By using the relation $A = N_{uscell}\, A_u$, we can obtain the total number of users that can be supported in the system.

a) Given $N_{chcell} = 5$, $A_u = 0.1$, GoS = 0.005. So, we obtain A = 1.13.
Therefore, total number of users, $N_{uscell} = A / A_u = 1.13 / 0.1 = 11$ user.

b) Given $N_{chcell} = 10$, $A_u = 0.1$, GoS = 0.005. So, we obtain A = 3.96.
Therefore, total number of users, $N_{uscell} = A / A_u = 3.96 / 0.1 = 39$ user.

c) Given $N_{chcell} = 20$, $A_u = 0.1$, GOS = 0.005. So, we obtain A = 11.1
Therefore, total number of users, $N_{uscell} = A / A_u = 11.1/0.1 = 110$ user.

d) Given $N_{chcell} = 100$, $A_u = 0.1$, GOS = 0.005. So, we obtain A = 80.9.
Therefore, total number of users, $N_{uscell} = A / A_u = 80.9/0.1 = 809$ user.

1.8 Trunking and Grade of Service (Continued)

Example 12 An urban area has a population of two million residents. Three competing trunked mobile networks (systems A, B, and C) provide cellular service in this area. System A has 394 cells ($N_{cell} = 394$) with $N_{chcell} = 20$ channels each, system B has 98 cells ($N_{cell} = 98$) with $N_{chcell} = 70$ channels each, and system C has 49 cells ($N_{cell} = 49$), each with $N_{chcell} = 100$ channels. Find the number of users that can be supported at 1% blocking if each user averages two calls per hour at an average call duration of three minutes,. Assuming that all three trunked systems are operated at maximum capacity, compute:

a) The total number of subscribes for each system

b) The percentage market penetration of each cellular provider.

Solution The request rate for each user, $\lambda = 2$ calls/Hour

The average duration of the call, H = 3 minutes

Probability of blocking, $P_b =$ GoS = 1 % = 0.01

a) For system A:

Number of channels per cell used in the system, $N_{chcell} = 20$ Channel

Traffic intensity per user, $A_u = \lambda H = 2 \times (3/60) = 0.1$ Erlangs

For GoS = 0.01 and $N_{chcell} = 20$, from the Erlang B Table 2.4:

The total carried traffic, A = 12 Erlangs.

Therefore, the number of users that can be supported per cell, N_{uscell} is:

$N_{uscell} = A / A_u = 12/0.1 = 95$ user.

1.8 Trunking and Grade of Service (Continued)

Example 12 Solution

Since there are 394 cells, the total number of subscribers that can be supported by system A is equal to 95 × 394 = 47280 subscribers.

For system B:

Number of channels per cell used in the system, N_{ch} = 70 Channel

Traffic intensity per user, $A_u = \lambda H = 2 \times (3/60) = 0.1$ Erlangs

For GoS = 0.01 and N_{chcell} = 70 Channel, from the Erlang B chart, Table 2.4:

The total carried traffic, A = 56.1 Erlangs. 51.0

Therefore, the number of users that can be supported per cell, N_{uscell} is:

$N_{uscell} = A / A_u = 56.1/0.1 = 561$

Since there are 98 cells, the total number of subscribers that can be supported by system B is equal to 561× 98 = 54,978 subscribers. 49,980

For system C:

Number of channels per cell used in the system, N_{chcell}= 100 Channel

Traffic intensity per user, $A_u = \lambda H = 2 \times (3/60) = 0.1$ Erlang.

For GOS = 0.01 and N_{chcell} = 100, from the Erlang B chart, Table 2.4:

The total carried traffic, A = 84.1 Erlang. 80.0

Therefore, the number of users that can be supported per cell, N_{uscell} is:

$N_{uscell} = A / A_u = 84.1/0.1 = 841$ subscribers.

1.8 Trunking and Grade of Service (Continued)

Example 12 Solution

Since there are 49 cells, the total number of subscribers that can be supported by system C is equal to 841 × 49 = 41,209 subscribers. 39,200

b) Therefore, total number of cellular subscribers that can be supported by these three systems are:

Total subscribers = 47,280 + 54,978 + 41,209 = 143,467 user or subscriber

- Since there are two million residents in the given urban area and the total number of cellular subscribers in system A is equal to 47280 therefore:

  System A of market penetration = 47,280 / 2,000,000 = 2.36 %

- Similarly, market penetration of system B is:

  System B of market penetration = 54,978 / 2,000,000 = 2.75 %

- The market penetration of system C is:

  System C of market penetration = 41,209 / 2,000,000 = 2.06 %

- The market penetration of the three systems combined is given by:

  The three-system market penetration = 129,626 / 2,000,000 = 7.17 %

1.8 Trunking and Grade of Service (Continued)

Example 13 A certain city has an area, $Area_{city}$ = 1,300 km^2 and is covered by a cellular system using reuse pattern, N =7. Each cell has a radius, R = 2 km and the city is allocated 40 MHz of spectrum with a full duplex channel bandwidth of 60 kHz. Assume a GoS of 1% for an Erlang B system is specified. If the offered traffic per user, A_u = 0.03 Erlangs, compute the following:

a) The number of clusters in the service area.
b) The number of channels per cell.
c) Traffic intensity of each cell.
d) The maximum carried traffic.
e) The total number of users that can be served for 1% GOS
f) The number of mobiles per unique channel (where it is understood that channels are reused).
g) the theoretical maximum number of users that could be served at one time by the system.

Solution a) For the system, Given:

Total coverage area, $Area_{city}$ = 1300 km^2 m, and cell radius, R = 2 km.

The cell area (hexagon) can be shown to be $2.5981R^2$, thus:

$$Area_{cell} = 2.5981 \times (2)^2 = 10.4 \text{ km}^2$$

Cluster area $Area_{cluster}$ = N x Cell area = 7 x 10.4 = 72.8 km^2

Hence, the total number of clusters are N_{cls}= 1300/72.8 = 18 cluster

1.8 Trunking and Grade of Service (Continued)

Example 13 Solution

b) # of channels per cluster, $N_{chcls} = N_{chnet} = 40{,}000{,}000/60{,}000 = 666$ channel

of channels per cell, $N_{chcell} = 666/7 = 95$ channel

c) Given: $N_{chcell} = 95$, and GoS = 0.01

From the Erlang B Table 2.4, we have traffic intensity per cell, A_{cell} is:

$A_{cell} = 84.1$ Erlang/cell

d) Maximum carried traffic = traffic intensity per cell x 7 x number of clusters

$A_{total} = 84.1 \times 7 \times 18 = 10596.6$ Erlang.

e) The traffic per user, $A_u = 0.03$ Erlangs

of users per cell (Cell Capacity), $N_{uscell} = A_{cell} / A_u = 84.1 / 0.03 = 2804$ user

Total number of users, N_{usnet} = Total traffic / traffic per user:

$N_{usnet} = A_{total} / A_u = 10596.6 / 0.03 = 353{,}220$ user

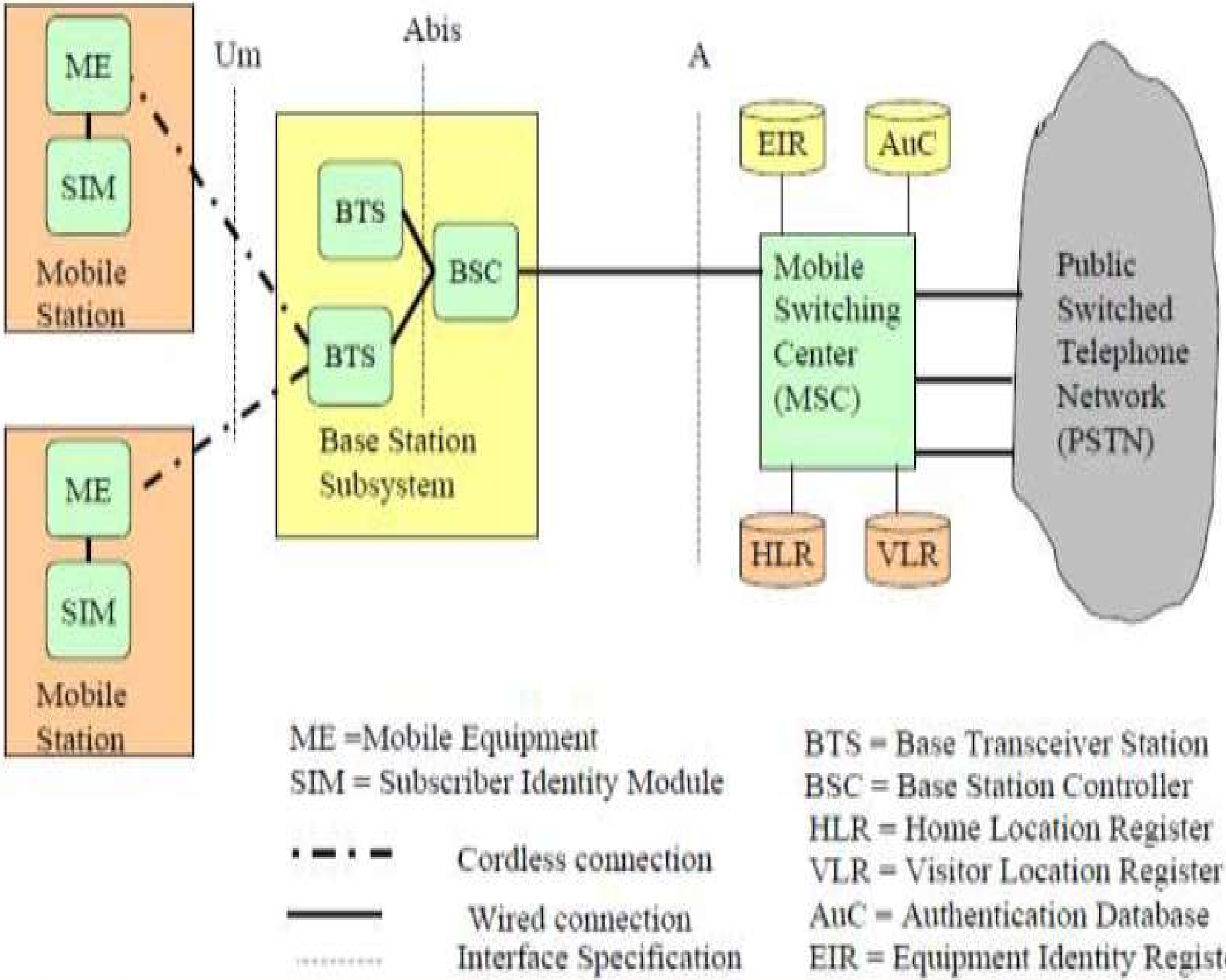
or $N_{usnet} = N_{uscell} \times N \times N_{cls} = 2804 \times 7 \times 18 = 353{,}220$ user.

f) Number of mobiles per channel = number of users/number of channels

$= 353{,}178 / 666 = 530$ mobiles/channel.

g) The maximum number of users that could be served at one time by this system: System capacity =

$N_{usnet} = N_{chcell} \times 8 \times N \times N_{cls} = 95 \times 8 \times 7 \times 18 = 95760$ user


ME
SIM
Mobile Station
ME
SIM
Mobile Station
Um
Abis
A
BTS
BTS
BSC
Base Station Subsystem
EIR
AuC
Mobile Switching Center (MSC)
HLR
VLR
Public Switched Telephone Network (PSTN)
ME =Mobile Equipment
SIM = Subscriber Identity Module
Cordless connection
Wired connection
Interface Specification
BTS = Base Transceiver Station
BSC = Base Station Controller
HLR = Home Location Register
VLR = Visitor Location Register
AuC = Authentication Database
EIR = Equipment Identity Register